No Global Gauge in Neural Weight Space: Branched Quotient Geometry and Atlas-Optimal Learning

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Abstract

Neural parameter spaces with exchangeable blocks are quotient spaces under mixed continuous and discrete symmetries, but existing analyses are largely confined to regular strata where the quotient is smooth. For a positively homogeneous two-layer base model, we identify the regular quotient exactly as an unordered configuration space. For a signature-complete exchangeable-block family, we then show that every simple pairwise collision has local normal form $(u, v) \mapsto (u, [v])$; on every transverse two-dimensional slice this is the branched double cover $z \mapsto z^2$. Hence linked punctured neighborhoods admit no global continuous gauge, and any canonicalization written in branched invariant coordinates has unavoidable $\epsilon^{-1/2}$ conditioning blow-up. We further prove that the ambient relative sectional category $\mathfrak{a}(K; q)$ is the exact chart threshold for canonicalization-based invariant learning, section-valued learning on associated bundles, and symmetry-respecting memoryless optimizer proposals. Controlled experiments recover the predicted monodromy, the exact conditioning law with leading constants, and the sharp phase transition at chart count 2. The resulting message is intrinsic: weight-space computation near singular strata should be atlas-based rather than globally gauge-fixed.

1. Introduction

Weight space is naturally a quotient: exchangeable units and internal gauge transformations identify many parameter vectors that realize the same predictor. On the regular locus this quotient is smooth, and recent work has used that smooth geometry to study loss landscapes and weight-space learning (Simsek et al., 2021; Navon et al., 2023; Dong & Cheng, 2026; Dayan et al., 2026). Mixed gauge-permutation groups also arise in modern architectures (Wang & Wang, 2025). The first singular step beyond that regime is the collision of two exchangeable blocks.

This paper shows that this first singularity already rules out a single global continuous gauge. After quotienting the continuous block symmetry, every simple pairwise collision has universal local model $(u, v) \mapsto (u, [v])$, hence $z \mapsto z^2$ on transverse two-dimensional slices. Three consequences follow: linked punctured neighborhoods admit no global continuous gauge, any canonicalizer written in branched invariant coordinates has exact $\epsilon^{-1/2}$ conditioning blow-up, and the minimal chart count for canonicalization-based learning is the ambient relative sectional category $\mathfrak{a}(K; q)$.

Our contributions are fourfold: an exact quotient identification for a homogeneous two-layer base model; a universal branch normal form for simple collisions; sharp chart-threshold theorems for invariant outputs, associated-bundle sections, and memoryless optimizer proposals; and theorem-validation experiments recovering the predicted monodromy, conditioning law, and two-chart phase transition. The resulting perspective is intrinsic: near singular strata, weight-space computation should be formulated in terms of atlases, transition maps, and associated bundles rather than a single global canonicalizer.

2. Setup and Problem Formulation

We model exchangeable-block parameter spaces as quotient objects under mixed continuous and discrete symmetries.

Assumption 2.1 (Exchangeable-block family)**.** The parameter space is a smooth G-invariant submanifold $\Theta \subseteq \Xi^m$, where $G := H^m \rtimes S_m$ acts by

$$((h_1, \dots, h_m), \pi) \cdot (\xi_1, \dots, \xi_m) = \big(h_1 \cdot \xi_{\pi^{-1}(1)}, \dots, h_m \cdot \xi_{\pi^{-1}(m)}\big). \quad (1)$$

and the realization map $\mathcal{R} : \Theta \to \mathcal{F}$ is G-invariant.

Assumption 2.2 (Signature-complete regular block quotient)**.** There exists an H-invariant open submanifold

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$\Xi_{\rm reg} \subseteq \Xi$ and a smooth signature map $\psi : \Xi_{\rm reg} \to M$ such that H acts freely and properly on $\Xi_{\rm reg}$, and the induced map $\bar{\psi} : \Xi_{\rm reg}/H \to M$ is a diffeomorphism.

Definition 2.3 (Regular locus, local gauges, and atlas complexity)**.** The regular locus is

$$\Theta_{\rm reg} := \Big\{(\xi_1, \dots, \xi_m) \in \Theta \cap \Xi^m_{\rm reg} : \psi(\xi_i) \neq \psi(\xi_j) \text{ for all } i \neq j\Big\},$$
$$\Sigma := \Theta \setminus \Theta_{\rm reg}.$$

Under Assumptions 2.1–2.2, $q : \Theta_{\rm reg} \to B_{\rm reg}$ is a principal G-bundle, where $B_{\rm reg} := \Theta_{\rm reg}/G$. A *local gauge* on $U \subseteq B_{\rm reg}$ is a continuous section $s : U \to \Theta_{\rm reg}$ with $q \circ s = \mathrm{id}_U$. For compact $K \subseteq B_{\rm reg}$,

$$\mathfrak{a}(K; q) := \mathrm{secat}_{B_{\rm reg}}(q; K),$$

i.e., the smallest number of ambient open neighborhoods of K that admit local gauges.

Definition 2.4 (Invariant and section-valued targets)**.** For compact $K \subseteq B_{\rm reg}$ and $p \in \mathbb{N}$, let $C_G(q^{-1}(K), \mathbb{R}^p)$ denote the continuous maps $f : q^{-1}(K) \to \mathbb{R}^p$ satisfying $f(g\cdot\theta) = f(\theta)$ for all $g \in G$ and $\theta \in q^{-1}(K)$. For a continuous finite-dimensional representation $\rho : G \to \mathrm{GL}(V)$, covariant outputs are continuous sections of the associated bundle

$$E_\rho := (\Theta_{\rm reg} \times V)/G \to B_{\rm reg}.$$

Section 3 studies the first singular failure of regularity; Section 4 turns the resulting obstruction into exact learning thresholds.

3. Branched Quotient Geometry and No-Global-Gauge Results

The first nonregular event is a transverse pairwise collision of exchangeable block signatures.

Theorem 3.1 (Exact quotient for the homogeneous base model)**.** *Consider the positively homogeneous two-layer network*

$$f_\theta(x) = \sum_{i=1}^m a_i \sigma(w_i^\top x), \qquad \sigma(ct) = c\,\sigma(t) \textit{ for } c > 0.$$

On

$$\Theta^{\rm hom}_{m,\rm reg} := \Big\{((a_i, w_i))_{i=1}^m : a_i \neq 0,\ w_i \neq 0,\ u_i \neq u_j \textit{ for all } i \neq j\Big\},$$

where $u_i := (a_i\|w_i\|, w_i/\|w_i\|)$, the canonical scaling action is

$$c_i \cdot (a_i, w_i) = (a_i/c_i, c_i w_i),$$

and the quotient by $(\mathbb{R}_{>0})^m \rtimes S_m$ is diffeomorphic to

$$\mathrm{UConf}_m\big((\mathbb{R} \setminus \{0\}) \times S^{d-1}\big).$$

Return now to the abstract setup. Write $\Theta^\sharp := \Theta \cap \Xi^m_{\rm reg}$, $\widetilde{\Theta} := \Theta^\sharp/H^m$, and let $\pi : \widetilde{\Theta} \to \widetilde{\Theta}/S_m$ be the residual permutation quotient. A *simple pairwise collision* is a point of $\Theta^\sharp$ where exactly one pair of signatures coincides transversely while all remaining signatures stay distinct.

Theorem 3.2 (Local branch normal form)**.** *Let $\theta^\star \in \Theta^\sharp$ be a simple pairwise collision, and set $r := \dim \widetilde{\Theta}$, $d := \dim M$, $x^\star := [\theta^\star]_{H^m}$, and $y^\star := [x^\star]_{S_m}$. Then some neighborhoods $U \ni x^\star$, $V \ni y^\star$, an open set $U_F \subseteq \mathbb{R}^{r-d}$, and a radius $\rho > 0$ admit coordinates*

$$\chi : U \cong U_F \times B_\rho(0), \qquad \overline{\chi} : V \cong U_F \times (B_\rho(0)/\{\pm 1\}),$$

for which $\overline{\chi} \circ \pi \circ \chi^{-1}(u, v) = (u, [v])$. Equivalently, after quotienting the continuous gauge and freezing the noncolliding blocks, the residual singularity is $\mathbb{R}^d/\{\pm 1\}$. If $d \geq 2$, every transverse two-dimensional slice is topologically equivalent to $z \mapsto z^2$.

Proof sketch. Reduce first by H^m. Near a simple collision, the only residual stabilizer is the transposition $\tau = (12)$; the reduced collision-difference map is a submersion whose zero set is $\mathrm{Fix}(\tau)$. Averaging a Riemannian metric over $\langle\tau\rangle$ and using a tubular neighborhood linearizes the involution to $(u, v) \mapsto (u, -v)$, and quotienting by $\langle\tau\rangle \cong \mathbb{Z}_2$ yields the model above. □

Corollary 3.3 (No global gauge and exact local atlas complexity)**.** *Assume $d \geq 2$. Every sufficiently small transverse two-dimensional slice through a simple pairwise collision contains compact annuli $K_{\epsilon,R} \subseteq B_{\rm reg}$ with $0 < \epsilon < R$ such that*

$$\mathfrak{a}(K_{\epsilon,R}; q) = 2.$$

Hence $q|_{q^{-1}(K_{\epsilon,R})}$ admits no continuous global section.

Proof sketch. On the transverse slice the quotient is $z \mapsto z^2$: the annulus has nontrivial monodromy, so no global section exists, while two slit annuli carry the two square-root branches. □

Corollary 3.4 (Branched-coordinate conditioning blow-up)**.** *For every sufficiently small slit-sector refinement $K^\delta_{\epsilon,R} \subseteq K_{\epsilon,R}$ with $0 < \delta < \pi$, there exist branched invariant coordinates $\zeta^\delta_{\epsilon,R}$ and a constant $C > 0$, independent of ϵ, such that every reduced local gauge $\widetilde{s}$ has branched-coordinate Lipschitz seminorm at least $C\,\epsilon^{-1/2}$. The same lower bound survives any fixed smooth normalization of the continuous gauge.*

4. Atlas-Optimal Learning and Optimization Consequences

Fix a compact set $K \subseteq B_{\rm reg}$, a smooth embedding $\iota : \Theta_{\rm reg} \hookrightarrow \mathbb{R}^D$, and a Euclidean model class $\mathcal{N}_{D,p} \subset$

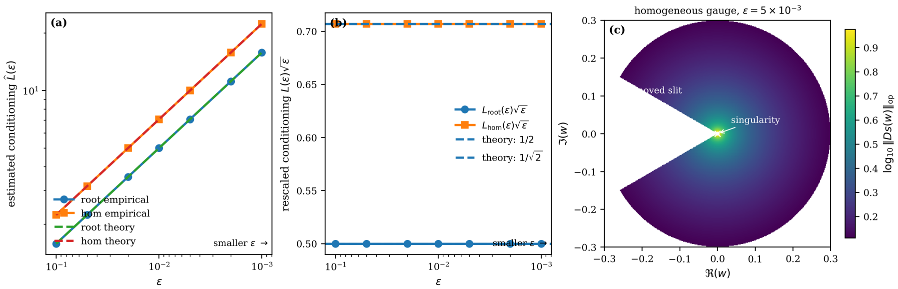

Figure 1. Conditioning near the branch singularity. Panel (a) shows the predicted $\epsilon^{-1/2}$ law for the exact square-root branch and the normalized homogeneous gauge; panel (b) recovers the constants $1/2$ and $1/\sqrt{2}$; panel (c) localizes the singularity near the branch point and removed slit.

Table 1. Fits recover the exact exponent and leading constant.

Metric	Root	Hom.
	-0.5	-0.5
$\hat{\beta}/\beta_{\mathrm{th}}$	-0.5	-0.5
	0.5	0.7071
$\hat{C}/C_{\mathrm{th}}$	0.5	0.7071
max rel. err.	4.88×10^{-6}	4.88×10^{-6}

$C(\mathbb{R}^D, \mathbb{R}^p)$ that is dense on compact subsets. An N-chart gauge-canonicalization scheme on K consists of open sets $U_1, \ldots, U_N \subseteq B_{\mathrm{reg}}$ covering K, local gauges $s_i : U_i \to \Theta_{\mathrm{reg}}$, and a continuous partition of unity $(\lambda_i)_{i=1}^N$ subordinate to $(U_i)_{i=1}^N$. Its invariant hypothesis class is

$$\mathcal{H}_{\mathcal{A},p} := \Big\{\theta \mapsto \sum_{i=1}^{N} \lambda_i(q(\theta))\, h_i(\iota(s_i(q(\theta)))) : h_i \in \mathcal{N}_{D,p}\Big\}.$$

Theorem 4.1 (Exact chart threshold for canonicalization-based invariant learning)**.** *Let $\mathfrak{U}_{\mathrm{can}}(K; q, p)$ be the smallest N for which an N-chart scheme is dense in $C_G(q^{-1}(K), \mathbb{R}^p)$ under $\|\cdot\|_\infty$. Then*

$$\mathfrak{U}_{\mathrm{can}}(K; q, p) = \mathfrak{a}(K; q).$$

Equivalently, a universal N-chart canonicalization-based invariant learner exists iff $N \geq \mathfrak{a}(K; q)$.

Proof sketch. The lower bound is immediate: any universal N-chart scheme already provides N local gauges over an ambient open cover of K. For the upper bound, choose a minimal atlas, descend the invariant target to K, approximate its chartwise representations in $\mathbb{R}^D$, and glue with a partition of unity. □

Theorem 4.2 (Exact chart threshold for section-valued canonicalization)**.** *For every compact $K \subseteq B_{\mathrm{reg}}$ and every finite-dimensional continuous representation $\rho : G \to \mathrm{GL}(V)$, the smallest chart count for density in the uniform section norm on $\Gamma(E_\rho|_K)$ is again*

$$\mathfrak{U}_{\mathrm{sec,can}}(K; q, \rho) = \mathfrak{a}(K; q).$$

Hence the same topological threshold governs canonicalization-based universality for bundle-valued covariant outputs.

Proof sketch. Local gauges trivialize E_ρ. The lower bound is unchanged; for the upper bound, approximate the local coefficient maps in those trivializations and glue the resulting local sections with the same partition of unity. □

Corollary 4.3 (No single global gauge-fixed universal optimizer)**.** *If Θ_{reg} is an open subset of a finite-dimensional representation W_{opt} of G, then symmetry-respecting memoryless optimizer proposals are sections of $E_{\mathrm{opt}} := (\Theta_{\mathrm{reg}} \times W_{\mathrm{opt}})/G \to B_{\mathrm{reg}}$, and their exact canonicalization-based chart threshold is again $\mathfrak{a}(K; q)$. In particular, if $\mathfrak{a}(K; q) > 1$, no single-chart gauge-fixed proposal class is universal on K.*

Combined with Corollary 3.3, every linked annulus $K_{\epsilon,R}$ has sharp local threshold 2 for invariant prediction, section-valued learning, and memoryless optimizer proposals.

5. Experiments

All experiments are theorem-validation studies on analytically controlled quotient domains. Figures 1 and Table 1 validate Corollary 3.4: dense finite-difference estimates recover the exact $\epsilon^{-1/2}$ law and the constants $1/2$ and $1/\sqrt{2}$. Figures 2 and Table 2 validate Theorem 4.1: on the linked annulus, one chart fails precisely at the branch cut, two charts fit the sampled atlas class to numerical precision, and a third chart brings no meaningful improvement. Whenever closed-form geometry is available we use exact lifts and dense-grid finite differences; the atlas-learning targets are drawn from smooth chart classes and fit by deterministic ridge regression in double precision. Appendix experiments isolate the monodromy loop test and the bundle-valued, optimizer, and local attention-head instantiations, all of which match the same predicted threshold.

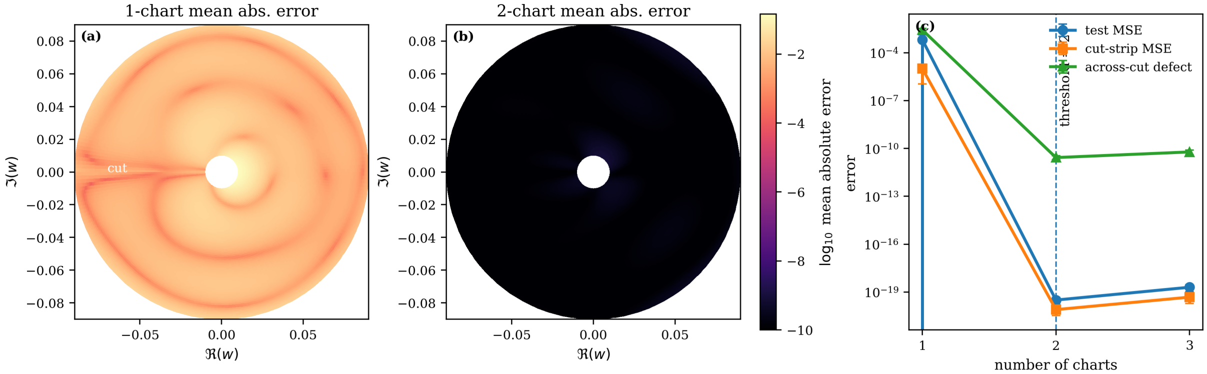

Figure 2. Exact chart threshold on the linked annulus. A one-chart canonicalizer leaves a cut-aligned error ridge, whereas a two-chart atlas reaches numerical precision; chart count 3 yields no material gain.

Table 2. Aggregate diagnostics (mean $\pm$ s.d. over target draws).

Metric	1 ch.	2 ch.	3 ch.
Features	21	20	30
Test MSE	6.54×10^{-4} $\pm 7.18 \times 10^{-4}$	3.12×10^{-20} $\pm 1.5 \times 10^{-20}$	1.99×10^{-19} $\pm 9.31 \times 10^{-20}$
Max abs. err.	0.1446 ± 0.1013	9.1×10^{-10} $\pm 2.29 \times 10^{-10}$	2.08×10^{-9} $\pm 5.28 \times 10^{-10}$
Cut-strip MSE	1.02×10^{-5} $\pm 9.09 \times 10^{-6}$	7.86×10^{-21} $\pm 4.52 \times 10^{-21}$	4.77×10^{-20} $\pm 2.84 \times 10^{-20}$
Across-cut defect	2.71×10^{-3} $\pm 6.48 \times 10^{-4}$	2.68×10^{-11} $\pm 7.63 \times 10^{-12}$	6.04×10^{-11} $\pm 1.97 \times 10^{-11}$

6. Conclusion

The central claim of this paper is that neural weight spaces with exchangeable blocks should not be modeled as Euclidean parameter sets with a nuisance relabeling superimposed afterward. On the regular locus they are principal quotients; near the first singular event—a simple pairwise collision—their local geometry is branched. We made this precise in two steps: first by identifying the regular quotient exactly in a homogeneous two-layer base model, and then by proving that every simple collision has the universal local normal form

$$(u, v) \mapsto (u, [v]), \qquad v \sim -v,$$

and hence reduces on transverse two-dimensional slices to the branched double cover $z \mapsto z^2$. From this branch model we obtained three consequences: there is, in general, no globally continuous gauge around linked collision loops; any canonicalization in branched invariant coordinates must become ill-conditioned with the exact $\epsilon^{-1/2}$ law; and the minimum number of local gauges required for canonicalization-based learning is not heuristic but exactly the ambient relative sectional category $\mathfrak{a}(K; q)$.

The second contribution is operational. For canonicalization-based invariant learning, section-valued learning, and symmetry-respecting memoryless optimizer proposals, we proved the sharp threshold $N \geq \mathfrak{a}(K; q)$. Thus the same topological invariant governs scalar outputs on the quotient, bundle-valued covariant outputs, and local optimizer fields. The obstruction is therefore not merely that a single global canonicalizer may fail; it has an exact replacement principle, namely an atlas of minimal size, with no smaller chart system capable of universality within the canonicalization-based paradigm. The experiments corroborate this theorem chain at three levels: exact monodromy in the homogeneous model, exact recovery of the $\epsilon^{-1/2}$ conditioning law and leading constants, and a sharp empirical phase transition at chart count 2 for invariant learning. The bundle-valued and local attention-head instantiations deferred to the appendix show the same atlas threshold in a modern architecture slice.

Conceptually, these results sharpen recent quotient-geometric and weight-space learning viewpoints into a genuinely singular perspective. Once simple collisions are allowed, the state space is no longer captured by principal bundle language alone. Methodologically, atlas-based and genuinely gauge-aware constructions are therefore not optional conveniences but mathematically forced objects. The next problems are global: orbit-type stratifications beyond simple pairwise collisions, singular quotient invariants beyond the regular principal bundle, and optimization procedures that evolve directly on quotient or associated-bundle state spaces. Those directions are not peripheral refinements; they are the natural extension of the atlas-based picture already forced by the first branch singularity.

Impact Statement

This paper is a foundational contribution to the theory of neural weight-space symmetries. Its primary goal is to improve the mathematical understanding of neural parameter spaces by showing that, near simple collision strata, the appropriate state space is a singular quotient with branched local geometry rather than a globally gauge-fixable Euclidean parameterization. We therefore expect its most immediate impact to be scientific and methodological: clarifying when canonicalization-based weight-space pipelines are intrinsically brittle, and providing a principled atlas-based alternative for invariant learning, section-valued learning, and symmetry-respecting optimizer design.

The positive downstream implications are mainly in reliability and rigor. Methods for model merging, weight-space learning, optimizer learning, and symmetry-aware analysis are increasingly being proposed for large models, yet many of these methods implicitly assume that one can choose a single globally well-behaved canonical representative. Our results identify a precise failure mode for that assumption and provide a mathematically grounded replacement in terms of local gauges, transition maps, and associated bundles. To the extent that future systems adopt this perspective, they may become more stable, interpretable, and less likely to suffer from hidden discontinuities induced by improper canonicalization.

Potential negative impacts are indirect but worth noting. Any theory that improves the efficiency or robustness of weight-space manipulation could, in principle, be used to accelerate model editing, model merging, or optimizer construction for systems that are later deployed in harmful settings. However, the present work does not itself introduce a deployment-ready method for training or modifying frontier models. Its results are theorem-level statements about geometric obstructions and expressive thresholds, and the empirical section validates these statements only on analytically controlled quotient domains and local architecture slices. The paper uses no human subjects data, no personal data, and no safety-critical application data.

Overall, we believe the main impact of this work is to raise the level of mathematical precision in weight-space learning and symmetry-aware optimization. By identifying topological obstructions rather than ignoring them, the paper argues for a safer and more principled design philosophy: when global canonicalization is impossible, algorithms should be built intrinsically on atlases and transition structure rather than on unstable single-chart representations.

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A. Topological Preliminaries and Notation

This appendix records the standard topological and bundle-theoretic notions used throughout the paper. We keep the discussion deliberately minimal: configuration spaces, free finite-group quotients, ambient relative sectional category, principal-bundle gauges, associated bundles, partitions of unity, and bundle norms. The only nonstandard convention is that, for compact subsets of the base manifold, we use an *ambient* relative sectional category, which is the exact quantity counted by gauge atlases and canonicalization-based local coordinate schemes.

Throughout this appendix, B denotes a smooth finite-dimensional manifold, G a Lie group, and $K \subseteq B$ a nonempty compact subset. All group actions are on the left.

A.1. Configuration spaces and free finite quotients

Definition A.1 (Ordered and unordered configuration spaces)**.** Let M be a Hausdorff topological space and let $m \in \mathbb{N}$. The ordered configuration space of m points in M is

$$\mathrm{Conf}_m(M) := \{(z_1, \ldots, z_m) \in M^m : z_i \neq z_j \text{ for all } i \neq j\}.$$

The symmetric group S_m acts on $\mathrm{Conf}_m(M)$ by coordinate permutation:

$$\pi \cdot (z_1, \ldots, z_m) := (z_{\pi^{-1}(1)}, \ldots, z_{\pi^{-1}(m)}).$$

The unordered configuration space is the orbit space

$$\mathrm{UConf}_m(M) := \mathrm{Conf}_m(M)/S_m.$$

Proposition A.2 (Configuration spaces as smooth free quotients)**.** *If M is a smooth manifold, then:*

1. $\mathrm{Conf}_m(M)$ *is an open smooth submanifold of* M^m*;*
2. *the action of* S_m *on* $\mathrm{Conf}_m(M)$ *is smooth, free, and proper;*
3. *the quotient* $\mathrm{UConf}_m(M)$ *is a smooth manifold;*
4. *the quotient map*
$$\pi_{\mathrm{conf}} : \mathrm{Conf}_m(M) \to \mathrm{UConf}_m(M)$$
is a finite regular covering map with deck group S_m*.*

Proof. For each pair $1 \leq i < j \leq m$, the diagonal

$$\Delta_{ij} := \{(z_1, \ldots, z_m) \in M^m : z_i = z_j\}$$

is closed in M^m because M is Hausdorff. Hence

$$\mathrm{Conf}_m(M) = M^m \setminus \bigcup_{1 \leq i < j \leq m} \Delta_{ij}$$

is open, so it is a smooth submanifold of M^m.

The permutation action is smooth because it is induced by smooth coordinate permutations of M^m. It is free: if

$$\pi \cdot (z_1, \ldots, z_m) = (z_1, \ldots, z_m)$$

for some $(z_1, \ldots, z_m) \in \mathrm{Conf}_m(M)$, then $z_{\pi^{-1}(i)} = z_i$ for all i. Since the coordinates are pairwise distinct, $\pi(i) = i$ for every i, hence $\pi = e$.

Properness is automatic because S_m is finite. Therefore the quotient $\mathrm{UConf}_m(M)$ is a smooth manifold.

It remains to prove that π_{conf} is a regular covering map. Fix $z = (z_1, \ldots, z_m) \in \mathrm{Conf}_m(M)$. Because M is Hausdorff and the points $z_1, \ldots, z_m$ are pairwise distinct, there exist pairwise disjoint open neighborhoods

$$V_1 \ni z_1, \ \ldots, \ V_m \ni z_m.$$

Set

$$U := V_1 \times \cdots \times V_m \subseteq \mathrm{Conf}_m(M).$$

If $\pi \neq e$, then $\pi(U) \cap U = \varnothing$: indeed, if $(w_1, \ldots, w_m) \in \pi(U) \cap U$, then for some i with $\pi^{-1}(i) \neq i$, the i-th coordinate belongs simultaneously to the disjoint sets V_i and $V_{\pi^{-1}(i)}$, a contradiction.

Therefore

$$\pi_{\mathrm{conf}}^{-1}\big(\pi_{\mathrm{conf}}(U)\big) = \bigsqcup_{\pi \in S_m} \pi(U),$$

and

$$\pi_{\mathrm{conf}}|_U : U \to \pi_{\mathrm{conf}}(U)$$

is a homeomorphism. Hence π_{conf} is a covering map. Since the action is transitive on each fiber, the cover is regular with deck group S_m. □

Proposition A.3 (Free finite-group quotients are coverings)**.** *Let* Γ *be a finite group acting freely by homeomorphisms on a Hausdorff space* X*. Then the quotient map*

$$p : X \to X/\Gamma$$

is a finite regular covering map with deck group Γ*.*

Proof. Fix $x \in X$. For each nontrivial $\gamma \in \Gamma$, the points x and γx are distinct, so by the Hausdorff property there exist disjoint open sets $U_\gamma \ni x$ and $V_\gamma \ni \gamma x$. Set

$$U := \bigcap_{\gamma \neq e} U_\gamma.$$

Then U is an open neighborhood of x, and for every $\gamma \neq e$,

$$\gamma U \cap U = \varnothing.$$

Indeed, if $y \in \gamma U \cap U$, then $y = \gamma u$ for some $u \in U$, hence $u \in U \cap \gamma^{-1} U$; but $U \subseteq U_{\gamma^{-1}}$ and $\gamma^{-1} U \subseteq V_{\gamma^{-1}}$, contradicting $U_{\gamma^{-1}} \cap V_{\gamma^{-1}} = \varnothing$.

Therefore

$$p^{-1}(p(U)) = \bigsqcup_{\gamma \in \Gamma} \gamma U,$$

and $p|_U : U \to p(U)$ is a homeomorphism. Hence p is a covering map. Regularity and the deck group description follow from the fact that the action of Γ is free and transitive on every fiber. □

A.2. Ambient relative sectional category

Classically, sectional category goes back to Schwarz's "genus of a fiber space" (Schwarz, 1966). For the present paper, the right notion is a relative version on compact subsets of the ambient base manifold.

Definition A.4 (Ambient relative sectional category)**.** Let $p : E \to B$ be a continuous surjection and let $K \subseteq B$ be compact. The *ambient relative sectional category* of p over K, denoted

$$\mathrm{secat}_B(p; K),$$

is the least integer $N \in \mathbb{N}$ such that there exist open sets

$$U_1, \ldots, U_N \subseteq B \qquad \text{with} \qquad K \subseteq U_1 \cup \cdots \cup U_N,$$

and continuous maps

$$s_i : U_i \to E \qquad (i = 1, \ldots, N)$$

satisfying

$$p \circ s_i = \mathrm{id}_{U_i} \qquad \text{for each } i.$$

If no such finite N exists, we set $\mathrm{secat}_B(p; K) = +\infty$.

Remark A.5 (Convention used in the paper). When the ambient base B is clear from context, we write

$$\operatorname{secat}(p; K)$$

for $\operatorname{secat}_B(p; K)$. In particular, for the principal quotient

$$q : \Theta_{\mathrm{reg}} \to B_{\mathrm{reg}}$$

and compact $K \subseteq B_{\mathrm{reg}}$, the paper's atlas complexity is

$$\mathfrak{a}(K; q) = \operatorname{secat}_{B_{\mathrm{reg}}}(q; K).$$

This is exactly the quantity counted by ambient gauge atlases and canonicalization-based local-coordinate learners.

Proposition A.6 (Basic properties of ambient relative sectional category)**.** *Let $p : E \to B$ be a continuous surjection.*

1. *If $K_1 \subseteq K_2 \subseteq B$ are compact, then*
$$\operatorname{secat}_B(p; K_1) \le \operatorname{secat}_B(p; K_2).$$
2. *If $U \subseteq B$ is open and admits a global section $s : U \to E$, then for every compact $K \subseteq U$,*
$$\operatorname{secat}_B(p; K) = 1.$$
3. *If $K \subseteq U_1 \cup \cdots \cup U_N$ with each $U_i \subseteq B$ open and $p|_{U_i}$ admitting a section, then*
$$\operatorname{secat}_B(p; K) \le N.$$

Proof. All three statements are immediate from Definition A.4. Item 1 follows by restricting an admissible open cover of K_2 to K_1. Item 2 is the special case of a one-set cover. Item 3 is exactly the defining upper-bound criterion. □

A.3. Principal bundles, local gauges, and associated bundles

Definition A.7 (Principal bundle and local gauge)**.** A *principal G-bundle* over a space B is a continuous surjection

$$q : P \to B$$

together with a free continuous left action of G on P such that every point $x \in B$ has an open neighborhood $U \subseteq B$ for which $q^{-1}(U)$ is G-equivariantly homeomorphic to $U \times G$, where G acts on $U \times G$ by left multiplication on the second factor. A *local gauge* on an open set $U \subseteq B$ is a continuous section

$$s : U \to P \qquad \text{with} \qquad q \circ s = \mathrm{id}_U\,.$$

Proposition A.8 (A local gauge induces a principal-bundle trivialization)**.** *Let $q : P \to B$ be a principal G-bundle and let $s : U \to P$ be a local gauge on an open set $U \subseteq B$. Then*

$$\alpha_s : U \times G \to q^{-1}(U), \qquad \alpha_s(x, g) := g \cdot s(x), \tag{2}$$

is a G-equivariant homeomorphism over U. In particular, for each $p \in q^{-1}(U)$ there exists a unique element $g_s(p) \in G$ such that

$$p = g_s(p) \cdot s(q(p)),$$

and the map

$$g_s : q^{-1}(U) \to G$$

is continuous.

Proof. For each $x \in U$, the fiber $q^{-1}(x)$ is a free transitive G-set. Hence every $p \in q^{-1}(U)$ can be written uniquely as

$$p = g \cdot s(q(p))$$

for a unique $g \in G$. Thus α_s is bijective, and the formula above defines its inverse:

$$\alpha_s^{-1}(p) = \big(q(p), g_s(p)\big).$$

It remains to prove continuity of g_s. Let $x_0 \in U$. By the principal bundle property, there exists an open neighborhood $W \subseteq U$ of x_0 and a G-equivariant homeomorphism

$$\beta : W \times G \xrightarrow{\cong} q^{-1}(W)$$

over W. Since $s(W) \subseteq q^{-1}(W)$, there exists a unique continuous map

$$a : W \to G$$

such that

$$s(x) = \beta(x, a(x)) \qquad \text{for all } x \in W.$$

If $p = \beta(x, h) \in q^{-1}(W)$, then

$$p = \beta(x, h) = h \cdot \beta(x, e) = h\, a(x)^{-1} \cdot s(x),$$

where we used G-equivariance of β and $s(x) = a(x) \cdot \beta(x, e)$. Hence

$$g_s(p) = h\, a(x)^{-1},$$

which is continuous in (x, h), and therefore continuous in p. Thus α_s^{-1} is continuous, so α_s is a homeomorphism. G-equivariance is immediate from the definition. □

Definition A.9 (Associated bundle)**.** Let $q : P \to B$ be a principal G-bundle and let

$$\rho : G \to \mathrm{GL}(V)$$

be a continuous finite-dimensional real representation. The associated vector bundle is the quotient

$$E_\rho := (P \times V)/G \to B,$$

where G acts diagonally by

$$g \cdot (p, v) := (g \cdot p, \rho(g)v).$$

The fiber over $x \in B$ is denoted $E_{\rho,x}$.

Proposition A.10 (A local gauge induces an associated-bundle trivialization)**.** *Let $q : P \to B$ be a principal G-bundle, let $\rho : G \to \mathrm{GL}(V)$ be a continuous representation, and let $s : U \to P$ be a local gauge on an open set $U \subseteq B$. Then*

$$\Phi_s : U \times V \to E_\rho|_U, \qquad \Phi_s(x, v) := [s(x), v], \tag{3}$$

is a vector-bundle homeomorphism over U. Its inverse is

$$\Phi_s^{-1}([p, v]) = \big(q(p), \rho(g_s(p))^{-1} v\big), \tag{4}$$

where g_s is the principal-bundle coordinate map from Proposition A.8.

Proof. Every element of $E_\rho|_U$ can be represented as $[p, v]$ with $q(p) \in U$. By Proposition A.8,

$$p = g_s(p) \cdot s(q(p)).$$

Hence, in the associated bundle,

$$[p, v] = [g_s(p) \cdot s(q(p)), v] = [s(q(p)), \rho(g_s(p))^{-1} v].$$

Therefore every class has the form $[s(x), w]$, so Φ_s is surjective. Injectivity follows from freeness: if

$$[s(x), v] = [s(x'), v'],$$

then $x = q(s(x)) = q(s(x')) = x'$, and there exists $g \in G$ with

$$s(x) = g \cdot s(x).$$

Freeness implies $g = e$, hence $v = v'$.

The inverse formula above is thus correct. Continuity follows from continuity of q, g_s, and ρ. Fiberwise linearity is immediate. $\square$

Corollary A.11 (Local coefficient representation of sections)**.** *Let $s : U \to P$ be a local gauge and let*

$$\sigma \in \Gamma(E_\rho|_U)$$

be a continuous section. Then there exists a unique continuous coefficient map

$$c_s^\sigma : U \to V$$

such that

$$\sigma(x) = [s(x), c_s^\sigma(x)] \qquad \textit{for all } x \in U.$$

Proof. This is immediate from the trivialization $\Phi_s : U \times V \to E_\rho|_U$ in Proposition A.10. Namely,

$$\Phi_s^{-1} \circ \sigma : U \to U \times V$$

must be of the form $x \mapsto (x, c_s^\sigma(x))$ for a unique continuous map c_s^σ. $\square$

Lemma A.12 (Transition functions and coefficient transformation law)**.** *Let $s_i : U_i \to P$ and $s_j : U_j \to P$ be local gauges on open sets $U_i, U_j \subseteq B$. Then:*

1. *there exists a unique continuous transition function*

$$t_{ij} : U_i \cap U_j \to G$$

such that

$$s_j(x) = t_{ij}(x) \cdot s_i(x) \qquad \textit{for all } x \in U_i \cap U_j;$$

2. *if $\sigma \in \Gamma(E_\rho|_{U_i \cup U_j})$ has local coefficient maps c_i and c_j in the gauges s_i and s_j, then*

$$c_j(x) = \rho(t_{ij}(x))\, c_i(x) \qquad \textit{for all } x \in U_i \cap U_j.$$

Proof. For each $x \in U_i \cap U_j$, the points $s_i(x)$ and $s_j(x)$ lie in the same fiber of the principal bundle q, hence there exists a unique element $t_{ij}(x) \in G$ with

$$s_j(x) = t_{ij}(x) \cdot s_i(x).$$

By Proposition A.8,

$$t_{ij}(x) = g_{s_i}(s_j(x)),$$

so t_{ij} is continuous.

Now let σ be a section. On $U_i \cap U_j$,

$$[s_i(x), c_i(x)] = \sigma(x) = [s_j(x), c_j(x)] = [t_{ij}(x) \cdot s_i(x), c_j(x)] = [s_i(x), \rho(t_{ij}(x))^{-1} c_j(x)].$$

By uniqueness of the local coefficient representation in the gauge s_i,

$$c_i(x) = \rho(t_{ij}(x))^{-1} c_j(x),$$

equivalently

$$c_j(x) = \rho(t_{ij}(x))\, c_i(x).$$

$\square$

A.4. Partitions of unity and bundle norms on compact subsets

Proposition A.13 (Partition of unity on a compact subset)**.** *Let $K \subseteq B$ be compact, and let*

$$K \subseteq U_1 \cup \cdots \cup U_N$$

with each $U_i \subseteq B$ open. Then there exist continuous functions

$$\lambda_1, \ldots, \lambda_N : K \to [0,1]$$

such that:

1. $$\sum_{i=1}^{N} \lambda_i(x) = 1 \qquad \textit{for all } x \in K;$$

2. *if $\mathrm{supp}_K(\lambda_i)$ denotes the support of λ_i taken in the subspace topology of K, then*

$$\mathrm{supp}_K(\lambda_i) \subseteq U_i \cap K \qquad \textit{for each } i.$$

In other words, $(\lambda_i)_{i=1}^N$ is a partition of unity on K subordinate to the ambient open cover $(U_i)_{i=1}^N$.

Proof. The manifold B is paracompact, hence admits smooth partitions of unity subordinate to any open cover. Consider the finite open cover

$$B = (U_1 \cup \cdots \cup U_N) \cup (B \setminus K).$$

Choose a smooth partition of unity

$$\widetilde{\lambda}_1, \ldots, \widetilde{\lambda}_N, \widetilde{\lambda}_{N+1}$$

subordinate to this cover, where $\widetilde{\lambda}_{N+1}$ is subordinate to $B \setminus K$. Restricting to K, we have

$$\widetilde{\lambda}_{N+1}|_K \equiv 0,$$

and therefore

$$\sum_{i=1}^{N} \widetilde{\lambda}_i(x) = 1 \qquad \text{for all } x \in K.$$

Set

$$\lambda_i := \widetilde{\lambda}_i|_K.$$

Then each λ_i is continuous, the sum is 1 on K, and because $\widetilde{\lambda}_i$ is subordinate to U_i, its support in K is contained in $U_i \cap K$. □

Proposition A.14 (Continuous bundle norms on compact bases)**.** *Let $\pi : E \to K$ be a finite-dimensional real topological vector bundle over a compact Hausdorff space K. Then:*

1. *E admits a continuous bundle norm;*

2. *if $\|\cdot\|_1$ and $\|\cdot\|_2$ are two continuous bundle norms on E, then there exist constants $0 < c \le C < \infty$ such that*

$$c\,\|v\|_1 \le \|v\|_2 \le C\,\|v\|_1 \qquad \textit{for all } v \in E.$$

Proof. Choose a finite trivializing cover

$$K \subseteq U_1 \cup \cdots \cup U_N,$$

and vector-bundle homeomorphisms

$$\Psi_i : U_i \times \mathbb{R}^r \xrightarrow{\cong} E|_{U_i}.$$

By Proposition A.13, there exists a continuous partition of unity $(\lambda_i)_{i=1}^N$ on K subordinate to $(U_i)_{i=1}^N$.

For $v \in E_x$, define

$$\|v\|^2 := \sum_{i=1}^{N} \lambda_i(x) \left\| \mathrm{pr}_2\big(\Psi_i^{-1}(v)\big) \right\|_2^2,$$

where the i-th term is interpreted as 0 whenever $x \notin U_i$. This is well defined because $\lambda_i(x) = 0$ for $x \notin U_i$. Continuity is immediate from continuity of the bundle charts and of the partition of unity. For $v \neq 0$, choose some index i with $\lambda_i(x) > 0$; then $\Psi_i^{-1}(v) = (x, u_i)$ with $u_i \neq 0$, hence

$$\|v\|^2 \geq \lambda_i(x)\|u_i\|_2^2 > 0.$$

Thus $\|\cdot\|$ is a continuous bundle norm, proving item 1.

For item 2, fix a finite trivializing cover

$$K \subseteq W_1 \cup \cdots \cup W_M$$

such that each closure $\overline{W_j}$ is compact and contained in a trivializing neighborhood for both norms. In such a common trivialization over $\overline{W_j}$, each bundle norm corresponds to a continuous norm on $\mathbb{R}^r$ depending continuously on the base point $x \in \overline{W_j}$. Consider the compact set

$$\overline{W_j} \times S^{r-1},$$

where $S^{r-1} \subseteq \mathbb{R}^r$ is the Euclidean unit sphere. The ratio of the two norms is a positive continuous function on this compact set, and therefore is bounded above and below by positive constants

$$0 < c_j \leq C_j < \infty.$$

By homogeneity, these bounds extend from unit vectors to all vectors in every fiber over $\overline{W_j}$:

$$c_j\, \|v\|_1 \leq \|v\|_2 \leq C_j\, \|v\|_1.$$

Setting

$$c := \min_{1 \leq j \leq M} c_j, \qquad C := \max_{1 \leq j \leq M} C_j$$

gives the desired global uniform equivalence on all of E. $\square$

Remark A.15 (Why these preliminaries are exactly the ones we need). The later appendices use these facts in a highly controlled way. Configuration spaces and free finite quotients drive the branch and monodromy arguments; ambient relative sectional category is the exact complexity measure for gauge atlases; principal and associated bundle trivializations underlie the local coordinate descriptions of invariant and section-valued learners; and partitions of unity plus bundle norms are the glueing mechanisms behind the universality theorems.

B. Exact Quotient Identification for the Homogeneous Two-Layer Base Model

Positive scaling and hidden-unit permutation are among the canonical sources of parameter non-identifiability in homogeneous shallow networks and already play a central role in modern analyses of symmetry-induced loss geometry and quotient-space optimization (Simsek et al., 2021; Dong & Cheng, 2026). In this appendix we verify the standing assumptions of Section 2 for a concrete width-m two-layer model and identify the resulting regular quotient exactly. The result is global on the regular locus: after modding out per-unit positive scalings, each hidden unit is represented by a nonzero signed amplitude together with a direction on the sphere, and the full regular quotient is the unordered configuration space of m such signatures.

B.1. Model, canonical symmetry group, and regular block manifold

Fix integers $d \geq 1$ and $m \geq 2$, let $\mathcal{X} \subseteq \mathbb{R}^d$ be a nonempty input domain, and let $\sigma : \mathbb{R} \to \mathbb{R}$ satisfy

$$\sigma(ct) = c\,\sigma(t) \qquad \text{for all } c > 0,\ t \in \mathbb{R}.$$

We consider the width-m positively homogeneous two-layer model

$$\mathcal{R}_m^{\mathrm{hom}}(\theta)(x) := \sum_{i=1}^{m} a_i\, \sigma(w_i^\top x), \qquad \theta = ((a_1, w_1), \ldots, (a_m, w_m)). \tag{5}$$

For the ambient block manifold and its regular part we set

$$\Xi^{\mathrm{hom}} := \mathbb{R} \times (\mathbb{R}^d \setminus \{0\}), \qquad \Xi^{\mathrm{hom}}_{\mathrm{reg}} := (\mathbb{R} \setminus \{0\}) \times (\mathbb{R}^d \setminus \{0\}).$$

Accordingly,

$$\Theta^{\mathrm{hom}}_m := (\Xi^{\mathrm{hom}})^m, \qquad (\Theta^{\mathrm{hom}}_m)_{\mathrm{reg},0} := (\Xi^{\mathrm{hom}}_{\mathrm{reg}})^m.$$

The canonical continuous symmetry group is

$$H := \mathbb{R}_{>0},$$

acting on a single block by

$$c \cdot (a, w) := \left(\frac{a}{c},\, c\, w\right), \qquad c \in \mathbb{R}_{>0}. \tag{6}$$

On m blocks we use the semidirect product

$$G_m := H^m \rtimes S_m,$$

with action

$$((c_1, \dots, c_m), \pi) \cdot ((a_1, w_1), \dots, (a_m, w_m)) = \big((a_{\pi^{-1}(1)}/c_1,\ c_1 w_{\pi^{-1}(1)}), \dots, (a_{\pi^{-1}(m)}/c_m,\ c_m w_{\pi^{-1}(m)})\big). \tag{7}$$

The basic block signature is

$$\psi_{\mathrm{hom}} : \Xi^{\mathrm{hom}}_{\mathrm{reg}} \to M_{\mathrm{hom}} := (\mathbb{R} \setminus \{0\}) \times S^{d-1}, \qquad \psi_{\mathrm{hom}}(a, w) := \left(a\|w\|,\ \frac{w}{\|w\|}\right). \tag{8}$$

The corresponding ordered signature map on m blocks is

$$\Psi^{\mathrm{hom}}_m(\theta) := \big(\psi_{\mathrm{hom}}(a_1, w_1), \dots, \psi_{\mathrm{hom}}(a_m, w_m)\big). \tag{9}$$

We will use the standard ordered and unordered configuration spaces

$$\mathrm{Conf}_m(M_{\mathrm{hom}}) := \left\{(z_1, \dots, z_m) \in M^m_{\mathrm{hom}} : z_i \neq z_j \text{ for all } i \neq j\right\},$$

$$\mathrm{UConf}_m(M_{\mathrm{hom}}) := \mathrm{Conf}_m(M_{\mathrm{hom}})/S_m,$$

where S_m acts on ordered m-tuples by coordinate permutation:

$$\pi \cdot (z_1, \dots, z_m) := (z_{\pi^{-1}(1)}, \dots, z_{\pi^{-1}(m)}).$$

The regular locus for the base model is

$$\Theta^{\mathrm{hom}}_{m,\mathrm{reg}} := \left\{\theta \in (\Theta^{\mathrm{hom}}_m)_{\mathrm{reg},0} : \Psi^{\mathrm{hom}}_m(\theta) \in \mathrm{Conf}_m(M_{\mathrm{hom}})\right\}. \tag{10}$$

Equivalently,

$$\Theta^{\mathrm{hom}}_{m,\mathrm{reg}} = \left\{((a_i, w_i))_{i=1}^m \in \big((\mathbb{R} \setminus \{0\}) \times (\mathbb{R}^d \setminus \{0\})\big)^m : \psi_{\mathrm{hom}}(a_i, w_i) \neq \psi_{\mathrm{hom}}(a_j, w_j)\ \forall i \neq j\right\}.$$

Lemma B.1 (Realization invariance under the canonical action)**.** *The realization map $\mathcal{R}^{\mathrm{hom}}_m$ is invariant under the G_m-action* (7)*. In particular,*

$$\mathcal{R}^{\mathrm{hom}}_m(g \cdot \theta) = \mathcal{R}^{\mathrm{hom}}_m(\theta) \qquad \textit{for all } g \in G_m,\ \theta \in \Theta^{\mathrm{hom}}_m.$$

Proof. It suffices to check invariance under the continuous part H^m and the permutation part S_m separately. Fix $c > 0$, $a \in \mathbb{R}$, $w \in \mathbb{R}^d \setminus \{0\}$, and $x \in \mathcal{X}$. By positive 1-homogeneity of σ,

$$\frac{a}{c}\, \sigma((cw)^\top x) = \frac{a}{c}\, \sigma(c\, w^\top x) = \frac{a}{c}\, c\, \sigma(w^\top x) = a\, \sigma(w^\top x).$$

Applying this coordinatewise shows invariance under H^m. Invariance under S_m is immediate because the network output is the sum of the unit contributions, hence is unchanged by reindexing the hidden units. Combining the two invariances proves the claim. □

Lemma B.2 (Global linearization of the block action)**.** *Define*

$$\Phi : \Xi^{\mathrm{hom}}_{\mathrm{reg}} \to M_{\mathrm{hom}} \times \mathbb{R}, \qquad \Phi(a, w) := \left(a\|w\|,\ \frac{w}{\|w\|},\ \log \|w\| \right). \tag{11}$$

Then Φ is a diffeomorphism with inverse

$$\Phi^{-1}(b, u, r) = \big(be^{-r},\ e^{r}u\big), \qquad (b, u, r) \in M_{\mathrm{hom}} \times \mathbb{R}.$$

Moreover, under the Lie-group isomorphism

$$\log : H = \mathbb{R}_{>0} \xrightarrow{\cong} (\mathbb{R}, +),$$

the block action (6) is conjugate via Φ to translation in the last coordinate:

$$\Phi(c \cdot (a, w)) = \big(b,\ u,\ r + \log c\big) \quad \textit{whenever } \Phi(a, w) = (b, u, r).$$

Proof. The inverse formula is immediate: if $(b, u, r) \in M_{\mathrm{hom}} \times \mathbb{R}$, then $u \in S^{d-1}$ and $e^{r}u \neq 0$, so $\Phi^{-1}(b, u, r) \in \Xi^{\mathrm{hom}}_{\mathrm{reg}}$. A direct calculation gives

$$\Phi\big(\Phi^{-1}(b, u, r)\big) = \left(be^{-r}\|e^{r}u\|,\ \frac{e^{r}u}{\|e^{r}u\|},\ \log \|e^{r}u\| \right) = (b, u, r),$$

since $\|u\| = 1$. Conversely, if $(a, w) \in \Xi^{\mathrm{hom}}_{\mathrm{reg}}$, then

$$\Phi^{-1}\big(\Phi(a, w)\big) = \left(a\|w\|\, e^{-\log \|w\|},\ e^{\log \|w\|} \frac{w}{\|w\|} \right) = (a, w).$$

Hence Φ is a bijection with smooth inverse, so it is a diffeomorphism.

Now let $c \in H$, write $\Phi(a, w) = (b, u, r)$, and compute

$$\Phi(c \cdot (a, w)) = \Phi\left(\frac{a}{c},\ cw\right) = \left(\frac{a}{c}\|cw\|,\ \frac{cw}{\|cw\|},\ \log \|cw\| \right) = \left(a\|w\|,\ \frac{w}{\|w\|},\ \log \|w\| + \log c \right) = (b, u, r + \log c).$$

This is exactly translation in the third coordinate. $\square$

Proposition B.3 (Exact block quotient)**.** *The H-action on $\Xi^{\mathrm{hom}}_{\mathrm{reg}}$ is free and proper, and the signature map ψ_{hom} induces a diffeomorphism*

$$\bar{\psi}_{\mathrm{hom}} : \Xi^{\mathrm{hom}}_{\mathrm{reg}}/H \xrightarrow{\cong} M_{\mathrm{hom}}.$$

Equivalently, two regular blocks (a, w) and (a', w') lie in the same H-orbit if and only if

$$\psi_{\mathrm{hom}}(a, w) = \psi_{\mathrm{hom}}(a', w').$$

Proof. By Lemma B.2, the H-action on $\Xi^{\mathrm{hom}}_{\mathrm{reg}}$ is conjugate to the translation action of $(\mathbb{R}, +)$ on $M_{\mathrm{hom}} \times \mathbb{R}$ given by

$$t \cdot (b, u, r) = (b, u, r + t).$$

This translation action is free. It is also proper: if

$$\big((b_n, u_n, r_n + t_n), (b_n, u_n, r_n)\big)$$

converges in $(M_{\mathrm{hom}} \times \mathbb{R})^2$, then the sequence (t_n) converges to the difference of the limiting third coordinates and is therefore bounded. Since properness is preserved under conjugation, the H-action on $\Xi^{\mathrm{hom}}_{\mathrm{reg}}$ is free and proper.

Still by Lemma B.2,

$$\psi_{\mathrm{hom}} = \mathrm{pr}_{M_{\mathrm{hom}}} \circ \Phi,$$

where $\mathrm{pr}_{M_{\mathrm{hom}}}$ denotes projection onto the first two coordinates. The fibers of $\mathrm{pr}_{M_{\mathrm{hom}}}$ are exactly the translation orbits in $M_{\mathrm{hom}} \times \mathbb{R}$, hence the fibers of ψ_{hom} are exactly the H-orbits in $\Xi^{\mathrm{hom}}_{\mathrm{reg}}$. Therefore ψ_{hom} descends to a bijection

$$\bar{\psi}_{\mathrm{hom}} : \Xi^{\mathrm{hom}}_{\mathrm{reg}}/H \to M_{\mathrm{hom}}.$$

Because the action is free and proper, the quotient carries the unique smooth structure making the orbit map a smooth submersion; with respect to that smooth structure, $\bar{\psi}_{\mathrm{hom}}$ is a diffeomorphism. $\square$

Corollary B.4 (Verification of the standing assumptions for the base model)**.** *The homogeneous two-layer model satisfies Assumptions 2.1 and 2.2 of Section 2 with*

$$\Xi = \Xi^{\mathrm{hom}}, \qquad \Xi_{\mathrm{reg}} = \Xi^{\mathrm{hom}}_{\mathrm{reg}}, \qquad H = \mathbb{R}_{>0}, \qquad M = M_{\mathrm{hom}}, \qquad \psi = \psi_{\mathrm{hom}}, \qquad \Theta = \Theta^{\mathrm{hom}}_m.$$

Proof. Lemma B.1 proves G_m-invariance of the realization map. Proposition B.3 proves that the H-action on $\Xi^{\mathrm{hom}}_{\mathrm{reg}}$ is free and proper and that the induced map

$$\Xi^{\mathrm{hom}}_{\mathrm{reg}}/H \to M_{\mathrm{hom}}$$

is a diffeomorphism. This is precisely the content required by Assumption 2.2. □

Proposition B.5 (Ordered quotient by positive scalings)**.** *The regular locus $\Theta^{\mathrm{hom}}_{m,\mathrm{reg}}$ is a G_m-invariant open submanifold of $(\Theta^{\mathrm{hom}}_m)_{\mathrm{reg},0}$. Moreover, the ordered signature map Ψ^{hom}_m is invariant under the H^m-action and descends to a diffeomorphism*

$$\bar{\Psi}^{\mathrm{hom}}_m : \Theta^{\mathrm{hom}}_{m,\mathrm{reg}}/H^m \xrightarrow{\cong} \mathrm{Conf}_m(M_{\mathrm{hom}}).$$

Proof. Since $\mathrm{Conf}_m(M_{\mathrm{hom}}) \subset M^m_{\mathrm{hom}}$ is the complement of the union of pairwise diagonals

$$\bigcup_{1 \le i < j \le m} \{(z_1, \ldots, z_m) \in M^m_{\mathrm{hom}} : z_i = z_j\},$$

it is open in M^m_{hom}. By definition

$$\Theta^{\mathrm{hom}}_{m,\mathrm{reg}} = (\Psi^{\mathrm{hom}}_m)^{-1}\big(\mathrm{Conf}_m(M_{\mathrm{hom}})\big),$$

so $\Theta^{\mathrm{hom}}_{m,\mathrm{reg}}$ is open in $(\Theta^{\mathrm{hom}}_m)_{\mathrm{reg},0}$. It is G_m-invariant because the block signatures are unchanged by the continuous scalings and merely permuted by S_m.

The map Ψ^{hom}_m is H^m-invariant because each block signature ψ_{hom} is H-invariant by Proposition B.3. Applying Proposition B.3 coordinatewise yields a diffeomorphism

$$(\Xi^{\mathrm{hom}}_{\mathrm{reg}})^m/H^m \xrightarrow{\cong} M^m_{\mathrm{hom}}$$

induced by Ψ^{hom}_m. Restricting this diffeomorphism to the preimage of $\mathrm{Conf}_m(M_{\mathrm{hom}})$ gives the claimed diffeomorphism

$$\bar{\Psi}^{\mathrm{hom}}_m : \Theta^{\mathrm{hom}}_{m,\mathrm{reg}}/H^m \xrightarrow{\cong} \mathrm{Conf}_m(M_{\mathrm{hom}}).$$

□

Theorem B.6 (Exact regular quotient for the homogeneous base model)**.** *Let*

$$B_{\mathrm{reg}}{}^{\mathrm{hom}}_m := \Theta^{\mathrm{hom}}_{m,\mathrm{reg}}/G_m.$$

Then the ordered signature diffeomorphism $\bar{\Psi}^{\mathrm{hom}}_m$ is S_m-equivariant and descends to a diffeomorphism

$$\widehat{\Psi}^{\mathrm{hom}}_m : B_{\mathrm{reg}}{}^{\mathrm{hom}}_m \xrightarrow{\cong} \mathrm{UConf}_m(M_{\mathrm{hom}}). \tag{12}$$

Equivalently, the regular quotient of the width-m positively homogeneous two-layer network by its canonical scaling-permutation symmetry is exactly the unordered configuration space of m distinct block signatures in

$$M_{\mathrm{hom}} = (\mathbb{R} \setminus \{0\}) \times S^{d-1}.$$

Proof. Let $[\theta]_{H^m} \in \Theta^{\mathrm{hom}}_{m,\mathrm{reg}}/H^m$. For $\pi \in S_m$,

$$\bar{\Psi}^{\mathrm{hom}}_m\big(\pi \cdot [\theta]_{H^m}\big) = \bar{\Psi}^{\mathrm{hom}}_m\big([\pi \cdot \theta]_{H^m}\big) = \pi \cdot \bar{\Psi}^{\mathrm{hom}}_m([\theta]_{H^m}),$$

because Ψ^{hom}_m records the ordered list of block signatures and permutation of hidden units simply permutes that list. Hence $\bar{\Psi}^{\mathrm{hom}}_m$ is S_m-equivariant.

Since H^m is a normal subgroup of $G_m = H^m \rtimes S_m$, there is a canonical identification of orbit spaces

$$\Theta^{\mathrm{hom}}_{m,\mathrm{reg}}/G_m \cong \left(\Theta^{\mathrm{hom}}_{m,\mathrm{reg}}/H^m\right)/S_m.$$

Under Proposition B.5, this becomes

$$B_{\mathrm{reg}}{}^{\mathrm{hom}}_{m} \cong \mathrm{Conf}_m(M_{\mathrm{hom}})/S_m.$$

Because the points of $\mathrm{Conf}_m(M_{\mathrm{hom}})$ have pairwise distinct coordinates, the permutation action of S_m on $\mathrm{Conf}_m(M_{\mathrm{hom}})$ is free; because S_m is finite, the action is also proper. Therefore the quotient $\mathrm{UConf}_m(M_{\mathrm{hom}})$ is a smooth manifold and the quotient of the S_m-equivariant diffeomorphism $\bar{\Psi}^{\mathrm{hom}}_m$ yields the diffeomorphism (12). □

Remark B.7 (Scope of the quotient identification). Theorem B.6 is an exact identification of the quotient by the canonical scaling-permutation group G_m. We do not need, and therefore do not assume, that G_m exhausts every possible function-preserving transformation of the realization map $\mathcal{R}^{\mathrm{hom}}_m$ for every activation and every input domain. For the purposes of the present paper, the relevant geometric object is exactly the regular quotient by the G_m-action.

Remark B.8 (Discriminant as the fat diagonal). Under the ordered quotient identification of Proposition B.5, the collision discriminant is the pullback of the fat diagonal in M^m_{hom}:

$$\Delta_{\mathrm{fat}} := M^m_{\mathrm{hom}} \setminus \mathrm{Conf}_m(M_{\mathrm{hom}}) = \bigcup_{1\le i<j\le m} \{(z_1,\dots,z_m) : z_i = z_j\}.$$

Thus, in the homogeneous base model, simple pairwise collisions are precisely the ordinary pairwise configuration collisions in M_{hom}. This is the geometric input behind the local branched normal form proved later in Appendix C.

C. Abstract Local Branch Normal Form

Recent work has made it increasingly clear that symmetry-reduced neural parameter spaces are the correct objects for studying intrinsic loss geometry and optimization, at least on regular loci where the quotient is smooth (Simsek et al., 2021; Dong & Cheng, 2026). The goal of the present appendix is to analyze the first singular event beyond that regular regime: a *simple pairwise collision* of two exchangeable blocks. We show that, after quotienting the continuous gauge symmetry, the residual local singularity is universal. More precisely, near a simple collision, the quotient by the full symmetry group is locally equivalent to the product of a smooth tangential factor and the sign quotient $\mathbb{R}^d/\{\pm 1\}$ on a d-dimensional relative coordinate. On every two-dimensional transverse slice this reduces to the classical branched double cover $z \mapsto z^2$.

Throughout this appendix we retain the notation of Section 2. In particular, $m \ge 2$ is the number of exchangeable blocks, H is the continuous per-block gauge group, $G = H^m \rtimes S_m$, $\psi : \Xi_{\mathrm{reg}} \to M$ is the block-signature map from Assumption 2.2, and $d := \dim M$.

C.1. Continuous-gauge reduction and residual permutation action

We first quotient out the continuous gauge symmetry and isolate the residual finite permutation action.

Definition C.1 (Reduced parameter manifold)**.** Define

$$\Theta^\sharp := \Theta \cap \Xi^m_{\mathrm{reg}}, \qquad \widetilde{\Theta} := \Theta^\sharp / H^m,$$

and let

$$p_H : \Theta^\sharp \to \widetilde{\Theta}$$

denote the quotient map by the H^m-action.

Proposition C.2 (Reduced manifold and residual S_m-action)**.** *The action of H^m on $\Theta^\sharp$ is free and proper. Consequently, $\widetilde{\Theta}$ is a smooth manifold and p_H is a smooth submersion. Moreover:*

1. *the quotient group $G/H^m \cong S_m$ acts smoothly on $\widetilde{\Theta}$ by*

$$\pi \cdot p_H(\theta) := p_H(\pi \cdot \theta), \qquad \pi \in S_m,\ \theta \in \Theta^\sharp;$$

2. *the coordinatewise signature map*

$$\Psi^\sharp : \Theta^\sharp \to M^m, \qquad \Psi^\sharp(\xi_1, \dots, \xi_m) := \big(\psi(\xi_1), \dots, \psi(\xi_m)\big)$$

is H^m-invariant and therefore descends to a smooth S_m-equivariant map

$$\widetilde{\Psi} : \widetilde{\Theta} \to M^m, \qquad \widetilde{\Psi} \circ p_H = \Psi^\sharp;$$

3. *the full quotient by G identifies canonically with the residual permutation quotient*

$$\mathcal{Q} := \widetilde{\Theta}/S_m \cong \Theta^\sharp/G.$$

Proof. By Assumption 2.2, the action of H on Ξ_{reg} is free and proper. Hence the product action of H^m on Ξ_{reg}^m is free and proper, and its restriction to the H^m-invariant submanifold $\Theta^\sharp \subseteq \Xi_{\mathrm{reg}}^m$ remains free and proper. Therefore $\widetilde{\Theta} = \Theta^\sharp/H^m$ is a smooth manifold and p_H is a smooth submersion.

Because H^m is a normal subgroup of $G = H^m \rtimes S_m$, the quotient group $G/H^m \cong S_m$ acts on $\widetilde{\Theta}$ by

$$\pi \cdot p_H(\theta) := p_H(\pi \cdot \theta).$$

This is well defined: if $p_H(\theta) = p_H(\theta')$, then $\theta' = h \cdot \theta$ for some $h \in H^m$, and

$$\pi \cdot \theta' = \pi \cdot (h \cdot \theta) = (\pi h \pi^{-1}) \cdot (\pi \cdot \theta),$$

with $\pi h \pi^{-1} \in H^m$, hence $p_H(\pi \cdot \theta') = p_H(\pi \cdot \theta)$. Smoothness follows from smoothness of the original G-action.

The map $\Psi^\sharp$ is H^m-invariant because each coordinate $\psi(\xi_i)$ is H-invariant. Thus $\Psi^\sharp$ descends uniquely to a smooth map $\widetilde{\Psi}$ satisfying $\widetilde{\Psi} \circ p_H = \Psi^\sharp$. S_m-equivariance follows from the fact that permutation of blocks simply permutes the corresponding signatures:

$$\widetilde{\Psi}(\pi \cdot p_H(\theta)) = \widetilde{\Psi}(p_H(\pi \cdot \theta)) = \Psi^\sharp(\pi \cdot \theta) = \pi \cdot \Psi^\sharp(\theta) = \pi \cdot \widetilde{\Psi}(p_H(\theta)).$$

Finally, since H^m is normal in G, quotienting $\Theta^\sharp$ first by H^m and then by the induced S_m-action yields the same orbit space as quotienting directly by G. This gives the canonical identification

$$\mathcal{Q} = \widetilde{\Theta}/S_m \cong \Theta^\sharp/G.$$

□

Definition C.3 (Simple pairwise collision)**.** A point $\theta^\star = (\xi_1^\star, \dots, \xi_m^\star) \in \Theta^\sharp$ is a *simple pairwise collision* if, after permuting the block indices if necessary, the following hold:

1. $\psi(\xi_1^\star) = \psi(\xi_2^\star) =: m^\star \in M$;
2. $\psi(\xi_k^\star) \neq m^\star$ for every $k = 3, \dots, m$, and $\psi(\xi_i^\star) \neq \psi(\xi_j^\star)$ for all $3 \le i < j \le m$;
3. there exist a coordinate chart $\varphi : U_M \to \mathbb{R}^d$ with $m^\star \in U_M$ and $\varphi(m^\star) = 0$, and a neighborhood $W \subseteq \Theta^\sharp$ of $\theta^\star$, such that $\psi(\xi_1), \psi(\xi_2) \in U_M$ for all $\theta = (\xi_1, \dots, \xi_m) \in W$ and the collision-difference map

$$\delta : W \to \mathbb{R}^d, \qquad \delta(\theta) := \varphi(\psi(\xi_1)) - \varphi(\psi(\xi_2)),$$

   has surjective differential at $\theta^\star$.

We now specialize to a simple pairwise collision point

$$\theta^\star = (\xi_1^\star, \dots, \xi_m^\star) \in \Theta^\sharp$$

in the sense of Definition C.3. Write

$$x^\star := p_H(\theta^\star) \in \widetilde{\Theta}, \qquad y^\star := [x^\star]_{S_m} \in \mathcal{Q}.$$

After relabeling if necessary, we may and will assume that the colliding pair is $(1, 2)$. Let

$$\tau := (12) \in S_m$$

denote the transposition of the colliding blocks, and let

$$m^\star := \psi(\xi_1^\star) = \psi(\xi_2^\star) \in M$$

be the collision signature.

C.2. Local reduction to the residual transposition

The first key point is that, near $x^\star$, the full permutation action reduces to the two-element group generated by τ.

Lemma C.4 (Stabilizer at a simple collision)**.** *The stabilizer of $x^\star$ under the residual S_m-action is exactly*

$$\mathrm{Stab}_{S_m}(x^\star) = \{e, \tau\}.$$

Proof. Write

$$\widetilde{\Psi}(x^\star) = (m^\star, m^\star, m_3^\star, \dots, m_m^\star),$$

where, by Definition C.3, the points $m_3^\star, \dots, m_m^\star$ are pairwise distinct and distinct from $m^\star$.

Let $\pi \in S_m$ satisfy $\pi \cdot x^\star = x^\star$. Since $\widetilde{\Psi}$ is S_m-equivariant,

$$\pi \cdot \widetilde{\Psi}(x^\star) = \widetilde{\Psi}(x^\star).$$

Equivalently, π leaves the ordered tuple $(m^\star, m^\star, m_3^\star, \dots, m_m^\star)$ unchanged. Because the entries $m_3^\star, \dots, m_m^\star$ are all distinct and all distinct from $m^\star$, every index $i \in \{3, \dots, m\}$ must be fixed by π, while the set $\{1, 2\}$ must be preserved. Thus $\pi \in \{e, \tau\}$.

Conversely, $\tau \cdot x^\star = x^\star$ because the first two signatures coincide. Indeed, if $\psi(\xi_1^\star) = \psi(\xi_2^\star)$, then by Assumption 2.2, $\xi_1^\star$ and $\xi_2^\star$ lie in the same H-orbit. Hence swapping the first two blocks changes $\theta^\star$ only by an element of H^m, so the image $x^\star = p_H(\theta^\star)$ is fixed by τ. Therefore $\mathrm{Stab}_{S_m}(x^\star) = \{e, \tau\}$. □

Lemma C.5 (Local permutation reduction)**.** *There exists an open neighborhood $U_0 \subseteq \widetilde{\Theta}$ of $x^\star$ such that*

$$\pi(U_0) \cap U_0 \neq \varnothing \qquad \Longrightarrow \qquad \pi \in \{e, \tau\}.$$

Consequently, the equivalence relation induced on U_0 by the full S_m-action coincides with the equivalence relation induced by the subgroup $\langle \tau \rangle \cong \mathbb{Z}_2$.

Proof. Because M is Hausdorff and the points

$$m^\star,\ m_3^\star, \dots, m_m^\star$$

are pairwise distinct, there exist pairwise disjoint open neighborhoods

$$N_{12} \ni m^\star, \qquad N_i \ni m_i^\star \quad (i = 3, \dots, m).$$

By continuity of the coordinate maps

$$\widetilde{\psi}_i := \mathrm{pr}_i \circ \widetilde{\Psi} : \widetilde{\Theta} \to M,$$

we may choose an open neighborhood $U_0 \ni x^\star$ such that

$$\widetilde{\psi}_1(U_0) \cup \widetilde{\psi}_2(U_0) \subseteq N_{12}, \qquad \widetilde{\psi}_i(U_0) \subseteq N_i \quad (i = 3, \dots, m).$$

Now let $\pi \in S_m$ satisfy $\pi(U_0) \cap U_0 \neq \varnothing$. Pick $x \in U_0$ such that $\pi \cdot x \in U_0$. For each $i \geq 3$,

$$\widetilde{\psi}_i(\pi \cdot x) = \widetilde{\psi}_{\pi^{-1}(i)}(x) \in N_i.$$

Since the sets $N_{12}, N_3, \dots, N_m$ are pairwise disjoint and $\widetilde{\psi}_1(x), \widetilde{\psi}_2(x) \in N_{12}$, the only coordinate of $\widetilde{\Psi}(x)$ that can lie in N_i is the i-th one. Hence $\pi^{-1}(i) = i$ for all $i = 3, \dots, m$. Therefore π fixes every index $3, \dots, m$ and preserves the set $\{1, 2\}$, so $\pi \in \{e, \tau\}$.

The final statement follows immediately: two points of U_0 lie in the same S_m-orbit if and only if they differ by either the identity or the transposition τ. □

C.3. Reduced collision map and fixed-point set

We next construct the local collision coordinate on the reduced space and identify the fixed-point set of τ.

Fix the chart

$$\varphi : U_M \to \mathbb{R}^d$$

centered at $m^\star$ from Definition C.3, and shrink U_0 further if necessary so that

$$\widetilde{\psi}_1(U_0) \cup \widetilde{\psi}_2(U_0) \subseteq U_M.$$

Define the *reduced collision map*

$$\overline{\delta} : U_0 \to \mathbb{R}^d, \qquad \overline{\delta}(x) := \varphi(\widetilde{\psi}_1(x)) - \varphi(\widetilde{\psi}_2(x)). \tag{13}$$

Lemma C.6 (Basic properties of the reduced collision map)**.** *The map $\overline{\delta}$ is smooth and satisfies:*

1. $\overline{\delta} \circ \tau = -\,\overline{\delta}$ *on* U_0*;*
2. $D\overline{\delta}(x^\star) : T_{x^\star}\widetilde{\Theta} \to \mathbb{R}^d$ *is surjective;*
3. *after shrinking* U_0 *further if necessary,*

$$\mathrm{Fix}(\tau) \cap U_0 = \overline{\delta}^{-1}(0),$$

and this set is a smooth embedded submanifold of codimension d*.*

Proof. Smoothness is immediate from smoothness of $\widetilde{\psi}_1$, $\widetilde{\psi}_2$, and φ.

For item 1, τ swaps the first two block signatures and fixes all others, so

$$\widetilde{\psi}_1(\tau \cdot x) = \widetilde{\psi}_2(x), \qquad \widetilde{\psi}_2(\tau \cdot x) = \widetilde{\psi}_1(x).$$

Therefore

$$\overline{\delta}(\tau \cdot x) = \varphi(\widetilde{\psi}_1(\tau \cdot x)) - \varphi(\widetilde{\psi}_2(\tau \cdot x)) = \varphi(\widetilde{\psi}_2(x)) - \varphi(\widetilde{\psi}_1(x)) = -\overline{\delta}(x).$$

For item 2, let δ be the collision-difference map from Definition C.3, defined on a neighborhood of $\theta^\star$ in $\Theta^\sharp$:

$$\delta(\theta) = \varphi(\psi(\xi_1)) - \varphi(\psi(\xi_2)).$$

By construction,

$$\delta = \overline{\delta} \circ p_H$$

near $\theta^\star$. Differentiating at $\theta^\star$ yields

$$D\delta(\theta^\star) = D\overline{\delta}(x^\star) \circ Dp_H(\theta^\star).$$

Since $Dp_H(\theta^\star)$ is surjective and $D\delta(\theta^\star)$ is surjective by Definition C.3, it follows that $D\overline{\delta}(x^\star)$ is also surjective.

For item 3, first suppose $x \in \mathrm{Fix}(\tau) \cap U_0$. Then $\widetilde{\psi}_1(x) = \widetilde{\psi}_2(x)$, hence $\overline{\delta}(x) = 0$. Therefore

$$\mathrm{Fix}(\tau) \cap U_0 \subseteq \overline{\delta}^{-1}(0).$$

Conversely, suppose $x \in U_0$ satisfies $\overline{\delta}(x) = 0$. Then $\widetilde{\psi}_1(x) = \widetilde{\psi}_2(x)$. Choose any representative $\theta = (\xi_1, \dots, \xi_m) \in \Theta^\sharp$ of x, so $x = p_H(\theta)$. Since $\psi(\xi_1) = \psi(\xi_2)$ and $\bar{\psi} : \Xi_{\mathrm{reg}}/H \to M$ is a diffeomorphism by Assumption 2.2, the blocks ξ_1 and ξ_2 lie in the same H-orbit. Hence there exists $h \in H$ such that $\xi_2 = h \cdot \xi_1$. Then

$$(\xi_2, \xi_1, \xi_3, \dots, \xi_m) = (h \cdot \xi_1, \xi_1, \xi_3, \dots, \xi_m)$$

and

$$(\xi_1, \xi_2, \xi_3, \dots, \xi_m) = (\xi_1, h \cdot \xi_1, \xi_3, \dots, \xi_m)$$

belong to the same H^m-orbit, so $\tau \cdot x = x$. Therefore

$$\overline{\delta}^{-1}(0) \subseteq \mathrm{Fix}(\tau) \cap U_0.$$

Hence

$$\mathrm{Fix}(\tau) \cap U_0 = \overline{\delta}^{-1}(0).$$

Finally, because surjectivity of the differential is an open condition, after shrinking U_0 if necessary, $D\overline{\delta}$ is surjective at every point of U_0. The submersion theorem then implies that $\overline{\delta}^{-1}(0)$ is a smooth embedded submanifold of codimension d. This completes the proof. □

C.4. Linearization of the residual involution

We now show that the residual transposition τ can be put into exact linear sign-reversal form near the collision stratum.

Proposition C.7 (Linearization near the fixed set)**.** *Let*

$$r := \dim \widetilde{\Theta}.$$

There exist an open neighborhood $U_1 \subseteq U_0$ of $x^\star$, an open neighborhood $U_F \subseteq \mathbb{R}^{r-d}$ of the origin, a radius $\rho > 0$, and a diffeomorphism

$$\chi : U_1 \xrightarrow{\cong} U_F \times B_\rho(0) \subseteq \mathbb{R}^{r-d} \times \mathbb{R}^d$$

such that:

1. $\chi(x^\star) = (0, 0)$*;*
2. $\chi\big(\mathrm{Fix}(\tau) \cap U_1\big) = U_F \times \{0\}$*;*
3. $$\chi \circ \tau \circ \chi^{-1}(u, v) = (u, -v) \qquad \textit{for all } (u, v) \in U_F \times B_\rho(0).$$

Proof. Set

$$F := \mathrm{Fix}(\tau) \cap U_0 = \overline{\delta}^{-1}(0),$$

which is a smooth embedded submanifold of codimension d by Lemma C.6. Choose any smooth Riemannian metric g_0 on U_0 and average it over the two-element group $\langle \tau \rangle$:

$$g := \frac{1}{2}\big(g_0 + \tau^* g_0\big).$$

Then g is a smooth Riemannian metric on U_0 with respect to which τ is an isometry.

Fix any $x \in F$. Because τ fixes F pointwise,

$$D\tau_x|_{T_x F} = I.$$

Conversely, suppose $v \in T_x\widetilde{\Theta}$ satisfies $D\tau_x v = v$. Using $\overline{\delta} \circ \tau = -\,\overline{\delta}$ and differentiating at $x \in F$, we obtain

$$D\overline{\delta}_x \circ D\tau_x = -\,D\overline{\delta}_x.$$

Applying this identity to v gives

$$D\overline{\delta}_x(v) = D\overline{\delta}_x(D\tau_x v) = -\,D\overline{\delta}_x(v),$$

hence $D\overline{\delta}_x(v) = 0$. Since $F = \overline{\delta}^{-1}(0)$ and $\overline{\delta}$ is a submersion on U_0, we have

$$T_x F = \ker D\overline{\delta}_x.$$

Therefore the $+1$-eigenspace of $D\tau_x$ is exactly $T_x F$.

Because $D\tau_x$ is a g_x-orthogonal involution, its eigenspaces for eigenvalues $+1$ and -1 are orthogonal and span $T_x\widetilde{\Theta}$. Since the $+1$-eigenspace is T_xF, the orthogonal complement

$$N_xF := (T_xF)^{\perp_g}$$

is exactly the -1-eigenspace. Thus

$$D\tau_x(\nu) = -\,\nu \qquad \text{for all } \nu \in N_xF,\ x \in F. \tag{14}$$

Let $NF \to F$ be the g-orthogonal normal bundle of F. By the tubular neighborhood theorem for Riemannian manifolds, after shrinking around $x^\star$ if necessary, there exists an open neighborhood $\Omega \subseteq NF$ of the zero section such that the exponential map

$$\mathrm{Exp} : \Omega \to U_1 \subseteq U_0$$

is a diffeomorphism onto an open neighborhood U_1 of $x^\star$. Because τ is a g-isometry fixing F pointwise, geodesics are mapped to geodesics and

$$\tau\big(\mathrm{Exp}_x(\nu)\big) = \mathrm{Exp}_x(D\tau_x\nu).$$

Using (14), this becomes

$$\tau\big(\mathrm{Exp}_x(\nu)\big) = \mathrm{Exp}_x(-\nu) \qquad \text{for all } (x,\nu) \in \Omega.$$

Now choose local coordinates

$$u : F \supseteq F_0 \to U_F \subseteq \mathbb{R}^{r-d}$$

centered at $x^\star$, and choose a smooth g-orthonormal frame

$$e_1, \dots, e_d$$

for the normal bundle NF over F_0. After shrinking F_0 and Ω if necessary, every $\nu \in \Omega|_{F_0}$ can be written uniquely as

$$\nu = \sum_{j=1}^{d} v_j e_j(x) \qquad \text{with } v = (v_1, \dots, v_d) \in B_\rho(0) \subseteq \mathbb{R}^d$$

for some $\rho > 0$. Define

$$\chi\big(\mathrm{Exp}_x(\nu)\big) := (u(x), v).$$

This is a diffeomorphism

$$\chi : U_1 \xrightarrow{\cong} U_F \times B_\rho(0).$$

By construction, $\chi(x^\star) = (0,0)$, the fixed set $F \cap U_1$ is sent to $U_F \times \{0\}$, and the equivariance of the exponential map gives

$$\chi \circ \tau \circ \chi^{-1}(u, v) = (u, -v).$$

This proves the proposition. $\square$

C.5. The branch normal form

We can now combine the local permutation reduction with the involution linearization to obtain the desired universal quotient model.

Theorem C.8 (Abstract local branch normal form)**.** *Let $\theta^\star \in \Theta^\sharp$ be a simple pairwise collision point and retain the notation introduced above. Then there exist:*

- *an open neighborhood $U \subseteq \widetilde{\Theta}$ of $x^\star$,*
- *an open neighborhood $V \subseteq \mathcal{Q}$ of $y^\star$,*
- *an open neighborhood $U_F \subseteq \mathbb{R}^{r-d}$ of the origin,*
- *a radius $\rho > 0$,*

- *a diffeomorphism*

$$\chi : U \xrightarrow{\cong} U_F \times B_\rho(0) \subseteq \mathbb{R}^{r-d} \times \mathbb{R}^d,$$

- *and a homeomorphism*

$$\overline{\chi} : V \xrightarrow{\cong} U_F \times \big(B_\rho(0)/\{\pm 1\}\big),$$

such that

$$\overline{\chi} \circ \pi \circ \chi^{-1}(u, v) = (u, [v]), \qquad (u, v) \in U_F \times B_\rho(0), \tag{15}$$

where

$$\pi : \widetilde{\Theta} \to \mathcal{Q} = \widetilde{\Theta}/S_m$$

is the quotient map and $[v] = \{v, -v\}$ denotes the orbit of v under the sign action of $\{\pm 1\}$ on $\mathbb{R}^d$. Moreover,

$$\chi\big(\mathrm{Fix}(\tau) \cap U\big) = U_F \times \{0\},$$

so the local singular stratum of the quotient corresponds exactly to the zero section of the relative coordinate.

Proof. Let U_0 be the neighborhood given by Lemma C.5, and let U_1 and χ be the neighborhood and coordinates from Proposition C.7. Set

$$U := U_0 \cap U_1.$$

After shrinking U_1 if necessary, we may assume that $\chi(U) = U_F \times B_\rho(0)$ for some open $U_F \subseteq \mathbb{R}^{r-d}$ and some radius $\rho > 0$.

By Lemma C.5, the orbit equivalence relation induced by the full S_m-action on U coincides with the orbit relation induced by the subgroup $\langle \tau \rangle = \{e, \tau\}$. Therefore the quotient of U inside $\mathcal{Q} = \widetilde{\Theta}/S_m$ agrees with the quotient of U by the involution τ. Concretely, if

$$V := \pi(U) \subseteq \mathcal{Q},$$

then V is canonically homeomorphic to $U/\langle \tau \rangle$.

Under the coordinates χ, the involution τ acts by

$$(u, v) \mapsto (u, -v)$$

by Proposition C.7. Hence

$$U/\langle \tau \rangle \cong \big(U_F \times B_\rho(0)\big)/\{\pm 1\} \cong U_F \times \big(B_\rho(0)/\{\pm 1\}\big),$$

where $\{\pm 1\}$ acts trivially on the u-coordinates and by sign reversal on the v-coordinates. Let

$$\overline{\chi} : V \to U_F \times \big(B_\rho(0)/\{\pm 1\}\big)$$

be the induced homeomorphism. Then, by construction,

$$\overline{\chi} \circ \pi \circ \chi^{-1}(u, v) = (u, [v]).$$

The statement about the singular stratum follows from $\chi(\mathrm{Fix}(\tau) \cap U) = U_F \times \{0\}$, proved in Proposition C.7. □

Corollary C.9 (Two-dimensional slice model)**.** *Assume the hypotheses of Theorem C.8. If $d \geq 2$, let $P \subseteq \mathbb{R}^d$ be any two-dimensional linear subspace. Choose a real linear isomorphism*

$$\ell : P \xrightarrow{\cong} \mathbb{C}.$$

Then the map

$$\vartheta : P/\{\pm 1\} \to \mathbb{C}, \qquad \vartheta([v]) := \ell(v)^2$$

is a well-defined homeomorphism. Consequently, on the slice $U_F \times (P \cap B_\rho(0))$, the local quotient map is topologically equivalent to

$$(u, z) \longmapsto (u, z^2), \qquad u \in U_F,\ z \in \mathbb{C}.$$

If $d = 1$, the transverse local model is the one-dimensional fold

$$t \longmapsto t^2.$$

Proof. The map ϑ is well defined because

$$\ell(-v)^2 = (-\ell(v))^2 = \ell(v)^2.$$

It is surjective because every $w \in \mathbb{C}$ has a square root $z \in \mathbb{C}$, and injective on the quotient because if $\ell(v)^2 = \ell(v')^2$, then $\ell(v') = \pm\ell(v)$, hence $v' = \pm v$, so $[v'] = [v]$. Continuity of ϑ is immediate. Continuity of ϑ^{-1} follows because ϑ factors the continuous proper map

$$P \to \mathbb{C}, \qquad v \mapsto \ell(v)^2$$

through the quotient $P/\{\pm 1\}$, and the induced bijection from the Hausdorff quotient $P/\{\pm 1\}$ to $\mathbb{C}$ is a homeomorphism.

The local quotient description from Theorem C.8 therefore becomes, on the slice with relative coordinate in P,

$$(u, v) \mapsto (u, [v]) \mapsto (u, \ell(v)^2).$$

Writing $z = \ell(v)$ yields the model $(u, z) \mapsto (u, z^2)$. When $d = 1$, the sign quotient $\mathbb{R}/\{\pm 1\}$ is represented by $t \mapsto t^2$, which is the standard one-dimensional fold. $\square$

Remark C.10 (What this theorem does and does not assert). Theorem C.8 is a *local* statement near a simple pairwise collision in the reduced manifold $\widetilde{\Theta}$. It does not assert that the full global quotient $\mathcal{Q}$ is everywhere modeled by a single $\mathbb{Z}_2$-quotient, nor does it assume that the complete singular stratification has only pairwise collisions. Its content is more precise: whenever a collision is locally generated by a single transverse pairwise coincidence of signatures, the local singularity type is universal and equal to the sign quotient on a d-dimensional relative coordinate.

Remark C.11 (Interpretation for the homogeneous base model). In the homogeneous two-layer model of Appendix B, the regular quotient was identified exactly with the unordered configuration space $\mathrm{UConf}_m(M_{\mathrm{hom}})$, and the discriminant is the pullback of the fat diagonal. In that setting, a simple pairwise collision is literally an ordinary pairwise configuration collision in M_{hom}, so Theorem C.8 gives the local singular chart near the first non-regular stratum of that explicit quotient.

D. Monodromy, No-Global-Gauge, and Exact Local Atlas Complexity

This appendix extracts the topological consequences of the branch normal form proved in Appendix C. The argument proceeds in two steps. First, we analyze the standard branched double cover

$$z \longmapsto z^2$$

on punctured annuli, prove that its monodromy is nontrivial, and compute its sectional category exactly. Second, we transfer that calculation to the neural quotient by means of the local branch chart from Theorem C.8. The outcome is a fully local and exact statement: near any simple pairwise collision of signature dimension $d \geq 2$, there exist compact regular subsets of the quotient on which a single continuous gauge is impossible and exactly two charts are necessary and sufficient.

D.1. The standard square cover on a punctured annulus

For $0 < \epsilon < R$, define the closed annuli

$$A_{\epsilon,R} := \{w \in \mathbb{C} : \epsilon \leq |w| \leq R\}, \qquad \widetilde{A}_{\epsilon,R} := \{z \in \mathbb{C} : \sqrt{\epsilon} \leq |z| \leq \sqrt{R}\}.$$

Let

$$\mathrm{sq}_{\epsilon,R} : \widetilde{A}_{\epsilon,R} \to A_{\epsilon,R}, \qquad \mathrm{sq}_{\epsilon,R}(z) := z^2.$$

Proposition D.1 (The square map is a connected two-sheeted cover)**.** *For every* $0 < \epsilon < R$*, the map*

$$\mathrm{sq}_{\epsilon,R} : \widetilde{A}_{\epsilon,R} \to A_{\epsilon,R}$$

is a connected two-sheeted covering map with deck group $\{\mathrm{id}, -\mathrm{id}\} \cong \mathbb{Z}_2$.

Proof. Since $0 \notin \widetilde{A}_{\epsilon,R}$, the derivative

$$D\,\mathrm{sq}_{\epsilon,R}(z) = 2z$$

is nonzero at every $z \in \widetilde{A}_{\epsilon,R}$. Hence $\mathrm{sq}_{\epsilon,R}$ is a local diffeomorphism, and therefore a covering map of manifolds with boundary. For every $w \in A_{\epsilon,R}$, the preimage $\mathrm{sq}_{\epsilon,R}^{-1}(w)$ consists of the two distinct square roots $\{\zeta, -\zeta\}$, so the cover has exactly two sheets.

The involution

$$\delta : \widetilde{A}_{\epsilon,R} \to \widetilde{A}_{\epsilon,R}, \qquad \delta(z) := -z$$

satisfies $\mathrm{sq}_{\epsilon,R} \circ \delta = \mathrm{sq}_{\epsilon,R}$, so it is a deck transformation. Since every fiber has exactly two points, the full deck group is $\{\mathrm{id}, \delta\} \cong \mathbb{Z}_2$.

Finally, $\widetilde{A}_{\epsilon,R}$ is an annulus and is therefore connected. $\square$

Proposition D.2 (Monodromy of the square cover)**.** *Fix $r \in [\epsilon, R]$ and consider the positively oriented loop*

$$\gamma_r : [0,1] \to A_{\epsilon,R}, \qquad \gamma_r(t) := re^{2\pi i t}.$$

Let $z_0 = \sqrt{r} \in \widetilde{A}_{\epsilon,R}$. Then the unique lift $\widetilde{\gamma}_r$ of γ_r with $\widetilde{\gamma}_r(0) = z_0$ is

$$\widetilde{\gamma}_r(t) = \sqrt{r}\, e^{\pi i t},$$

and in particular

$$\widetilde{\gamma}_r(1) = -z_0.$$

Equivalently, the monodromy action of the positive generator of $\pi_1(A_{\epsilon,R}) \cong \mathbb{Z}$ on a fiber of the cover $\mathrm{sq}_{\epsilon,R}$ is the nontrivial transposition of the two sheets.

Proof. Define

$$\widetilde{\gamma}_r(t) := \sqrt{r}\, e^{\pi i t}.$$

Then

$$\mathrm{sq}_{\epsilon,R}(\widetilde{\gamma}_r(t)) = \left(\sqrt{r}\, e^{\pi i t}\right)^2 = re^{2\pi i t} = \gamma_r(t),$$

so $\widetilde{\gamma}_r$ is a lift of γ_r. Moreover, $\widetilde{\gamma}_r(0) = \sqrt{r} = z_0$, so by uniqueness of path lifting for covering maps, $\widetilde{\gamma}_r$ is the unique such lift. Evaluating at $t = 1$ gives

$$\widetilde{\gamma}_r(1) = \sqrt{r}\, e^{\pi i} = -\sqrt{r} = -z_0.$$

Thus one full loop in the base exchanges the two sheets of the fiber. $\square$

Proposition D.3 (No global section of the square cover on the annulus)**.** *For every $0 < \epsilon < R$, the covering map*

$$\mathrm{sq}_{\epsilon,R} : \widetilde{A}_{\epsilon,R} \to A_{\epsilon,R}$$

admits no continuous global section.

Proof. Assume for contradiction that there exists a continuous section

$$s : A_{\epsilon,R} \to \widetilde{A}_{\epsilon,R} \qquad \text{such that} \qquad \mathrm{sq}_{\epsilon,R} \circ s = \mathrm{id}_{A_{\epsilon,R}}.$$

Fix $r \in [\epsilon, R]$. Let

$$C_r := \{w \in \mathbb{C} : |w| = r\} \subset A_{\epsilon,R}, \qquad \widetilde{C}_r := \{z \in \mathbb{C} : |z| = \sqrt{r}\} \subset \widetilde{A}_{\epsilon,R}.$$

Then the restriction

$$s_r := s|_{C_r} : C_r \to \widetilde{C}_r$$

satisfies

$$\mathrm{sq}_{\epsilon,R}|_{\widetilde{C}_r} \circ s_r = \mathrm{id}_{C_r}.$$

Identify C_r and $\widetilde{C}_r$ with S^1 by radial rescaling. Under these identifications, $\mathrm{sq}_{\epsilon,R}|_{\widetilde{C}_r}$ becomes the map

$$\omega : S^1 \to S^1, \qquad \omega(\zeta) = \zeta^2.$$

On fundamental groups, ω_* is multiplication by 2 on $\pi_1(S^1) \cong \mathbb{Z}$. Since

$$\omega_* \circ (s_r)_* = (\mathrm{id}_{C_r})_* = \mathrm{id}_{\mathbb{Z}},$$

we would obtain an endomorphism $(s_r)_* : \mathbb{Z} \to \mathbb{Z}$ satisfying

$$2\,(s_r)_* = \mathrm{id}_{\mathbb{Z}},$$

which is impossible because multiplication by 2 is not invertible in $\mathrm{End}(\mathbb{Z})$. This contradiction proves that no global section exists. $\square$

Theorem D.4 (Exact sectional category of the square cover)**.** *For every* $0 < \epsilon < R$*,*

$$\mathrm{secat}(\mathrm{sq}_{\epsilon,R}) = 2.$$

Equivalently, exactly two local sections are necessary and sufficient to cover the annulus $A_{\epsilon,R}$*.*

Proof. By Proposition D.3, the covering has no global section, hence

$$\mathrm{secat}(\mathrm{sq}_{\epsilon,R}) > 1.$$

It remains to prove the upper bound $\mathrm{secat}(\mathrm{sq}_{\epsilon,R}) \le 2$. Define the two open subsets of $A_{\epsilon,R}$ (open in the subspace topology of $A_{\epsilon,R}$)

$$U_+ := A_{\epsilon,R} \cap \big(\mathbb{C} \setminus (-\infty, 0]\big), \qquad U_- := A_{\epsilon,R} \cap \big(\mathbb{C} \setminus [0, \infty)\big).$$

Clearly

$$A_{\epsilon,R} = U_+ \cup U_-.$$

On U_+, the principal argument

$$\mathrm{Arg}_+ : U_+ \to (-\pi, \pi)$$

is continuous, so every $w \in U_+$ admits a unique representation

$$w = |w| e^{i\,\mathrm{Arg}_+(w)}.$$

Define

$$s_+(w) := |w|^{1/2} \exp\left(\frac{i}{2}\,\mathrm{Arg}_+(w)\right), \qquad w \in U_+.$$

Then s_+ is continuous and

$$\mathrm{sq}_{\epsilon,R}(s_+(w)) = |w| \exp\big(i\,\mathrm{Arg}_+(w)\big) = w.$$

Similarly, on U_-, choose the continuous branch

$$\mathrm{Arg}_- : U_- \to (0, 2\pi)$$

and define

$$s_-(w) := |w|^{1/2} \exp\left(\frac{i}{2}\,\mathrm{Arg}_-(w)\right), \qquad w \in U_-.$$

Again s_- is continuous and satisfies

$$\mathrm{sq}_{\epsilon,R} \circ s_- = \mathrm{id}_{U_-}.$$

Thus $A_{\epsilon,R}$ admits an open cover by two sets, each carrying a continuous local section. Therefore $\mathrm{secat}(\mathrm{sq}_{\epsilon,R}) \le 2$. The lower and upper bounds together yield

$$\mathrm{secat}(\mathrm{sq}_{\epsilon,R}) = 2.$$

$\square$

D.2. Transfer to the neural quotient

From now on we assume $d = \dim M \geq 2$, so that the two-dimensional slice model of Corollary C.9 is available.

Let $\theta^\star \in \Theta^\sharp$ be a simple pairwise collision point and retain the notation of Appendix C. In particular, Theorem C.8 provides:

- an open neighborhood $U \subseteq \widetilde{\Theta}$ of $x^\star = p_H(\theta^\star)$,
- an open neighborhood $V \subseteq \mathcal{Q}$ of $y^\star = [x^\star]$,
- a homeomorphism
$$\overline{\chi} : V \xrightarrow{\cong} U_F \times \big(B_\rho(0)/\{\pm 1\}\big),$$
- and a diffeomorphism
$$\chi : U \xrightarrow{\cong} U_F \times B_\rho(0)$$
intertwining the local quotient with $(u, v) \mapsto (u, [v])$.

Shrinking U and V if necessary, we may and will assume that

$$\chi(U) = B_\eta^{r-d}(0) \times B_\rho(0)$$

for some $\eta, \rho > 0$, where $r = \dim \widetilde{\Theta}$. In particular, U is contractible. Write

$$\widetilde{\Theta}_{\mathrm{reg}} := p_H(\Theta_{\mathrm{reg}}) \subseteq \widetilde{\Theta}, \qquad \pi_{\mathrm{reg}} : \widetilde{\Theta}_{\mathrm{reg}} \to B_{\mathrm{reg}}$$

for the residual quotient map on the regular locus, so that

$$q = \pi_{\mathrm{reg}} \circ p_H|_{\Theta_{\mathrm{reg}}}.$$

Fix once and for all:

- the base tangential coordinate $u = 0 \in B_\eta^{r-d}(0)$,
- a two-dimensional linear subspace $P \subseteq \mathbb{R}^d$,
- a real-linear isomorphism $\ell : P \xrightarrow{\cong} \mathbb{C}$,
- the induced homeomorphism
$$\vartheta : P/\{\pm 1\} \xrightarrow{\cong} \mathbb{C}, \qquad \vartheta([v]) = \ell(v)^2$$
from Corollary C.9.

For $0 < \epsilon < R < \rho^2$, define the compact *transverse neural annulus*

$$K_{\epsilon,R} := \overline{\chi}^{-1}\big(\{0\} \times \vartheta^{-1}(A_{\epsilon,R})\big) \subseteq V \cap B_{\mathrm{reg}}, \tag{16}$$

and its local lifted annulus

$$\widetilde{K}^{\mathrm{loc}}_{\epsilon,R} := \chi^{-1}\big(\{0\} \times \ell^{-1}(\widetilde{A}_{\epsilon,R})\big) \subseteq U \cap \widetilde{\Theta}_{\mathrm{reg}}. \tag{17}$$

Proposition D.5 (Local preimage decomposition and square-cover equivalence)**.** *For every $0 < \epsilon < R < \rho^2$, the following hold.*

1. *The restriction*
$$\pi_{\mathrm{reg}}|_{\widetilde{K}^{\mathrm{loc}}_{\epsilon,R}} : \widetilde{K}^{\mathrm{loc}}_{\epsilon,R} \to K_{\epsilon,R}$$
is homeomorphic to the standard square cover
$$\mathrm{sq}_{\epsilon,R} : \widetilde{A}_{\epsilon,R} \to A_{\epsilon,R}.$$

2. *The full preimage of $K_{\epsilon,R}$ under π_{reg} decomposes as the disjoint union*

$$\pi_{\mathrm{reg}}^{-1}(K_{\epsilon,R}) = \bigsqcup_{[\sigma]\in S_m/\langle\tau\rangle} \sigma \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R},$$

where $\langle\tau\rangle = \{e, \tau\}$.

3. *Each set $\sigma \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R}$ is a connected component of $\pi_{\mathrm{reg}}^{-1}(K_{\epsilon,R})$, and on each component the restricted map π_{reg} is homeomorphic to $\mathrm{sq}_{\epsilon,R}$.*

Proof. For item 1, by the defining intertwining relation from Theorem C.8,

$$\overline{\chi} \circ \pi \circ \chi^{-1}(u, v) = (u, [v]).$$

Restricting to the slice $u = 0$ and the plane $P \subseteq \mathbb{R}^d$, then composing with $\vartheta([v]) = \ell(v)^2$, yields the commutative diagram

$$\begin{tikzcd} \widetilde{K}^{\mathrm{loc}}_{\epsilon,R} \arrow[r, "\chi"] \arrow[d, "\pi_{\mathrm{reg}}|_{\widetilde{K}^{\mathrm{loc}}_{\epsilon,R}}"'] & \{0\} \times \ell^{-1}(\widetilde{A}_{\epsilon,R}) \arrow[d, "(0,v)\mapsto(0,\ell(v)^2)"] \\ K_{\epsilon,R} \arrow[r, "\overline{\chi}"] & \{0\} \times \vartheta^{-1}(A_{\epsilon,R}), \end{tikzcd}$$

which identifies $\pi_{\mathrm{reg}}|_{\widetilde{K}^{\mathrm{loc}}_{\epsilon,R}}$ with $\mathrm{sq}_{\epsilon,R}$.

For item 2, first note that

$$\widetilde{K}^{\mathrm{loc}}_{\epsilon,R} = \pi_{\mathrm{reg}}^{-1}(K_{\epsilon,R}) \cap U.$$

Indeed, inside the chart U, the quotient map is exactly $(u, v) \mapsto (u, [v])$, so the only points of U projecting to $K_{\epsilon,R}$ are those with $u = 0$, $v \in P$, and $\ell(v) \in \widetilde{A}_{\epsilon,R}$.

Now let $x \in \pi_{\mathrm{reg}}^{-1}(K_{\epsilon,R})$. Since $K_{\epsilon,R} \subseteq \pi(U)$, there exists $u_x \in U$ such that $\pi(u_x) = \pi(x)$. Hence $x = \sigma \cdot u_x$ for some $\sigma \in S_m$. Because $u_x \in \pi_{\mathrm{reg}}^{-1}(K_{\epsilon,R}) \cap U$, we have $u_x \in \widetilde{K}^{\mathrm{loc}}_{\epsilon,R}$. Therefore

$$x \in \sigma \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R},$$

showing

$$\pi_{\mathrm{reg}}^{-1}(K_{\epsilon,R}) = \bigcup_{\sigma\in S_m} \sigma \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R}.$$

Because τ acts in the local coordinates by $v \mapsto -v$, and because $\widetilde{A}_{\epsilon,R}$ is invariant under $z \mapsto -z$, we have

$$\tau \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R} = \widetilde{K}^{\mathrm{loc}}_{\epsilon,R}.$$

Thus the union depends only on cosets in $S_m/\langle\tau\rangle$.

It remains to prove disjointness of distinct cosets. Suppose

$$\sigma_1 \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R} \cap \sigma_2 \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R} \neq \varnothing.$$

Then

$$(\sigma_2^{-1}\sigma_1) \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R} \cap \widetilde{K}^{\mathrm{loc}}_{\epsilon,R} \neq \varnothing.$$

Since $\widetilde{K}^{\mathrm{loc}}_{\epsilon,R} \subseteq U$, this implies

$$(\sigma_2^{-1}\sigma_1)(U) \cap U \neq \varnothing.$$

By Lemma C.5, we must have

$$\sigma_2^{-1}\sigma_1 \in \{e, \tau\} = \langle\tau\rangle.$$

Hence $[\sigma_1] = [\sigma_2]$ in $S_m/\langle\tau\rangle$. This proves the disjoint union formula.

For item 3, each set $\sigma \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R}$ is connected because $\widetilde{K}^{\mathrm{loc}}_{\epsilon,R}$ is connected and permutation acts by homeomorphisms. Moreover,

$$\sigma \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R} = \pi_{\mathrm{reg}}^{-1}(K_{\epsilon,R}) \cap \sigma(U),$$

so it is open in the subspace topology of $\pi_{\mathrm{reg}}^{-1}(K_{\epsilon,R})$. Being also closed by the disjoint union description, it is a connected component. Since π_{reg} commutes with the S_m-action, the restriction of π_{reg} to each component is conjugate to the restriction on $\widetilde{K}^{\mathrm{loc}}_{\epsilon,R}$, hence again homeomorphic to the square cover. □

Corollary D.6 (No global gauge on a transverse neural annulus)**.** *Assume $d \geq 2$ and let $K_{\epsilon,R} \subseteq B_{\mathrm{reg}}$ be the compact transverse annulus defined in (16). Then there does not exist a continuous global gauge*

$$s : K_{\epsilon,R} \to \Theta_{\mathrm{reg}} \qquad \textit{such that} \qquad q \circ s = \mathrm{id}_{K_{\epsilon,R}}.$$

Equivalently,

$$\mathfrak{a}(K_{\epsilon,R}; q) > 1.$$

Proof. Assume for contradiction that there exists a continuous section

$$s : K_{\epsilon,R} \to \Theta_{\mathrm{reg}} \qquad \text{with} \qquad q \circ s = \mathrm{id}_{K_{\epsilon,R}}.$$

Set

$$s^{\mathrm{red}} := p_H \circ s : K_{\epsilon,R} \to \widetilde{\Theta}_{\mathrm{reg}}.$$

Since $q = \pi_{\mathrm{reg}} \circ p_H|_{\Theta_{\mathrm{reg}}}$, we have

$$\pi_{\mathrm{reg}} \circ s^{\mathrm{red}} = \mathrm{id}_{K_{\epsilon,R}},$$

so s^{red} is a continuous section of π_{reg} over $K_{\epsilon,R}$.

By Proposition D.5,

$$\pi_{\mathrm{reg}}^{-1}(K_{\epsilon,R}) = \bigsqcup_{[\sigma] \in S_m/\langle \tau \rangle} \sigma \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R}$$

is a disjoint union of connected components. Since $K_{\epsilon,R}$ is connected and s^{red} is continuous, the image $s^{\mathrm{red}}(K_{\epsilon,R})$ must lie in a single connected component, say

$$s^{\mathrm{red}}(K_{\epsilon,R}) \subseteq \sigma \cdot \widetilde{K}^{\mathrm{loc}}_{\epsilon,R}$$

for some $\sigma \in S_m$.

Define

$$\widetilde{s} : K_{\epsilon,R} \to \widetilde{K}^{\mathrm{loc}}_{\epsilon,R}, \qquad \widetilde{s}(x) := \sigma^{-1} \cdot s^{\mathrm{red}}(x).$$

Because π_{reg} is S_m-invariant,

$$\pi_{\mathrm{reg}} \circ \widetilde{s} = \pi_{\mathrm{reg}} \circ s^{\mathrm{red}} = \mathrm{id}_{K_{\epsilon,R}}.$$

Thus $\widetilde{s}$ is a continuous section of $\pi_{\mathrm{reg}}|_{\widetilde{K}^{\mathrm{loc}}_{\epsilon,R}}$. By item 1 of Proposition D.5, this is equivalent to a continuous section of the square cover $\mathrm{sq}_{\epsilon,R}$, contradicting Proposition D.3. Therefore no such global gauge s can exist.

The final inequality $\mathfrak{a}(K_{\epsilon,R}; q) > 1$ is exactly the statement that the restricted quotient map $q|_{q^{-1}(K_{\epsilon,R})}$ admits no single-chart section. □

Corollary D.7 (Exact local atlas complexity on a transverse neural annulus)**.** *Under the hypotheses of Corollary D.6,*

$$\mathfrak{a}(K_{\epsilon,R}; q) = 2.$$

That is, on the compact annulus $K_{\epsilon,R}$, exactly two local gauges are necessary and sufficient.

Proof. By Corollary D.6,

$$\mathfrak{a}(K_{\epsilon,R}; q) > 1.$$

It remains to prove the upper bound $\mathfrak{a}(K_{\epsilon,R}; q) \leq 2$.

Because U is contractible and

$$p_H : p_H^{-1}(U) \to U$$

is a principal H^m-bundle, it is trivial. Hence there exists a continuous section

$$\sigma_H : U \to p_H^{-1}(U) \subseteq \Theta^\sharp.$$

Since $U \cap \widetilde{\Theta}_{\mathrm{reg}}$ consists of reduced regular points, the image of σ_H over that set lies in Θ_{reg}.

Now let $U_+, U_- \subseteq A_{\epsilon,R}$ be the two slit-annulus cover from the proof of Theorem D.4. Define

$$W_\pm := K_{\epsilon,R} \cap \overline{\chi}^{-1}\big(\{0\} \times \vartheta^{-1}(U_\pm)\big).$$

These are open in the subspace topology of $K_{\epsilon,R}$, and

$$K_{\epsilon,R} = W_+ \cup W_-.$$

Via the homeomorphism from item 1 of Proposition D.5, the square-root branches s_+ and s_- on U_+ and U_- transport to continuous local sections

$$s_\pm^{\mathrm{red}} : W_\pm \to \widetilde{K}^{\mathrm{loc}}_{\epsilon,R} \subseteq U \cap \widetilde{\Theta}_{\mathrm{reg}}$$

of π_{reg}, i.e.

$$\pi_{\mathrm{reg}} \circ s_\pm^{\mathrm{red}} = \mathrm{id}_{W_\pm}.$$

Define

$$s_\pm := \sigma_H \circ s_\pm^{\mathrm{red}} : W_\pm \to \Theta_{\mathrm{reg}}.$$

Then

$$q \circ s_\pm = \pi_{\mathrm{reg}} \circ p_H \circ \sigma_H \circ s_\pm^{\mathrm{red}} = \pi_{\mathrm{reg}} \circ s_\pm^{\mathrm{red}} = \mathrm{id}_{W_\pm}.$$

Thus each $W_\pm$ admits a continuous local gauge for q, so

$$\mathfrak{a}(K_{\epsilon,R}; q) \leq 2.$$

Combining the lower and upper bounds yields

$$\mathfrak{a}(K_{\epsilon,R}; q) = 2.$$

□

Remark D.8 (Why the exact value is genuinely local). Corollary D.7 does not depend on any global model of the full neural quotient. It is a purely local consequence of the branch normal form: once a simple pairwise collision is present and the local two-dimensional transverse slice exists, the compact punctured neighborhood $K_{\epsilon,R}$ already forces the exact chart complexity 2.

Remark D.9 (The $d = 1$ case). When $d = 1$, the transverse local model from Corollary C.9 is the fold $t \mapsto t^2$ rather than the branched double cover $z \mapsto z^2$. In that case a punctured neighborhood is disconnected and the monodromy argument of the present appendix does not apply. The exact value 2 in Corollary D.7 is therefore a genuinely two-dimensional phenomenon.

E. Conditioning Lower Bound

This appendix derives the quantitative instability law behind the branch normal form. The main point is subtle but fundamental: the blow-up occurs when the base space is parameterized by a *branched invariant coordinate* of the form $w = z^2$, which is precisely the coordinate system induced by local collision invariants. By contrast, if one works in the orbifold coordinate $[v] \in \mathbb{R}^d/\{\pm 1\}$ itself, the square-root singularity disappears. The conditioning theorem below is therefore a statement about the instability of *canonicalization in branched coordinates*, not about an intrinsic Lipschitz obstruction in the quotient orbifold metric.

We begin with the model sector for the square map and then transfer the result to the local neural quotient. Throughout this appendix we fix:

- a smooth Riemannian metric $g_{\widetilde{\Theta}}$ on the reduced manifold $\widetilde{\Theta}$, with induced geodesic distance $d_{\widetilde{\Theta}}$;
- a smooth Riemannian metric $g_{\Theta^\sharp}$ on $\Theta^\sharp = \Theta \cap \Xi^m_{\mathrm{reg}}$, with induced geodesic distance $d_{\Theta^\sharp}$.

All constants below are allowed to depend on these fixed metrics and on the local branch chart, but never on ϵ.

E.1. Square-root rigidity and conditioning on a slit sector

Fix parameters

$$0 < \delta < \pi, \qquad 0 < \epsilon < R.$$

Define the closed sector annulus

$$S^{\delta}_{\epsilon,R} := \left\{ w = re^{i\theta} \in \mathbb{C} : \epsilon \le r \le R,\ |\theta| \le \pi - \delta \right\}. \tag{18}$$

This is a compact, connected, simply connected subset of $\mathbb{C} \setminus \{0\}$. Its principal square-root image is

$$\widetilde{S}^{\delta}_{\epsilon,R} := \left\{ z = \rho e^{i\phi} \in \mathbb{C} : \sqrt{\epsilon} \le \rho \le \sqrt{R},\ |\phi| \le \frac{\pi - \delta}{2} \right\}. \tag{19}$$

Let

$$\mathrm{sq}(z) := z^2.$$

Then sq maps $\widetilde{S}^{\delta}_{\epsilon,R}$ homeomorphically onto $S^{\delta}_{\epsilon,R}$. The principal argument branch on $S^{\delta}_{\epsilon,R}$ is the continuous map

$$\mathrm{Arg}_\delta : S^{\delta}_{\epsilon,R} \to [-\pi + \delta, \pi - \delta]$$

defined by $w = |w| e^{i \, \mathrm{Arg}_\delta(w)}$.

Definition E.1 (Principal square-root branch on the slit sector)**.** Define

$$\mathfrak{s}^{\delta}_{\epsilon,R} : S^{\delta}_{\epsilon,R} \to \widetilde{S}^{\delta}_{\epsilon,R}, \qquad \mathfrak{s}^{\delta}_{\epsilon,R}(w) := |w|^{1/2} \exp\!\left(\frac{i}{2} \, \mathrm{Arg}_\delta(w) \right). \tag{20}$$

Then

$$\mathrm{sq} \circ \mathfrak{s}^{\delta}_{\epsilon,R} = \mathrm{id}_{S^{\delta}_{\epsilon,R}}.$$

Proposition E.2 (Rigidity of continuous sections on a slit sector)**.** *Let*

$$t : S^{\delta}_{\epsilon,R} \to \widetilde{A}_{\epsilon,R}$$

be a continuous section of the square map, i.e.

$$t(w)^2 = w \qquad \textit{for all } w \in S^{\delta}_{\epsilon,R}.$$

Then either

$$t = \mathfrak{s}^{\delta}_{\epsilon,R} \qquad \textit{or} \qquad t = -\,\mathfrak{s}^{\delta}_{\epsilon,R}.$$

Proof. Because $\mathfrak{s}^{\delta}_{\epsilon,R}(w) \neq 0$ for all $w \in S^{\delta}_{\epsilon,R}$, the pointwise ratio

$$\eta(w) := \frac{t(w)}{\mathfrak{s}^{\delta}_{\epsilon,R}(w)}$$

is well defined and continuous. Since both $t(w)$ and $\mathfrak{s}^{\delta}_{\epsilon,R}(w)$ are square roots of w, we have

$$\eta(w)^2 = \frac{t(w)^2}{(\mathfrak{s}^{\delta}_{\epsilon,R}(w))^2} = 1.$$

Hence

$$\eta(w) \in \{+1, -1\} \qquad \text{for all } w \in S^{\delta}_{\epsilon,R}.$$

The codomain $\{+1, -1\}$ is discrete, while $S^{\delta}_{\epsilon,R}$ is connected. Therefore η is constant. If $\eta \equiv +1$, then $t = \mathfrak{s}^{\delta}_{\epsilon,R}$; if $\eta \equiv -1$, then $t = -\,\mathfrak{s}^{\delta}_{\epsilon,R}$. ∎

Proposition E.3 (Sharp square-root conditioning on a slit sector)**.** *For any continuous section*

$$t : S^{\delta}_{\epsilon,R} \to \widetilde{A}_{\epsilon,R} \qquad \textit{with} \qquad t(w)^2 = w,$$

its Euclidean Lipschitz constant satisfies

$$\mathrm{Lip}_{\mathbb{C}}(t) \geq \frac{1}{2\sqrt{\epsilon}}, \tag{21}$$

where

$$\mathrm{Lip}_{\mathbb{C}}(t) := \sup_{w \neq w' \in S^{\delta}_{\epsilon,R}} \frac{|t(w) - t(w')|}{|w - w'|}.$$

Proof. By Proposition E.2, either $t = \mathfrak{s}^{\delta}_{\epsilon,R}$ or $t = -\mathfrak{s}^{\delta}_{\epsilon,R}$. It is therefore enough to prove the estimate for the principal branch, since the Lipschitz constant is unchanged by multiplication by -1.

The positive real interval $[\epsilon, R]$ lies inside $S^{\delta}_{\epsilon,R}$. On this interval,

$$\mathfrak{s}^{\delta}_{\epsilon,R}(r) = \sqrt{r} \qquad \text{for } r \in [\epsilon, R].$$

Hence

$$\mathrm{Lip}_{\mathbb{C}}(\mathfrak{s}^{\delta}_{\epsilon,R}) \geq \sup_{x \neq y \in [\epsilon,R]} \frac{|\sqrt{x} - \sqrt{y}|}{|x - y|}.$$

For $x \neq y$,

$$\frac{|\sqrt{x} - \sqrt{y}|}{|x - y|} = \frac{1}{\sqrt{x} + \sqrt{y}}.$$

Taking $x = \epsilon$ and letting $y \downarrow \epsilon$ gives

$$\sup_{x \neq y \in [\epsilon,R]} \frac{|\sqrt{x} - \sqrt{y}|}{|x - y|} = \frac{1}{2\sqrt{\epsilon}}.$$

Therefore

$$\mathrm{Lip}_{\mathbb{C}}(t) = \mathrm{Lip}_{\mathbb{C}}(\mathfrak{s}^{\delta}_{\epsilon,R}) \geq \frac{1}{2\sqrt{\epsilon}}.$$

□

Remark E.4 (Why the exponent $1/2$ is unavoidable). The exponent $1/2$ is not an artifact of the proof. It is exactly the branch order of the local quotient map $z \mapsto z^2$. Higher-order branch models would produce different exponents. In the present paper, the simple pairwise collision is universally of order 2, so the conditioning exponent is universally $1/2$.

E.2. Reduced neural sectors and branched-coordinate conditioning

We now transfer the preceding calculation to the local neural quotient.

Fix the branch chart from Theorem C.8 and the transverse complex-line model from Corollary C.9. Thus we have:

- a neighborhood $U \subseteq \widetilde{\Theta}$ of the collision point $x^\star$,
- a neighborhood $V \subseteq \mathcal{Q}$ of the quotient point $y^\star$,
- a diffeomorphism
$$\chi : U \xrightarrow{\cong} U_F \times B_\rho(0) \subseteq \mathbb{R}^{r-d} \times \mathbb{R}^d,$$
- a homeomorphism
$$\overline{\chi} : V \xrightarrow{\cong} U_F \times \big(B_\rho(0)/\{\pm 1\}\big),$$
- a fixed two-dimensional linear subspace $P \subseteq \mathbb{R}^d$,
- a real-linear isomorphism
$$\ell : P \xrightarrow{\cong} \mathbb{C},$$

- and the induced homeomorphism

$$\vartheta : P/\{\pm 1\} \xrightarrow{\cong} \mathbb{C}, \qquad \vartheta([v]) = \ell(v)^2.$$

Fix once and for all

$$0 < \delta < \pi, \qquad 0 < R_0 < \rho^2.$$

For every $0 < \epsilon < R \leq R_0$, define the compact transverse neural sector

$$K^{\delta}_{\epsilon,R} := \overline{\chi}^{-1}\big(\{0\} \times \vartheta^{-1}(S^{\delta}_{\epsilon,R})\big) \subseteq V \cap B_{\mathrm{reg}}. \tag{22}$$

Its principal local reduced lift is

$$\widetilde{K}^{\delta,+}_{\epsilon,R} := \chi^{-1}\big(\{0\} \times \ell^{-1}(\widetilde{S}^{\delta}_{\epsilon,R})\big) \subseteq U \cap \widetilde{\Theta}_{\mathrm{reg}}. \tag{23}$$

The opposite local lift is

$$\widetilde{K}^{\delta,-}_{\epsilon,R} := \tau \cdot \widetilde{K}^{\delta,+}_{\epsilon,R},$$

where τ is the residual transposition from Appendix C.

Definition E.5 (Branched coordinate on the transverse neural sector)**.** Define

$$\zeta^{\delta}_{\epsilon,R} : K^{\delta}_{\epsilon,R} \to S^{\delta}_{\epsilon,R}, \qquad \zeta^{\delta}_{\epsilon,R}(x) := \vartheta\big(\mathrm{pr}_2\big(\overline{\chi}(x)\big)\big)\,. \tag{24}$$

By construction, $\zeta^{\delta}_{\epsilon,R}$ is a homeomorphism.

Definition E.6 (Branched-coordinate Lipschitz seminorm)**.** Let

$$\widetilde{s} : K^{\delta}_{\epsilon,R} \to \widetilde{\Theta}_{\mathrm{reg}}$$

be a continuous reduced local gauge, i.e.

$$\pi_{\mathrm{reg}} \circ \widetilde{s} = \mathrm{id}_{K^{\delta}_{\epsilon,R}}\,.$$

Its branched-coordinate Lipschitz seminorm is

$$\mathrm{Lip}_{\mathrm{br}}(\widetilde{s}; K^{\delta}_{\epsilon,R}) := \sup_{x \neq y \in K^{\delta}_{\epsilon,R}} \frac{d_{\widetilde{\Theta}}(\widetilde{s}(x), \widetilde{s}(y))}{|\zeta^{\delta}_{\epsilon,R}(x) - \zeta^{\delta}_{\epsilon,R}(y)|}. \tag{25}$$

Lemma E.7 (Local sheet structure over a transverse sector)**.** *For every $0 < \epsilon < R \leq R_0$, the restriction*

$$\pi_{\mathrm{reg}}|_{\widetilde{K}^{\delta,+}_{\epsilon,R}} : \widetilde{K}^{\delta,+}_{\epsilon,R} \to K^{\delta}_{\epsilon,R}$$

is a homeomorphism. The same is true for $\widetilde{K}^{\delta,-}_{\epsilon,R}$. Moreover, the local preimage in U decomposes as the disjoint union

$$\pi^{-1}_{\mathrm{reg}}(K^{\delta}_{\epsilon,R}) \cap U = \widetilde{K}^{\delta,+}_{\epsilon,R} \ \sqcup\ \widetilde{K}^{\delta,-}_{\epsilon,R}.$$

Proof. Inside the branch chart, the quotient map is

$$(u, v) \mapsto (u, [v]).$$

On the transverse slice $u = 0$ and the chosen plane P, the principal sector $\ell^{-1}(\widetilde{S}^{\delta}_{\epsilon,R})$ is mapped homeomorphically onto its image in the sign quotient, because the sector width is strictly smaller than π and therefore contains no pair $v, -v$ with both points in the sector. Hence

$$\pi_{\mathrm{reg}}|_{\widetilde{K}^{\delta,+}_{\epsilon,R}}$$

is a homeomorphism. Applying τ, which acts by $v \mapsto -v$, gives the same statement for the opposite sheet $\widetilde{K}^{\delta,-}_{\epsilon,R}$.

Finally, still inside U, the only lifts of the base set $K^{\delta}_{\epsilon,R}$ are the two sign choices v and $-v$, so the local preimage is exactly the disjoint union of the principal and opposite sheets. $\square$

Proposition E.8 (Transport of reduced gauges to square-root branches)**.** *Let*

$$\widetilde{s} : K^{\delta}_{\epsilon,R} \to \widetilde{\Theta}_{\mathrm{reg}}$$

be a continuous reduced local gauge. Then there exist

$$\sigma \in S_m, \qquad \eta \in \{+1, -1\},$$

such that the map

$$t_{\widetilde{s}}(w) := \eta\, \ell\big(\mathrm{pr}_2\big(\chi\big(\sigma^{-1} \cdot \widetilde{s}\big((\zeta^{\delta}_{\epsilon,R})^{-1}(w)\big)\big)\big)\big), \qquad w \in S^{\delta}_{\epsilon,R}, \tag{26}$$

is a continuous section of the square map on $S^{\delta}_{\epsilon,R}$, i.e.

$$t_{\widetilde{s}}(w)^2 = w \qquad \textit{for all } w \in S^{\delta}_{\epsilon,R}.$$

Consequently,

$$t_{\widetilde{s}} = \mathfrak{s}^{\delta}_{\epsilon,R} \quad \textit{or} \quad t_{\widetilde{s}} = -\,\mathfrak{s}^{\delta}_{\epsilon,R}.$$

Proof. Because $\pi_{\mathrm{reg}} : \widetilde{\Theta}_{\mathrm{reg}} \to B_{\mathrm{reg}}$ is the quotient by the free action of the finite group S_m, it is a covering map. Since $K^{\delta}_{\epsilon,R}$ is connected and $\widetilde{s}$ is a section of π_{reg}, the image $\widetilde{s}(K^{\delta}_{\epsilon,R})$ lies in a single sheet of the covering over $K^{\delta}_{\epsilon,R}$.

The finite S_m-action is transitive on the sheets of this covering. Since $K^{\delta}_{\epsilon,R} \subseteq \pi_{\mathrm{reg}}(U)$, there exists $\sigma \in S_m$ such that

$$\sigma^{-1} \cdot \widetilde{s}(K^{\delta}_{\epsilon,R}) \subseteq \widetilde{K}^{\delta,+}_{\epsilon,R} \quad \text{or} \quad \sigma^{-1} \cdot \widetilde{s}(K^{\delta}_{\epsilon,R}) \subseteq \widetilde{K}^{\delta,-}_{\epsilon,R}.$$

If the image lands in the principal sheet, choose $\eta = +1$; if it lands in the opposite sheet, choose $\eta = -1$. By Lemma E.7, this sign normalization places the relative coordinate in the principal square-root sector $\widetilde{S}^{\delta}_{\epsilon,R}$.

It remains to verify the square-root identity. Let $x = (\zeta^{\delta}_{\epsilon,R})^{-1}(w) \in K^{\delta}_{\epsilon,R}$. Write

$$\chi\big(\sigma^{-1} \cdot \widetilde{s}(x)\big) = (0, v) \qquad \text{with } v \in P.$$

Because $\widetilde{s}$ is a section,

$$\pi_{\mathrm{reg}}\big(\sigma^{-1} \cdot \widetilde{s}(x)\big) = x.$$

Applying the branch chart and the definition of $\zeta^{\delta}_{\epsilon,R}$, we obtain

$$w = \zeta^{\delta}_{\epsilon,R}(x) = \vartheta([v]) = \ell(v)^2.$$

By construction,

$$t_{\widetilde{s}}(w) = \eta\, \ell(v),$$

hence

$$t_{\widetilde{s}}(w)^2 = \ell(v)^2 = w.$$

Therefore $t_{\widetilde{s}}$ is a continuous section of the square map. The final claim follows from Proposition E.2. □

Theorem E.9 (Reduced branched-coordinate conditioning lower bound)**.** *Fix $0 < \delta < \pi$ and $0 < R_0 < \rho^2$. Then there exists a constant*

$$C_{\mathrm{red}} > 0,$$

depending only on the fixed local branch chart $(U, V, \chi, \overline{\chi})$, the choice of plane P and linear isomorphism ℓ, and the fixed metric $g_{\widetilde{\Theta}}$, such that for every $0 < \epsilon < R \le R_0$ and every continuous reduced local gauge

$$\widetilde{s} : K^{\delta}_{\epsilon,R} \to \widetilde{\Theta}_{\mathrm{reg}} \qquad \textit{with} \qquad \pi_{\mathrm{reg}} \circ \widetilde{s} = \mathrm{id}_{K^{\delta}_{\epsilon,R}},$$

one has

$$\mathrm{Lip}_{\mathrm{br}}(\widetilde{s}; K^{\delta}_{\epsilon,R}) \ge C_{\mathrm{red}}\, \epsilon^{-1/2}. \tag{27}$$

Proof. Define the closed principal master sector

$$\widetilde{S}^{\delta}_{0,R_0} := \left\{ z = \rho e^{i\phi} \in \mathbb{C} : 0 \le \rho \le \sqrt{R_0},\ |\phi| \le \frac{\pi - \delta}{2} \right\},$$

and its local lift

$$\widetilde{K}^{\delta,+}_{0,R_0} := \chi^{-1}\big(\{0\} \times \ell^{-1}(\widetilde{S}^{\delta}_{0,R_0})\big) \subseteq U.$$

This is compact in $\widetilde{\Theta}$.

For each $\sigma \in S_m$, define

$$\Gamma_\sigma : \sigma \cdot \widetilde{K}^{\delta,+}_{0,R_0} \to \mathbb{C}, \qquad \Gamma_\sigma(x) := \ell\big(\mathrm{pr}_2\big(\chi(\sigma^{-1} \cdot x)\big)\big)\,.$$

Each Γ_σ is smooth on a neighborhood of the compact set $\sigma \cdot \widetilde{K}^{\delta,+}_{0,R_0}$, hence Lipschitz with respect to $d_{\widetilde{\Theta}}$. Because S_m is finite, the constant

$$L_{\max} := \max_{\sigma \in S_m} \mathrm{Lip}\big(\Gamma_\sigma;\, \sigma \cdot \widetilde{K}^{\delta,+}_{0,R_0}\big)$$

is finite.

Now let $\widetilde{s}$ be a reduced local gauge over $K^{\delta}_{\epsilon,R}$. By Proposition E.8, there exist $\sigma \in S_m$ and $\eta \in \{\pm 1\}$ such that

$$t_{\widetilde{s}} = \eta\, \Gamma_\sigma \circ \widetilde{s} \circ (\zeta^{\delta}_{\epsilon,R})^{-1} : S^{\delta}_{\epsilon,R} \to \widetilde{A}_{\epsilon,R}$$

is a continuous section of the square map. Since multiplication by η does not change the Lipschitz constant,

$$\mathrm{Lip}_{\mathbb{C}}(t_{\widetilde{s}}) \le L_{\max}\ \mathrm{Lip}_{\mathrm{br}}(\widetilde{s}; K^{\delta}_{\epsilon,R}).$$

By Proposition E.3,

$$\mathrm{Lip}_{\mathbb{C}}(t_{\widetilde{s}}) \ge \frac{1}{2\sqrt{\epsilon}}.$$

Combining the two inequalities gives

$$\mathrm{Lip}_{\mathrm{br}}(\widetilde{s}; K^{\delta}_{\epsilon,R}) \ge \frac{1}{2L_{\max}}\, \epsilon^{-1/2}.$$

Thus (27) holds with

$$C_{\mathrm{red}} := \frac{1}{2L_{\max}}.$$

□

E.3. Lift to normalized full gauges

The reduced theorem above is the intrinsic statement. To obtain an unreduced version, one must choose a specific smooth normalization of the continuous H^m-gauge. This is exactly what canonicalization methods do in practice.

Assumption E.10 (Fixed smooth local continuous-gauge normalization)**.** Let $U \subseteq \widetilde{\Theta}$ be the neighborhood from Theorem C.8. Fix a smooth section

$$\sigma_H : U \to \Theta^\sharp \qquad \text{such that} \qquad p_H \circ \sigma_H = \mathrm{id}_U\,.$$

Remark E.11. Such a section exists because $p_H : \Theta^\sharp \to \widetilde{\Theta}$ is a principal H^m-bundle and U is contractible; principal bundles over contractible smooth manifolds are smoothly trivial.

Definition E.12 (Normalized full gauge associated with a reduced gauge)**.** Let

$$\widetilde{s} : K^{\delta}_{\epsilon,R} \to \widetilde{K}^{\delta,+}_{\epsilon,R} \subseteq U$$

be a reduced local gauge whose image lies in the principal local sheet. The associated normalized full gauge is

$$s_H := \sigma_H \circ \widetilde{s} : K^{\delta}_{\epsilon,R} \to \Theta^\sharp.$$

Its branched-coordinate Lipschitz seminorm is

$$\mathrm{Lip}^{\Theta^\sharp}_{\mathrm{br}}(s_H; K^{\delta}_{\epsilon,R}) := \sup_{x \ne y \in K^{\delta}_{\epsilon,R}} \frac{d_{\Theta^\sharp}(s_H(x), s_H(y))}{|\zeta^{\delta}_{\epsilon,R}(x) - \zeta^{\delta}_{\epsilon,R}(y)|}. \tag{28}$$

Corollary E.13 (Conditioning lower bound for normalized full gauges)**.** *Under Assumption E.10, there exists a constant*

$$C_{\text{full}} > 0,$$

depending only on the fixed local branch chart, the chosen smooth normalization σ_H, the choice of (P, ℓ), and the fixed metrics $g_{\widetilde{\Theta}}, g_{\Theta^\sharp}$, such that for every $0 < \epsilon < R \le R_0$ and every reduced local gauge

$$\widetilde{s} : K^\delta_{\epsilon,R} \to \widetilde{K}^{\delta,+}_{\epsilon,R}, \qquad \pi_{\text{reg}} \circ \widetilde{s} = \mathrm{id}_{K^\delta_{\epsilon,R}},$$

the associated normalized full gauge $s_H = \sigma_H \circ \widetilde{s}$ satisfies

$$\mathrm{Lip}^{\Theta^\sharp}_{\text{br}}(s_H; K^\delta_{\epsilon,R}) \ge C_{\text{full}}\, \epsilon^{-1/2}. \tag{29}$$

Proof. Set

$$K^{\delta,+}_{0,R_0} := \widetilde{K}^{\delta,+}_{0,R_0} \subseteq U.$$

Its image $\sigma_H(K^{\delta,+}_{0,R_0})$ is compact in $\Theta^\sharp$. Since p_H is smooth, its restriction to this compact set is Lipschitz: there exists $L_H < \infty$ such that

$$d_{\widetilde{\Theta}}(x, y) = d_{\widetilde{\Theta}}(p_H(\sigma_H(x)), p_H(\sigma_H(y))) \le L_H\, d_{\Theta^\sharp}(\sigma_H(x), \sigma_H(y))$$

for all $x, y \in K^{\delta,+}_{0,R_0}$.

Applying this to $x = \widetilde{s}(u)$, $y = \widetilde{s}(v)$, we obtain

$$d_{\widetilde{\Theta}}(\widetilde{s}(u), \widetilde{s}(v)) \le L_H\, d_{\Theta^\sharp}(s_H(u), s_H(v))$$

for all $u, v \in K^\delta_{\epsilon,R}$. Dividing by $|\zeta^\delta_{\epsilon,R}(u) - \zeta^\delta_{\epsilon,R}(v)|$ and taking the supremum yields

$$\mathrm{Lip}_{\text{br}}(\widetilde{s}; K^\delta_{\epsilon,R}) \le L_H\ \mathrm{Lip}^{\Theta^\sharp}_{\text{br}}(s_H; K^\delta_{\epsilon,R}).$$

By Theorem E.9,

$$\mathrm{Lip}_{\text{br}}(\widetilde{s}; K^\delta_{\epsilon,R}) \ge C_{\text{red}}\, \epsilon^{-1/2}.$$

Therefore

$$\mathrm{Lip}^{\Theta^\sharp}_{\text{br}}(s_H; K^\delta_{\epsilon,R}) \ge \frac{C_{\text{red}}}{L_H}\, \epsilon^{-1/2}.$$

This proves (29) with

$$C_{\text{full}} := \frac{C_{\text{red}}}{L_H}.$$

□

Remark E.14 (Why the reduced theorem is the intrinsic one). The reduced lower bound in Theorem E.9 is canonical: it depends only on the local branch chart and on the chosen branched coordinate on the base. By contrast, any unreduced statement necessarily depends on how one chooses a representative inside each noncompact H^m-orbit. Corollary E.13 therefore applies to *normalized* full gauges, i.e. canonicalization procedures that commit to a specific smooth gauge-fixing convention on the continuous fibers.

Remark E.15 (Branched coordinates versus orbifold coordinates). The $\epsilon^{-1/2}$ blow-up is specific to the branched invariant coordinate $w = z^2$. It is not a contradiction that the sign quotient $\mathbb{R}^d/\{\pm 1\}$ itself admits smooth local coordinates away from the branch stratum: the singularity is precisely the failure of those smooth coordinates to extend through the collision in the invariant variable w. This is exactly the coordinate pathology exploited by canonicalization-based pipelines and is the mechanism quantified by the present appendix.

F. Sharp Threshold for Invariant Learning

Recent work on weight-space learning has established strong equivariant architectures and, more recently, broad universality guarantees for permutation-equivariant models (Navon et al., 2023; Dayan et al., 2026). The theorem proved in the present appendix addresses a different question. We study *gauge-canonicalization-based invariant learners*: models that first select local representatives of weight-space equivalence classes and then apply ordinary Euclidean function approximators to those canonicalized representatives. For this class of learners, the topological invariant $\mathfrak{a}(K;q) = \mathrm{secat}(q_K)$ is the exact minimum number of charts required for universality. In particular, the obstruction identified earlier is not merely qualitative: it is the precise complexity threshold for canonicalization-based invariant learning.

Throughout this appendix, $K \subseteq B_{\mathrm{reg}}$ denotes a nonempty compact set,

$$q_K := q|_{q^{-1}(K)} : q^{-1}(K) \to K$$

is the restricted quotient map, and $p \in \mathbb{N}$ is the output dimension.

F.1. Canonicalization-based invariant learners

We begin by fixing an ambient Euclidean encoding of the regular parameter manifold and an abstract local universal approximator family.

Assumption F.1 (Ambient Euclidean embedding)**.** Fix once and for all a smooth embedding

$$\iota : \Theta_{\mathrm{reg}} \hookrightarrow \mathbb{R}^D$$

into some finite-dimensional Euclidean space $\mathbb{R}^D$.

Remark F.2. The role of ι is purely representational: it turns a chosen local gauge $s : U \to \Theta_{\mathrm{reg}}$ into an ordinary Euclidean coordinate representation $\iota \circ s : U \to \mathbb{R}^D$. The learning theorem below depends only on the existence of some fixed embedding, not on its particular choice.

Assumption F.3 (Local Euclidean universal approximator family)**.** For each output dimension $p \in \mathbb{N}$, fix a class

$$\mathcal{N}_{D,p} \subseteq C(\mathbb{R}^D, \mathbb{R}^p)$$

such that for every compact set $C \subseteq \mathbb{R}^D$, the restricted class

$$\mathcal{N}_{D,p}|_C := \{h|_C : h \in \mathcal{N}_{D,p}\} \subseteq C(C, \mathbb{R}^p)$$

is dense in $C(C, \mathbb{R}^p)$ with respect to the uniform norm.

Remark F.4. Assumption F.3 abstracts the local Euclidean approximation step and separates it cleanly from the topology of the quotient. Standard multilayer perceptrons with classical nonpolynomial activations satisfy this assumption by the usual Euclidean universal approximation theorems; see, for example, Cybenko (1989); Hornik (1991).

Definition F.5 (N-chart gauge-canonicalization scheme)**.** Let $N \in \mathbb{N}$. An *N-chart gauge-canonicalization scheme* on K is a tuple

$$\mathcal{A} = \big((U_i, s_i, \lambda_i)\big)_{i=1}^{N}$$

such that:

1. $U_1, \dots, U_N \subseteq B_{\mathrm{reg}}$ are open and cover K;
2. each $s_i : U_i \to \Theta_{\mathrm{reg}}$ is a continuous local gauge of q, i.e.

$$q \circ s_i = \mathrm{id}_{U_i};$$

3. $\lambda_1, \dots, \lambda_N : K \to [0,1]$ form a continuous partition of unity subordinate to the cover $\{U_i\}_{i=1}^N$, i.e.

$$\sum_{i=1}^{N} \lambda_i(x) = 1 \quad \text{for all } x \in K, \qquad \mathrm{supp}(\lambda_i) \subseteq U_i.$$

Definition F.6 (Hypothesis class associated with a gauge atlas)**.** Let $\mathcal{A} = ((U_i, s_i, \lambda_i))_{i=1}^N$ be an N-chart gauge-canonicalization scheme on K. The associated hypothesis class

$$\mathcal{H}_{\mathcal{A},p} \subseteq C_G(q^{-1}(K), \mathbb{R}^p)$$

consists of all functions $F : q^{-1}(K) \to \mathbb{R}^p$ of the form

$$F(\theta) = \sum_{i=1}^{N} \bar{H}_i(q(\theta)), \qquad \theta \in q^{-1}(K), \tag{30}$$

where for each i there exists $h_i \in \mathcal{N}_{D,p}$ such that

$$\bar{H}_i(x) := \begin{cases} \lambda_i(x)\, h_i(\iota(s_i(x))), & x \in U_i, \\ 0, & x \notin U_i. \end{cases}$$

Lemma F.7 (Well-posedness of the atlas hypothesis class)**.** *For every N-chart gauge-canonicalization scheme $\mathcal{A}$ on K, every function $F \in \mathcal{H}_{\mathcal{A},p}$ is continuous and G-invariant. Hence*

$$\mathcal{H}_{\mathcal{A},p} \subseteq C_G(q^{-1}(K), \mathbb{R}^p).$$

Proof. Fix $i \in \{1, \dots, N\}$. Since $\operatorname{supp}(\lambda_i) \subseteq U_i$, the closed sets $\operatorname{supp}(\lambda_i)$ and $K \setminus U_i$ are disjoint. Therefore every point $x \in K \setminus U_i$ has a neighborhood in K on which $\lambda_i \equiv 0$. It follows that the piecewise-defined map

$$\bar{H}_i(x) = \begin{cases} \lambda_i(x)\, h_i(\iota(s_i(x))), & x \in U_i, \\ 0, & x \notin U_i \end{cases}$$

is continuous on all of K: continuity on U_i is immediate, and near every point of $K \setminus U_i$ the function is identically zero.

Consequently, the finite sum

$$\bar{F}(x) := \sum_{i=1}^{N} \bar{H}_i(x)$$

is continuous on K. By definition,

$$F = \bar{F} \circ q_K.$$

Since q_K is continuous, F is continuous on $q^{-1}(K)$. Moreover, $q_K(g \cdot \theta) = q_K(\theta)$ for every $g \in G$, so

$$F(g \cdot \theta) = \bar{F}(q_K(g \cdot \theta)) = \bar{F}(q_K(\theta)) = F(\theta).$$

Hence F is G-invariant. □

F.2. Exact chart threshold

We now prove the main result of the appendix.

Theorem F.8 (Sharp threshold for canonicalization-based invariant learning)**.** *Let $K \subseteq B_{\mathrm{reg}}$ be compact and let $p \in \mathbb{N}$. Define*

$$\mathfrak{U}_{\mathrm{can}}(K; q, p) := \inf\Big\{ N \in \mathbb{N} : \exists \text{ an } N\text{-chart gauge-canonicalization scheme } \mathcal{A} \text{ on } K \text{ such that } \overline{\mathcal{H}_{\mathcal{A},p}}^{\|\cdot\|_\infty} = C_G(q^{-1}(K), \mathbb{R}^p) \Big\},$$

with the convention that the infimum of the empty set is $+\infty$. Then

$$\mathfrak{U}_{\mathrm{can}}(K; q, p) = \mathfrak{a}(K; q). \tag{31}$$

Equivalently, a universal N-chart gauge-canonicalization scheme on K exists if and only if

$$N \geq \mathfrak{a}(K; q).$$

Proof. We prove the lower and upper bounds separately.

Lower bound. Assume there exists an N-chart gauge-canonicalization scheme

$$\mathcal{A} = ((U_i, s_i, \lambda_i))_{i=1}^{N}$$

on K. By Definition F.5, the sets $U_1, \dots, U_N$ form an open cover of K, and each U_i admits a continuous local gauge s_i. By the definition of sectional category,

$$\mathfrak{a}(K; q) = \operatorname{secat}(q_K) \leq N.$$

Since this is true for every N-chart scheme, we obtain

$$\mathfrak{U}_{\mathrm{can}}(K; q, p) \geq \mathfrak{a}(K; q).$$

Upper bound. Set

$$a := \mathfrak{a}(K; q).$$

By Definition 2.3, there exists an open cover

$$K \subseteq U_1 \cup \cdots \cup U_a,$$

together with continuous local gauges

$$s_i : U_i \to \Theta_{\mathrm{reg}}, \qquad q \circ s_i = \mathrm{id}_{U_i}, \qquad i = 1, \dots, a.$$

Because K is a compact Hausdorff space and the cover is finite and open, there exists a continuous partition of unity

$$\lambda_1, \dots, \lambda_a : K \to [0, 1]$$

subordinate to $\{U_i\}_{i=1}^{a}$.

Let

$$\mathcal{A}^{\star} := ((U_i, s_i, \lambda_i))_{i=1}^{a}.$$

We claim that

$$\overline{\mathcal{H}_{\mathcal{A}^{\star}, p}}^{\|\cdot\|_{\infty}} = C_G(q^{-1}(K), \mathbb{R}^p).$$

Fix any target

$$F \in C_G(q^{-1}(K), \mathbb{R}^p)$$

and any accuracy $\varepsilon > 0$. By Definition 2.4, F descends uniquely to a continuous map

$$\bar{F} : K \to \mathbb{R}^p \qquad \text{such that} \qquad F = \bar{F} \circ q_K.$$

For each $i \in \{1, \dots, a\}$, define the compact set

$$C_i := \iota\big(s_i(\operatorname{supp}(\lambda_i))\big) \subseteq \mathbb{R}^D.$$

Because $\operatorname{supp}(\lambda_i)$ is compact and s_i, ι are continuous, C_i is compact. Moreover,

$$\iota \circ s_i : \operatorname{supp}(\lambda_i) \to C_i$$

is a homeomorphism. Indeed, s_i is injective because $q \circ s_i = \mathrm{id}_{U_i}$, and ι is injective by assumption; thus $\iota \circ s_i$ is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism onto its image.

Define the local target

$$g_i : C_i \to \mathbb{R}^p, \qquad g_i\big(\iota(s_i(x))\big) := \bar{F}(x), \qquad x \in \operatorname{supp}(\lambda_i).$$

This is well defined and continuous because $(\iota \circ s_i)|_{\operatorname{supp}(\lambda_i)}$ is a homeomorphism.

By Assumption F.3, for each i there exists $h_i \in \mathcal{N}_{D,p}$ such that

$$\sup_{u \in C_i} \|h_i(u) - g_i(u)\| < \varepsilon.$$

Using these h_i, define

$$F_\varepsilon \in \mathcal{H}_{\mathcal{A}^\star,p}$$

via Definition F.6. We now estimate the uniform error. Fix any $\theta \in q^{-1}(K)$ and write

$$x := q_K(\theta) \in K.$$

Then

$$F_\varepsilon(\theta) = \sum_{i=1}^{a} \lambda_i(x)\, h_i(\iota(s_i(x))),$$

where terms with $x \notin U_i$ vanish because then $\lambda_i(x) = 0$. Hence

$$\begin{aligned}\|F_\varepsilon(\theta) - F(\theta)\| &= \left\| \sum_{i=1}^{a} \lambda_i(x) \left(h_i(\iota(s_i(x))) - \bar{F}(x) \right) \right\| \\ &\leq \sum_{i=1}^{a} \lambda_i(x)\, \|h_i(\iota(s_i(x))) - \bar{F}(x)\|.\end{aligned}$$

If $\lambda_i(x) \neq 0$, then $x \in \operatorname{supp}(\lambda_i)$, so $\iota(s_i(x)) \in C_i$ and therefore

$$\|h_i(\iota(s_i(x))) - \bar{F}(x)\| = \|h_i(\iota(s_i(x))) - g_i(\iota(s_i(x)))\| < \varepsilon.$$

Thus

$$\|F_\varepsilon(\theta) - F(\theta)\| \leq \sum_{i=1}^{a} \lambda_i(x)\, \varepsilon = \varepsilon.$$

Taking the supremum over $\theta \in q^{-1}(K)$ yields

$$\|F_\varepsilon - F\|_\infty \leq \varepsilon.$$

Since $F \in C_G(q^{-1}(K), \mathbb{R}^p)$ and $\varepsilon > 0$ were arbitrary,

$$\overline{\mathcal{H}_{\mathcal{A}^\star,p}}^{\|\cdot\|_\infty} = C_G(q^{-1}(K), \mathbb{R}^p).$$

Therefore

$$\mathfrak{U}_{\mathrm{can}}(K; q, p) \leq a = \mathfrak{a}(K; q).$$

Combining the lower and upper bounds proves (31). The equivalent "if and only if" statement follows immediately. Indeed, if $N > \mathfrak{a}(K; q)$, one can append redundant zero-weight charts to any universal $\mathfrak{a}(K; q)$-chart scheme. □

Corollary F.9 (Exact threshold on the transverse neural annulus)**.** *Assume the hypotheses of Appendix D, and let*

$$K_{\epsilon,R} \subseteq B_{\mathrm{reg}}$$

be the compact transverse neural annulus defined in (16). Then, for every output dimension $p \in \mathbb{N}$,

$$\mathfrak{U}_{\mathrm{can}}(K_{\epsilon,R}; q, p) = 2.$$

Equivalently, a universal one-chart gauge-canonicalization scheme on $K_{\epsilon,R}$ does not exist, whereas a universal two-chart scheme does.

Proof. By Corollary D.7,

$$\mathfrak{a}(K_{\epsilon,R}; q) = 2.$$

The conclusion therefore follows immediately from Theorem F.8. □

Remark F.10 (What the theorem does and does not say). Theorem F.8 is a theorem about *canonicalization-based* invariant learning. It does not claim that every possible invariant architecture must explicitly use $\mathfrak{a}(K; q)$ charts. Rather, it proves that any architecture whose mechanism is "choose local representatives, process them with ordinary Euclidean models, and glue the results" has exact minimum chart complexity equal to $\operatorname{secat}(q_K)$. This is precisely the design paradigm implemented by local gauge fixing and re-basin-style canonicalization procedures.

Remark F.11 (Interpretation for weight-space learning). Theorem F.8 should be read as a topological complement to existing expressivity results in weight-space learning. Prior universality theorems show that carefully designed equivariant architectures can represent large classes of weight-space maps (Navon et al., 2023; Dayan et al., 2026). Our result isolates a different phenomenon: when the learner is forced to operate through local canonical representatives, the topology of the quotient imposes an exact chart threshold, and that threshold is the sectional category of the quotient map.

G. Section-Valued Universality

The previous appendix established an exact chart-complexity threshold for *canonicalization-based invariant learning*, namely for models whose outputs are ordinary Euclidean quantities associated with quotient points. Many weight-space tasks, however, are naturally *covariant* rather than invariant: examples include update directions, editing directions, transport fields, or other outputs that transform under a representation of the symmetry group. The mathematically correct output object is then not a scalar function on the quotient, but a continuous section of an associated vector bundle. This appendix proves that the same topological invariant $\mathfrak{a}(K;q)$ is again the exact chart threshold for canonicalization-based learning, now in the bundle-valued setting.

Throughout, $K \subseteq B_{\mathrm{reg}}$ is a nonempty compact set,

$$q_K := q|_{q^{-1}(K)} : q^{-1}(K) \to K$$

is the restricted quotient map, $\rho : G \to \mathrm{GL}(V)$ is a fixed continuous finite-dimensional real representation, and

$$E_\rho := (\Theta_{\mathrm{reg}} \times V)/G \to B_{\mathrm{reg}}$$

is the associated vector bundle from Definition 2.4.

G.1. Local trivializations induced by local gauges

We first record the standard but crucial fact that a local gauge canonically induces a local trivialization of the associated bundle.

Definition G.1 (Fiber notation and section norm)**.** For $x \in K$, write $E_{\rho,x}$ for the fiber of E_ρ over x. Fix once and for all a continuous bundle norm

$$\|\cdot\|_{E_\rho}$$

on $E_\rho|_K$. For a continuous section

$$\sigma \in \Gamma(E_\rho|_K),$$

define its uniform norm by

$$\|\sigma\|_{\Gamma,\infty} := \sup_{x\in K} \|\sigma(x)\|_{E_{\rho,x}}. \tag{32}$$

Remark G.2. Because K is compact and $E_\rho|_K$ is a finite-dimensional continuous vector bundle, continuous bundle norms exist, and any two such norms induce equivalent supremum norms on $\Gamma(E_\rho|_K)$. Hence the topology used below does not depend on the particular norm chosen in Definition G.1.

Definition G.3 (Linear model space for local coefficients)**.** Set

$$d_\rho := \dim V.$$

Fix once and for all a linear isomorphism

$$J_\rho : V \xrightarrow{\cong} \mathbb{R}^{d_\rho},$$

and define the norm on V by

$$\|v\|_V := \|J_\rho(v)\|_2.$$

Lemma G.4 (Bundle trivialization induced by a local gauge)**.** *Let $U \subseteq B_{\mathrm{reg}}$ be open and let*

$$s : U \to \Theta_{\mathrm{reg}}$$

be a continuous local gauge, i.e. $q \circ s = \mathrm{id}_U$. *Then the map*

$$\Phi_s : U \times V \to E_\rho|_U, \qquad \Phi_s(x, v) := [s(x), v], \tag{33}$$

is a vector-bundle homeomorphism over U. *Its inverse is given as follows: for every* $[\,\theta, v\,] \in E_\rho|_U$*, there exists a unique* $g_s(\theta) \in G$ *such that*

$$\theta = g_s(\theta) \cdot s(q(\theta)),$$

and

$$\Phi_s^{-1}([\,\theta, v\,]) = \big(q(\theta),\ \rho(g_s(\theta))^{-1} v\big). \tag{34}$$

Proof. Since $q : \Theta_{\mathrm{reg}} \to B_{\mathrm{reg}}$ is a principal G-bundle by Definition 2.3, the map

$$\alpha_s : U \times G \to q^{-1}(U), \qquad \alpha_s(x, g) := g \cdot s(x)$$

is a homeomorphism. Indeed, it is continuous, surjective by definition of a principal bundle, and injective because the G-action on Θ_{reg} is free. The inverse map is of the form

$$\alpha_s^{-1}(\theta) = \big(q(\theta), g_s(\theta)\big),$$

where $g_s(\theta)$ is uniquely defined by $\theta = g_s(\theta) \cdot s(q(\theta))$. Continuity of g_s follows from continuity of α_s^{-1}.

Define Φ_s by (33). This map is well defined because $[s(x), v] \in E_{\rho,x}$. To define the inverse, let $[\,\theta, v\,] \in E_\rho|_U$. Set

$$x := q(\theta) \in U, \qquad g := g_s(\theta) \in G.$$

Then $\theta = g \cdot s(x)$. By the defining equivalence relation of the associated bundle,

$$[\theta, v] = [g \cdot s(x), v] = [s(x), \rho(g)^{-1} v].$$

Hence (34) is well defined and is clearly the inverse of Φ_s. Fiberwise linearity is immediate. Continuity of Φ_s^{-1} follows from continuity of q, g_s, and ρ. $\square$

Corollary G.5 (Local coefficient representation of sections)**.** *Let* $U \subseteq B_{\mathrm{reg}}$ *be open and* $s : U \to \Theta_{\mathrm{reg}}$ *a continuous local gauge. Then every section*

$$\sigma \in \Gamma(E_\rho|_U)$$

admits a unique continuous coefficient map

$$c_s^\sigma : U \to V$$

such that

$$\sigma(x) = [s(x), c_s^\sigma(x)] \qquad \textit{for all } x \in U. \tag{35}$$

Proof. By Lemma G.4, $\Phi_s : U \times V \to E_\rho|_U$ is a bundle trivialization. Therefore

$$\Phi_s^{-1} \circ \sigma : U \to U \times V$$

is continuous and must be of the form

$$x \mapsto (x, c_s^\sigma(x))$$

for a unique continuous map $c_s^\sigma : U \to V$. Reapplying Φ_s gives (35). $\square$

Lemma G.6 (Transition law on chart overlaps)**.** *Let* $U_i, U_j \subseteq B_{\mathrm{reg}}$ *be open and let*

$$s_i : U_i \to \Theta_{\mathrm{reg}}, \qquad s_j : U_j \to \Theta_{\mathrm{reg}}$$

be continuous local gauges. On the overlap $U_i \cap U_j$*, there exists a unique continuous transition function*

$$t_{ij} : U_i \cap U_j \to G$$

such that

$$s_j(x) = t_{ij}(x) \cdot s_i(x) \qquad \textit{for all } x \in U_i \cap U_j. \tag{36}$$

If $\sigma \in \Gamma(E_\rho|_{U_i \cup U_j})$ *has local coefficient maps* c_i *and* c_j *in the gauges* s_i *and* s_j*, respectively, then*

$$c_j(x) = \rho(t_{ij}(x))\, c_i(x) \qquad \textit{for all } x \in U_i \cap U_j. \tag{37}$$

Proof. For each $x \in U_i \cap U_j$, the points $s_i(x)$ and $s_j(x)$ lie in the same fiber of the principal bundle q, hence there exists a unique $t_{ij}(x) \in G$ satisfying (36). Continuity follows from the trivialization map of the principal bundle exactly as in the proof of Lemma G.4.

Now let σ be a section and write

$$\sigma(x) = [s_i(x), c_i(x)] = [s_j(x), c_j(x)].$$

Using $s_j(x) = t_{ij}(x) \cdot s_i(x)$ and the associated-bundle equivalence relation,

$$[s_j(x), c_j(x)] = [t_{ij}(x) \cdot s_i(x), c_j(x)] = [s_i(x), \rho(t_{ij}(x))^{-1} c_j(x)].$$

Since the representation of a point in the trivialization induced by s_i is unique, we obtain

$$c_i(x) = \rho(t_{ij}(x))^{-1} c_j(x),$$

equivalently (37). □

G.2. Canonicalization-based section learners

We now define the class of local-canonicalization learners for bundle-valued outputs.

Recall from Appendix F that an N-chart gauge-canonicalization scheme on K is a tuple

$$\mathcal{A} = ((U_i, s_i, \lambda_i))_{i=1}^N$$

consisting of an open cover of K, continuous local gauges s_i, and a continuous partition of unity $(\lambda_i)_{i=1}^N$ subordinate to that cover.

Definition G.7 (Section hypothesis class associated with a gauge atlas)**.** Let $\mathcal{A} = ((U_i, s_i, \lambda_i))_{i=1}^N$ be an N-chart gauge-canonicalization scheme on K. For each i, write

$$\Phi_i := \Phi_{s_i} : U_i \times V \to E_\rho|_{U_i}$$

for the local trivialization from Lemma G.4. The associated section hypothesis class

$$\mathcal{S}_{\mathcal{A},\rho} \subseteq \Gamma(E_\rho|_K)$$

consists of all sections Σ of the form

$$\Sigma(x) = \sum_{i=1}^N \Sigma_i(x), \qquad x \in K, \tag{38}$$

where for each i there exists

$$h_i \in \mathcal{N}_{D,d_\rho}$$

such that

$$\Sigma_i(x) := \begin{cases} \Phi_i\big(x, \lambda_i(x)\, J_\rho^{-1}\big(h_i(\iota(s_i(x)))\big)\big), & x \in U_i, \\ 0_x, & x \notin U_i, \end{cases} \tag{39}$$

with $0_x \in E_{\rho,x}$ denoting the zero vector in the fiber over x.

Lemma G.8 (Well-posedness of the section hypothesis class)**.** *For every N-chart gauge-canonicalization scheme $\mathcal{A}$ on K, every element of $\mathcal{S}_{\mathcal{A},\rho}$ is a continuous section of $E_\rho|_K$.*

Proof. Fix $i \in \{1, \dots, N\}$. On U_i, the map

$$x \mapsto \Phi_i\big(x, \lambda_i(x)\, J_\rho^{-1}(h_i(\iota(s_i(x))))\big)$$

is continuous because Φ_i, λ_i, h_i, ι, and s_i are continuous. Since

$$\mathrm{supp}(\lambda_i) \subseteq U_i,$$

every point $x \in K \setminus U_i$ has a neighborhood in K on which $\lambda_i \equiv 0$. On that neighborhood, the expression above is the zero section, so the piecewise definition (39) extends continuously across the boundary of U_i. Hence each Σ_i is a continuous section of $E_\rho|_K$. The finite fiberwise sum $\Sigma = \sum_{i=1}^N \Sigma_i$ is therefore again a continuous section. □

G.3. Exact chart threshold for section-valued universality

We now prove the section-valued analogue of Theorem F.8.

Definition G.9 (Section-valued universal chart complexity)**.** Define

$$\mathfrak{U}_{\mathrm{sec,can}}(K;q,\rho) := \inf\Big\{N \in \mathbb{N} : \exists \text{ an } N\text{-chart gauge-canonicalization scheme } \mathcal{A} \text{ on } K \text{ such that } \overline{\mathcal{S}_{\mathcal{A},\rho}}^{\|\cdot\|_{\Gamma,\infty}} = \Gamma(E_\rho|_K)\Big\},$$

with the convention that the infimum of the empty set is $+\infty$.

Theorem G.10 (Exact threshold for canonicalization-based section learning)**.** *For every compact $K \subseteq B_{\mathrm{reg}}$ and every finite-dimensional continuous representation $\rho : G \to \mathrm{GL}(V)$,*

$$\mathfrak{U}_{\mathrm{sec,can}}(K;q,\rho) = \mathfrak{a}(K;q). \tag{40}$$

Equivalently, a universal N-chart gauge-canonicalization scheme for section-valued outputs exists if and only if

$$N \geq \mathfrak{a}(K;q).$$

Proof. We again prove the lower and upper bounds separately.

Lower bound. Assume there exists an N-chart gauge-canonicalization scheme

$$\mathcal{A} = ((U_i, s_i, \lambda_i))_{i=1}^N$$

on K such that

$$\overline{\mathcal{S}_{\mathcal{A},\rho}}^{\|\cdot\|_{\Gamma,\infty}} = \Gamma(E_\rho|_K).$$

Then, by the very definition of an N-chart scheme, the open sets $U_1, \dots, U_N$ cover K and each U_i admits a continuous local gauge s_i. Therefore

$$\mathfrak{a}(K;q) = \mathrm{secat}(q_K) \leq N.$$

Since this holds for every universal N-chart scheme,

$$\mathfrak{U}_{\mathrm{sec,can}}(K;q,\rho) \geq \mathfrak{a}(K;q).$$

Upper bound. Let

$$a := \mathfrak{a}(K;q).$$

Choose an a-chart gauge-canonicalization scheme

$$\mathcal{A}^\star = ((U_i, s_i, \lambda_i))_{i=1}^a$$

on K; this exists by Definition 2.3. We claim that

$$\overline{\mathcal{S}_{\mathcal{A}^\star,\rho}}^{\|\cdot\|_{\Gamma,\infty}} = \Gamma(E_\rho|_K).$$

Fix an arbitrary target section

$$\sigma \in \Gamma(E_\rho|_K)$$

and an accuracy $\varepsilon > 0$. By Corollary G.5, each chart U_i admits a unique continuous local coefficient function

$$c_i : U_i \to V \qquad \text{such that} \qquad \sigma(x) = \Phi_i(x, c_i(x)) \quad \text{for } x \in U_i.$$

For each i, define the compact set

$$C_i := \iota\big(s_i(\operatorname{supp}\lambda_i)\big) \subseteq \mathbb{R}^D.$$

As in the proof of Theorem F.8,

$$\iota \circ s_i : \operatorname{supp}(\lambda_i) \to C_i$$

is a homeomorphism. Define the Euclidean local target

$$g_i : C_i \to \mathbb{R}^{d_\rho}, \qquad g_i(\iota(s_i(x))) := J_\rho(c_i(x)), \qquad x \in \operatorname{supp}(\lambda_i).$$

This is well defined and continuous.

Next, because $\Phi_i(x, \cdot) : V \to E_{\rho,x}$ is linear and continuous, the operator norm

$$x \longmapsto \|\Phi_i(x, \cdot)\|_{\mathrm{op}}$$

is continuous on $\operatorname{supp}(\lambda_i)$. Since that set is compact,

$$M_i := \sup_{x \in \operatorname{supp}(\lambda_i)} \|\Phi_i(x, \cdot)\|_{\mathrm{op}} < \infty.$$

Set

$$M := \max_{1 \le i \le a} M_i.$$

By Assumption F.3, for each i there exists $h_i \in \mathcal{N}_{D,d_\rho}$ satisfying

$$\sup_{u \in C_i} \|h_i(u) - g_i(u)\|_2 < \frac{\varepsilon}{M}.$$

Using these h_i, form the approximant

$$\Sigma_\varepsilon \in \mathcal{S}_{\mathcal{A}^\star,\rho}$$

according to Definition G.7.

We now estimate the error. Fix $x \in K$. Since $\sum_{i=1}^a \lambda_i(x) = 1$ and scalar multiplication is fiberwise linear,

$$\sigma(x) = \sum_{i=1}^a \lambda_i(x)\sigma(x).$$

For every index i with $\lambda_i(x) \neq 0$, we have $x \in \operatorname{supp}(\lambda_i) \subseteq U_i$, and therefore

$$\lambda_i(x)\sigma(x) = \Phi_i(x, \lambda_i(x)c_i(x)).$$

By definition of Σ_ε,

$$\Sigma_\varepsilon(x) - \sigma(x) = \sum_{i=1}^a \Phi_i\big(x, \lambda_i(x)\big(J_\rho^{-1}(h_i(\iota(s_i(x)))) - c_i(x)\big)\big).$$

Hence, using the fiber norm and the operator-norm bounds,

$$\begin{aligned}
\|\Sigma_\varepsilon(x) - \sigma(x)\|_{E_{\rho,x}} &\le \sum_{i=1}^a \lambda_i(x) \left\|\Phi_i\big(x, J_\rho^{-1}(h_i(\iota(s_i(x)))) - c_i(x)\big)\right\|_{E_{\rho,x}} \\
&\le \sum_{i=1}^a \lambda_i(x)\, M_i \left\|J_\rho^{-1}(h_i(\iota(s_i(x)))) - c_i(x)\right\|_V.
\end{aligned}$$

Because $\|v\|_V = \|J_\rho(v)\|_2$,

$$\left\|J_\rho^{-1}(h_i(\iota(s_i(x)))) - c_i(x)\right\|_V = \|h_i(\iota(s_i(x))) - g_i(\iota(s_i(x)))\|_2 < \frac{\varepsilon}{M}.$$

Therefore

$$\|\Sigma_\varepsilon(x) - \sigma(x)\|_{E_{\rho,x}} < \sum_{i=1}^a \lambda_i(x)\, M_i\, \frac{\varepsilon}{M} \le \varepsilon.$$

Taking the supremum over $x \in K$ yields

$$\|\Sigma_\varepsilon - \sigma\|_{\Gamma,\infty} \le \varepsilon.$$

Since σ and ε were arbitrary,

$$\overline{\mathcal{S}_{\mathcal{A}^\star,\rho}}^{\|\cdot\|_{\Gamma,\infty}} = \Gamma(E_\rho|_K).$$

Thus

$$\mathfrak{U}_{\mathrm{sec,can}}(K;q,\rho) \le a = \mathfrak{a}(K;q).$$

Combining the lower and upper bounds proves (40). The equivalent "if and only if" formulation follows immediately. ☐

Corollary G.11 (Exact threshold on the transverse neural annulus)**.** *Assume the hypotheses of Appendix D, and let*

$$K_{\epsilon,R} \subseteq B_{\mathrm{reg}}$$

be the compact transverse neural annulus defined in (16)*. Then, for every finite-dimensional continuous representation $\rho : G \to \mathrm{GL}(V)$,*

$$\mathfrak{U}_{\mathrm{sec,can}}(K_{\epsilon,R};q,\rho) = 2.$$

Equivalently, no one-chart gauge-canonicalization scheme is universal for section-valued targets on $K_{\epsilon,R}$, whereas a two-chart scheme is.

Proof. By Corollary D.7,

$$\mathfrak{a}(K_{\epsilon,R};q) = 2.$$

The claim follows immediately from Theorem G.10. ☐

Remark G.12 (Why bundle-valued outputs matter). Invariant scalar outputs are not the only natural objects in weight-space learning. Any target that should transform covariantly under weight-space symmetries is, mathematically, a section of an associated bundle rather than a function on the quotient. Theorem G.10 shows that the same topological obstruction governing scalar canonicalization also governs this broader covariant regime.

Remark G.13 (Position relative to existing WSL expressivity theory). Existing WSL expressivity results establish universality for carefully designed equivariant architectures operating directly on weight tensors (Navon et al., 2023; Dayan et al., 2026). Theorem G.10 addresses a different axis of the problem: it gives the exact minimum number of local gauges required by the *canonicalization-based* route to bundle-valued universality. In this sense, it is a topological lower-and-upper-bound theorem about local-coordinate architectures, not a universality theorem for arbitrary equivariant networks.

H. Optimizer Corollary

Symmetry does not merely affect representation; it directly shapes optimization dynamics. Recent work shows that quotient geometry isolates the intrinsic horizontal component of gradient flow on regular neural quotients, while SGD can systematically drift along continuous symmetry directions because of gradient noise (Dong & Cheng, 2026; Ziyin et al., 2024). In parallel, learned optimizers have begun to exploit parameter symmetries explicitly (Zamir et al., 2025). The purpose of this appendix is to derive the optimizer-facing consequence of Appendix G: for the canonicalization-based route to symmetry-respecting optimization, the exact minimum number of local gauges is again $\mathfrak{a}(K;q)$.

The correct object here is a *memoryless optimizer proposal field*: a rule that assigns to each equivalence class of parameters a symmetry-compatible update vector. Mathematically, such proposal fields are sections of a vector bundle over the quotient.

H.1. Ambient linear optimizer model

To speak about ambient update vectors, we impose an additional assumption that matches the usual tensor-parameterized neural-network setting.

Assumption H.1 (Ambient linear optimizer representation)**.** There exists a finite-dimensional real vector space W_{opt} and a continuous linear representation

$$\rho_{\mathrm{opt}} : G \to \mathrm{GL}(W_{\mathrm{opt}})$$

such that:

1. Θ_{reg} is a G-invariant open subset of W_{opt};

2. the given action of G on Θ_{reg} is the restriction of the linear action ρ_{opt}, i.e.

$$g \cdot \theta = \rho_{\mathrm{opt}}(g)\theta \qquad \text{for all } g \in G,\ \theta \in \Theta_{\mathrm{reg}}.$$

Remark H.2. Assumption H.1 is satisfied in the standard parameter-tensor setting for many architectures of interest: weights, biases, low-rank factors, or attention matrices live in finite-dimensional Euclidean spaces, and the regular locus is obtained by removing a discriminant from an open G-invariant subset. In this regime, $T_\theta \Theta_{\mathrm{reg}} \cong W_{\mathrm{opt}}$ canonically for every $\theta \in \Theta_{\mathrm{reg}}$, so ambient update vectors and tangent update vectors coincide.

Definition H.3 (Optimizer proposal bundle)**.** Under Assumption H.1, define the associated vector bundle

$$E_{\mathrm{opt}} := (\Theta_{\mathrm{reg}} \times W_{\mathrm{opt}})/G \longrightarrow B_{\mathrm{reg}}, \tag{41}$$

where the diagonal action is

$$g \cdot (\theta, v) := (g \cdot \theta, \rho_{\mathrm{opt}}(g)v).$$

Definition H.4 (Symmetry-respecting memoryless optimizer proposal)**.** A *symmetry-respecting memoryless optimizer proposal on $K \subseteq B_{\mathrm{reg}}$* is a continuous section

$$\Sigma \in \Gamma(E_{\mathrm{opt}}|_K).$$

This definition is intrinsic to the quotient. It says that the proposal is not a single globally gauge-fixed ambient vector field, but a bundle-valued object whose local representatives must transform covariantly under changes of parameter representative.

H.2. Equivalent ambient formulation

We now identify the bundle-based definition with the more concrete ambient equivariance condition.

Definition H.5 (Ambient equivariant proposal field)**.** Let $K \subseteq B_{\mathrm{reg}}$ be compact. An *ambient equivariant proposal field* on $q^{-1}(K)$ is a continuous map

$$\Delta : q^{-1}(K) \to W_{\mathrm{opt}}$$

such that

$$\Delta(g \cdot \theta) = \rho_{\mathrm{opt}}(g)\, \Delta(\theta) \qquad \text{for all } g \in G,\ \theta \in q^{-1}(K). \tag{42}$$

Proposition H.6 (Sections and ambient equivariant proposal fields are equivalent)**.** *For every compact $K \subseteq B_{\mathrm{reg}}$, there is a natural bijection between*

1. *continuous sections $\Sigma \in \Gamma(E_{\mathrm{opt}}|_K)$, and*
2. *ambient equivariant proposal fields $\Delta : q^{-1}(K) \to W_{\mathrm{opt}}$ satisfying* (42).

Proof. We construct inverse maps in both directions.

From ambient equivariant fields to sections. Let

$$\Delta : q^{-1}(K) \to W_{\mathrm{opt}}$$

be continuous and satisfy (42). Define

$$\Sigma_\Delta : K \to E_{\mathrm{opt}}|_K$$

by

$$\Sigma_\Delta(x) := [\theta, \Delta(\theta)], \qquad \theta \in q^{-1}(x). \tag{43}$$

We first check that this is well defined. If $\theta' \in q^{-1}(x)$, then $\theta' = g \cdot \theta$ for a unique $g \in G$ because q is a principal G-bundle. Using (42),

$$[\theta', \Delta(\theta')] = [g \cdot \theta, \rho_{\mathrm{opt}}(g)\Delta(\theta)] = [\theta, \Delta(\theta)]$$

in the associated bundle. Hence Σ_Δ is well defined.

To prove continuity, choose any finite open cover

$$K \subseteq U_1 \cup \cdots \cup U_N$$

by sets admitting local gauges

$$s_i : U_i \to \Theta_{\mathrm{reg}}.$$

On U_i,

$$\Sigma_\Delta(x) = [s_i(x), \Delta(s_i(x))],$$

and $x \mapsto \Delta(s_i(x))$ is continuous. By Lemma G.4, this is precisely the local coefficient representation of a continuous section. Therefore Σ_Δ is a continuous section of $E_{\mathrm{opt}}|_K$.

From sections to ambient equivariant fields. Conversely, let

$$\Sigma \in \Gamma(E_{\mathrm{opt}}|_K).$$

Choose a finite open cover

$$K \subseteq U_1 \cup \cdots \cup U_N$$

and continuous local gauges

$$s_i : U_i \to \Theta_{\mathrm{reg}}.$$

By Corollary G.5, on each U_i there exists a unique continuous coefficient map

$$c_i : U_i \to W_{\mathrm{opt}}$$

such that

$$\Sigma(x) = [s_i(x), c_i(x)].$$

For each i, let

$$g_i : q^{-1}(U_i) \to G$$

be the unique continuous map characterized by

$$\theta = g_i(\theta) \cdot s_i(q(\theta)) \qquad \text{for all } \theta \in q^{-1}(U_i).$$

Such a map exists because q is a principal G-bundle and s_i is a local gauge; it is exactly the group-coordinate function from the principal bundle trivialization.

Define

$$\Delta_i : q^{-1}(U_i) \to W_{\mathrm{opt}}, \qquad \Delta_i(\theta) := \rho_{\mathrm{opt}}(g_i(\theta))\, c_i(q(\theta)).$$

Each Δ_i is continuous.

We claim that these local definitions agree on overlaps. Let $\theta \in q^{-1}(U_i \cap U_j)$ and write $x = q(\theta)$. By Lemma G.6, there exists a unique transition function

$$t_{ij} : U_i \cap U_j \to G$$

such that

$$s_j(x) = t_{ij}(x) \cdot s_i(x), \qquad c_j(x) = \rho_{\mathrm{opt}}(t_{ij}(x))\, c_i(x).$$

Since

$$\theta = g_i(\theta) \cdot s_i(x) = g_j(\theta) \cdot s_j(x) = g_j(\theta)\, t_{ij}(x) \cdot s_i(x),$$

freeness of the G-action implies

$$g_i(\theta) = g_j(\theta)\, t_{ij}(x).$$

Hence

$$\begin{aligned}
\Delta_i(\theta) &= \rho_{\mathrm{opt}}(g_i(\theta))\, c_i(x) \\
&= \rho_{\mathrm{opt}}(g_j(\theta) t_{ij}(x))\, c_i(x) \\
&= \rho_{\mathrm{opt}}(g_j(\theta))\, \rho_{\mathrm{opt}}(t_{ij}(x))\, c_i(x) \\
&= \rho_{\mathrm{opt}}(g_j(\theta))\, c_j(x) = \Delta_j(\theta).
\end{aligned}$$

Therefore the Δ_i glue to a global continuous map

$$\Delta_\Sigma : q^{-1}(K) \to W_{\mathrm{opt}}.$$

It remains to verify equivariance. Fix $g \in G$ and $\theta \in q^{-1}(K)$. Choose i with $q(\theta) \in U_i$. Then $q(g \cdot \theta) = q(\theta) \in U_i$ as well, and

$$g \cdot \theta = g\, g_i(\theta) \cdot s_i(q(\theta)).$$

By uniqueness of $g_i(g \cdot \theta)$,

$$g_i(g \cdot \theta) = g\, g_i(\theta).$$

Thus

$$\Delta_\Sigma(g \cdot \theta) = \rho_{\mathrm{opt}}(g_i(g \cdot \theta))\, c_i(q(\theta)) = \rho_{\mathrm{opt}}(g)\, \rho_{\mathrm{opt}}(g_i(\theta))\, c_i(q(\theta)) = \rho_{\mathrm{opt}}(g)\, \Delta_\Sigma(\theta).$$

So Δ_Σ is an ambient equivariant proposal field.

Inverse property. Starting from Δ, constructing Σ_Δ, and then applying the second construction recovers Δ because in any gauge s_i the local coefficient of Σ_Δ is exactly $x \mapsto \Delta(s_i(x))$. Conversely, starting from Σ, constructing Δ_Σ, and then applying (43) recovers Σ by construction. Thus the two maps are inverse bijections. □

Corollary H.7 (Local coefficient law for optimizer proposals)**.** *Let $\Sigma \in \Gamma(E_{\mathrm{opt}}|_K)$, let*

$$s_i : U_i \to \Theta_{\mathrm{reg}}, \qquad s_j : U_j \to \Theta_{\mathrm{reg}}$$

be local gauges, and let $u_i : U_i \to W_{\mathrm{opt}}$, $u_j : U_j \to W_{\mathrm{opt}}$ be the corresponding local coefficient maps, so that

$$\Sigma(x) = [s_i(x), u_i(x)] = [s_j(x), u_j(x)].$$

Then on the overlap $U_i \cap U_j$,

$$u_j(x) = \rho_{\mathrm{opt}}(t_{ij}(x))\, u_i(x),$$

where t_{ij} is the gauge transition function from Lemma G.6.

Proof. This is exactly the transition law (37) from Lemma G.6, specialized to the representation ρ_{opt}. □

Proposition H.8 (Gauge compatibility of local one-step proposals)**.** *Assume $\Sigma \in \Gamma(E_{\mathrm{opt}}|_K)$, let u_i, u_j be local coefficient maps as above, and fix a step size $\eta > 0$. Define the local one-step proposal maps*

$$T_{i,\eta}(x) := s_i(x) - \eta\, u_i(x), \qquad T_{j,\eta}(x) := s_j(x) - \eta\, u_j(x),$$

whenever these values lie in Θ_{reg}. Then on $U_i \cap U_j$,

$$T_{j,\eta}(x) = \rho_{\mathrm{opt}}(t_{ij}(x))\, T_{i,\eta}(x). \tag{44}$$

In particular, the local proposal transforms equivariantly under changes of gauge.

Proof. By Assumption H.1,

$$s_j(x) = \rho_{\mathrm{opt}}(t_{ij}(x))\, s_i(x).$$

By Corollary H.7,

$$u_j(x) = \rho_{\mathrm{opt}}(t_{ij}(x))\, u_i(x).$$

Since $\rho_{\mathrm{opt}}(t_{ij}(x))$ is linear,

$$\begin{aligned} T_{j,\eta}(x) &= s_j(x) - \eta\, u_j(x) \\ &= \rho_{\mathrm{opt}}(t_{ij}(x))\, s_i(x) - \eta\, \rho_{\mathrm{opt}}(t_{ij}(x))\, u_i(x) \\ &= \rho_{\mathrm{opt}}(t_{ij}(x))\big(s_i(x) - \eta\, u_i(x)\big) \\ &= \rho_{\mathrm{opt}}(t_{ij}(x))\, T_{i,\eta}(x). \end{aligned}$$

This proves (44). □

Remark H.9 (Why proposals rather than full optimization trajectories). Proposition H.8 concerns the *proposal field* of a memoryless optimizer. To obtain a globally defined iterative algorithm, one must also choose a step-size rule or retraction that keeps the update inside Θ_{reg}. The topological obstruction proved below is therefore an obstruction to globally gauge-fixed *proposal fields*, which is the right static object for local-coordinate optimizer design.

H.3. Exact chart threshold for canonicalization-based optimizer proposals

We now specialize the general section-valued universality theorem to optimizer proposal fields.

Definition H.10 (Canonicalization-based optimizer chart complexity)**.** Define

$$\mathfrak{U}_{\mathrm{opt,can}}(K; q, \rho_{\mathrm{opt}}) := \mathfrak{U}_{\mathrm{sec,can}}(K; q, \rho_{\mathrm{opt}}),$$

where the right-hand side is the section-valued universal chart complexity from Definition G.9, specialized to the representation ρ_{opt}.

Corollary H.11 (Exact chart threshold for canonicalization-based optimizer proposals)**.** *Under Assumption H.1,*

$$\mathfrak{U}_{\mathrm{opt,can}}(K; q, \rho_{\mathrm{opt}}) = \mathfrak{a}(K; q). \tag{45}$$

Equivalently, a universal N-chart gauge-canonicalization architecture for symmetry-respecting memoryless optimizer proposals on K exists if and only if

$$N \geq \mathfrak{a}(K; q).$$

In particular, if

$$\mathfrak{a}(K; q) > 1,$$

then no single global gauge-fixed canonicalization scheme can be universal for optimizer proposal fields on K.

Proof. By Definition H.10, the quantity $\mathfrak{U}_{\mathrm{opt,can}}(K; q, \rho_{\mathrm{opt}})$ is exactly the section-valued universal chart complexity for the associated bundle E_{opt}. Therefore (45) is an immediate specialization of Theorem G.10. The final statement follows directly. □

Corollary H.12 (Exact optimizer threshold on the transverse neural annulus)**.** *Assume the hypotheses of Appendix D, and let*

$$K_{\epsilon,R} \subseteq B_{\mathrm{reg}}$$

be the compact transverse neural annulus defined in (16). Under Assumption H.1,

$$\mathfrak{U}_{\mathrm{opt,can}}(K_{\epsilon,R}; q, \rho_{\mathrm{opt}}) = 2.$$

Equivalently, no one-chart canonicalization-based optimizer proposal class is universal on $K_{\epsilon,R}$, whereas a two-chart class is.

Proof. By Corollary D.7,

$$\mathfrak{a}(K_{\epsilon,R}; q) = 2.$$

Apply Corollary H.11. □

Remark H.13 (State-augmented or learned optimizers). The memoryless assumption is made only for clarity. If an optimizer carries a finite-dimensional internal state that transforms under a linear representation of G, one can enlarge W_{opt} to include the state variables and apply the same argument to the augmented proposal bundle. Thus the topological chart obstruction is not specific to bare gradient-like updates.

Remark H.14 (Intrinsic formulation via the quotient tangent bundle). Without Assumption H.1, one may formulate an intrinsic version of the present corollary using the quotient vector bundle $T\Theta_{\mathrm{reg}}/G \to B_{\mathrm{reg}}$, sometimes called the Atiyah bundle of the principal G-bundle $q : \Theta_{\mathrm{reg}} \to B_{\mathrm{reg}}$. Continuous G-equivariant vector fields on Θ_{reg} correspond to sections of that quotient bundle. The ambient-linear formulation adopted here is more concrete and is the one most directly relevant to practical parameter-space optimizers.

I. Local Attention-Head Corollary

Recent work has characterized the generic gauge group of standard multi-head attention as

$$\big((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h\big) \rtimes S_h,$$

with independent query–key and value–output gauge factors per head and a residual head-permutation symmetry (Wang & Wang, 2025). The goal of the present appendix is not to re-prove that global maximality theorem. Instead, we verify that standard multi-head attention fits the abstract framework of the paper on a concrete full-rank stratum, and we derive the local pairwise-head-collision consequence needed by our main theorem package.

I.1. Standard multi-head attention as an exchangeable-block family

Fix integers

$$h \geq 2, \qquad 1 \leq d_k, d_v \leq d_{\mathrm{model}}.$$

For a sequence length n, an input is a matrix

$$X \in \mathbb{R}^{n \times d_{\mathrm{model}}}.$$

A single attention head is parameterized by a quadruple

$$\xi = (Q, K, V, O),$$

with shapes

$$Q, K \in \mathbb{R}^{d_{\mathrm{model}} \times d_k}, \qquad V \in \mathbb{R}^{d_{\mathrm{model}} \times d_v}, \qquad O \in \mathbb{R}^{d_v \times d_{\mathrm{model}}}.$$

Definition I.1 (Single-head attention map)**.** For $\xi = (Q, K, V, O)$, define

$$\mathsf{Head}_\xi(X) := \mathrm{softmax}\left(\frac{XQK^\top X^\top}{\sqrt{d_k}}\right) XVO \in \mathbb{R}^{n \times d_{\mathrm{model}}}, \tag{46}$$

where softmax acts row-wise on the score matrix.

Definition I.2 (Multi-head attention block)**.** For

$$\theta = (\xi_1, \ldots, \xi_h), \qquad \xi_i = (Q_i, K_i, V_i, O_i),$$

define

$$\mathsf{MHA}_\theta(X) := \sum_{i=1}^{h} \mathsf{Head}_{\xi_i}(X). \tag{47}$$

Define the full-rank single-head stratum by

$$\Xi_{\mathrm{att}}^{\mathrm{fr}} := \{(Q, K, V, O) : \mathrm{rank}(Q) = \mathrm{rank}(K) = d_k,\ \mathrm{rank}(V) = \mathrm{rank}(O) = d_v\}. \tag{48}$$

The continuous per-head gauge group is

$$H_{\mathrm{att}} := \mathrm{GL}(d_k) \times \mathrm{GL}(d_v),$$

acting on $\Xi_{\mathrm{att}}^{\mathrm{fr}}$ by

$$(A, C) \cdot (Q, K, V, O) := \big(QA,\ KA^{-\top},\ VC,\ C^{-1}O\big). \tag{49}$$

For h heads, the full symmetry group is

$$G_{\mathrm{att},h} := H_{\mathrm{att}}^h \rtimes S_h,$$

acting on

$$\Theta^\sharp_{\mathrm{att},h} := \big(\Xi_{\mathrm{att}}^{\mathrm{fr}}\big)^h$$

by the general exchangeable-block rule from (1).

Lemma I.3 (Gauge invariance of standard multi-head attention)**.** *The realization map*

$$\mathcal{R}_{\mathrm{att},h} : \Theta^\sharp_{\mathrm{att},h} \to \{X \mapsto \mathsf{MHA}_\theta(X)\}$$

is invariant under the action of $G_{\mathrm{att},h}$.

Proof. It suffices to verify invariance under the continuous head-wise action and under head permutation.

Fix one head $\xi = (Q, K, V, O)$, one pair $(A, C) \in H_{\mathrm{att}}$, and one input X. Then

$$\begin{aligned}
\mathsf{Head}_{(A,C)\cdot\xi}(X) &= \mathrm{softmax}\left(\frac{X(QA)(KA^{-\top})^\top X^\top}{\sqrt{d_k}}\right) X(VC)(C^{-1}O) \\
&= \mathrm{softmax}\left(\frac{XQAA^{-1}K^\top X^\top}{\sqrt{d_k}}\right) XVO \\
&= \mathrm{softmax}\left(\frac{XQK^\top X^\top}{\sqrt{d_k}}\right) XVO \\
&= \mathsf{Head}_\xi(X).
\end{aligned}$$

Thus each head is invariant under its own continuous gauge action, and hence the whole multi-head block is invariant under H_{att}^{h}.

Invariance under S_h is immediate from the summation in (47): permuting the heads only reorders the terms of the sum. Therefore $\mathcal{R}_{\mathrm{att},h}$ is $G_{\mathrm{att},h}$- invariant. □

I.2. Head-signature completeness on the full-rank stratum

Define the fixed-rank matrix manifolds

$$\mathcal{R}_r(m,m) := \{M \in \mathbb{R}^{m\times m} : \mathrm{rank}(M) = r\}.$$

Set

$$M_{\mathrm{att}} := \mathcal{R}_{d_k}(d_{\mathrm{model}}, d_{\mathrm{model}}) \times \mathcal{R}_{d_v}(d_{\mathrm{model}}, d_{\mathrm{model}}). \tag{50}$$

Its dimension is

$$D_{\mathrm{att}} := \dim M_{\mathrm{att}} = d_k(2d_{\mathrm{model}} - d_k) + d_v(2d_{\mathrm{model}} - d_v). \tag{51}$$

Definition I.4 (Head signature)**.** Define

$$\psi_{\mathrm{att}} : \Xi_{\mathrm{att}}^{\mathrm{fr}} \to M_{\mathrm{att}}, \qquad \psi_{\mathrm{att}}(Q,K,V,O) := \big(QK^\top,\ VO\big). \tag{52}$$

Lemma I.5 (A head depends only on its signature)**.** *For every $\xi = (Q,K,V,O) \in \Xi_{\mathrm{att}}^{\mathrm{fr}}$,*

$$\mathsf{Head}_\xi(X) = \mathrm{softmax}\!\left(\frac{XMX^\top}{\sqrt{d_k}}\right) XN, \qquad (M,N) := \psi_{\mathrm{att}}(\xi).$$

In particular, two head parameters with the same signature define the same single-head function.

Proof. This is immediate from the definitions:

$$M = QK^\top, \qquad N = VO,$$

hence

$$\mathsf{Head}_\xi(X) = \mathrm{softmax}\!\left(\frac{XQK^\top X^\top}{\sqrt{d_k}}\right) XVO = \mathrm{softmax}\!\left(\frac{XMX^\top}{\sqrt{d_k}}\right) XN.$$

□

Proposition I.6 (Single-head signature completeness)**.** *The action of H_{att} on $\Xi_{\mathrm{att}}^{\mathrm{fr}}$ is free and proper. Moreover, the signature map ψ_{att} is constant on H_{att}-orbits, its fibers are exactly those orbits, and the induced quotient map*

$$\bar{\psi}_{\mathrm{att}} : \Xi_{\mathrm{att}}^{\mathrm{fr}}/H_{\mathrm{att}} \longrightarrow M_{\mathrm{att}}$$

is a diffeomorphism.

Proof. **Step 1: orbit invariance.** Let $(A,C) \in H_{\mathrm{att}}$. Then

$$\begin{aligned}
\psi_{\mathrm{att}}\big((A,C)\cdot(Q,K,V,O)\big) &= \psi_{\mathrm{att}}(QA,\ KA^{-\top},\ VC,\ C^{-1}O) \\
&= \big(QA(KA^{-\top})^\top,\ (VC)(C^{-1}O)\big) \\
&= \big(QAA^{-1}K^\top,\ VO\big) \\
&= (QK^\top,\ VO) \\
&= \psi_{\mathrm{att}}(Q,K,V,O).
\end{aligned}$$

So ψ_{att} is constant on H_{att}-orbits.

Step 2: fibers are exactly the H_{att}-orbits. Suppose

$$\psi_{\mathrm{att}}(Q,K,V,O) = \psi_{\mathrm{att}}(Q',K',V',O').$$

Then

$$QK^\top = Q'K'^\top, \qquad VO = V'O'.$$

We first analyze the query–key sector. Because Q and K both have full column rank d_k,

$$\operatorname{rank}(QK^\top) = d_k.$$

Hence

$$\operatorname{col}(QK^\top) = \operatorname{col}(Q), \qquad \operatorname{col}(Q'K'^\top) = \operatorname{col}(Q').$$

Since the products are equal, $\operatorname{col}(Q) = \operatorname{col}(Q')$, so there exists a unique $A \in \mathrm{GL}(d_k)$ such that

$$Q' = QA.$$

Using $QK^\top = Q'K'^\top = QAK'^\top$ and a left inverse L of Q ($LQ = I_{d_k}$), we obtain

$$K^\top = AK'^\top,$$

hence

$$K' = KA^{-\top}.$$

Now analyze the value–output sector. Because V has full column rank and O has full row rank,

$$\operatorname{rank}(VO) = d_v.$$

Therefore

$$\operatorname{col}(VO) = \operatorname{col}(V), \qquad \operatorname{col}(V'O') = \operatorname{col}(V').$$

Since $VO = V'O'$, we get $\operatorname{col}(V) = \operatorname{col}(V')$, hence there exists a unique $C \in \mathrm{GL}(d_v)$ such that

$$V' = VC.$$

Using $VO = V'O' = VCO'$ and a left inverse L_V of V, we obtain

$$O = CO',$$

equivalently

$$O' = C^{-1}O.$$

Thus

$$(Q', K', V', O') = (A, C) \cdot (Q, K, V, O).$$

So the fibers of ψ_{att} are exactly the H_{att}- orbits.

Step 3: freeness. If

$$(A, C) \cdot (Q, K, V, O) = (Q, K, V, O),$$

then

$$QA = Q, \qquad VC = V.$$

Since Q and V have full column rank, we must have

$$A = I_{d_k}, \qquad C = I_{d_v}.$$

Hence the action is free.

Step 4: properness. Let

$$(A_n, C_n) \in H_{\mathrm{att}}, \qquad \xi_n = (Q_n, K_n, V_n, O_n) \in \Xi^{\mathrm{fr}}_{\mathrm{att}},$$

and assume

$$\xi_n \to \xi = (Q, K, V, O), \qquad (A_n, C_n) \cdot \xi_n \to \xi' = (Q', K', V', O').$$

We show that (A_n, C_n) has a convergent subsequence in H_{att}.

Because $Q_n \to Q$ and Q has full column rank, for all sufficiently large n, the Moore–Penrose left inverse

$$Q_n^\dagger := (Q_n^\top Q_n)^{-1} Q_n^\top$$

is well defined and uniformly bounded. Since

$$A_n = Q_n^\dagger (Q_n A_n),$$

and $Q_n A_n \to Q'$, the sequence (A_n) is bounded.

Likewise, because $K_n \to K$ with K full column rank, the left inverses

$$K_n^\dagger := (K_n^\top K_n)^{-1} K_n^\top$$

are eventually well defined and uniformly bounded. Since

$$A_n^{-\top} = K_n^\dagger (K_n A_n^{-\top}),$$

and $K_n A_n^{-\top} \to K'$, the sequence (A_n^{-1}) is bounded.

Similarly, using

$$V_n^\dagger := (V_n^\top V_n)^{-1} V_n^\top,$$

we get

$$C_n = V_n^\dagger (V_n C_n),$$

so (C_n) is bounded. Finally, because $O_n \to O$ and O has full row rank, the right inverses

$$O_n^\sharp := O_n^\top (O_n O_n^\top)^{-1}$$

are eventually well defined and uniformly bounded. Since

$$C_n^{-1} = (C_n^{-1} O_n)\, O_n^\sharp,$$

and $C_n^{-1} O_n \to O'$, the sequence (C_n^{-1}) is bounded.

Thus (A_n, C_n) is bounded in the finite-dimensional vector space

$$\mathbb{R}^{d_k \times d_k} \times \mathbb{R}^{d_v \times d_v}.$$

It therefore has a convergent subsequence

$$(A_{n_j}, C_{n_j}) \to (A_\infty, C_\infty).$$

Because $(A_{n_j}^{-1})$ and $(C_{n_j}^{-1})$ are bounded, the limits A_∞ and C_∞ are invertible. Hence

$$(A_\infty, C_\infty) \in H_{\mathrm{att}}.$$

So the action is proper.

Step 5: quotient identification. Because the action is free and proper, the orbit space

$$\Xi_{\mathrm{att}}^{\mathrm{fr}} / H_{\mathrm{att}}$$

is a smooth manifold, and the quotient map is a smooth principal H_{att}-bundle. By Steps 1 and 2, ψ_{att} descends to a smooth bijection

$$\bar{\psi}_{\mathrm{att}} : \Xi_{\mathrm{att}}^{\mathrm{fr}} / H_{\mathrm{att}} \to M_{\mathrm{att}}.$$

It remains to prove that the inverse is smooth. Fix

$$(M_0, N_0) \in M_{\mathrm{att}}.$$

Choose index sets $I_k, J_k \subseteq \{1, \ldots, d_{\mathrm{model}}\}$ of size d_k such that the minor $(M_0)_{I_k,J_k}$ is invertible, and choose $I_v, J_v \subseteq \{1, \ldots, d_{\mathrm{model}}\}$ of size d_v such that $(N_0)_{I_v,J_v}$ is invertible. On the open neighborhood of M_{att} where these minors remain invertible, define

$$Q_\star(M) := M_{:,J_k}, \qquad K_\star(M)^\top := M_{I_k,J_k}^{-1} M_{I_k,:},$$

$$V_\star(N) := N_{:,J_v}, \qquad O_\star(N) := N_{I_v,J_v}^{-1} N_{I_v,:}.$$

These maps are smooth. Because M has rank d_k and $M_{I_k,J_k} \in \mathrm{GL}(d_k)$, the columns indexed by J_k form a basis of the column space of M, and the coefficients expressing each column of M in that basis are determined by the pivot rows indexed by I_k. Therefore

$$M = Q_\star(M) K_\star(M)^\top.$$

Similarly,

$$N = V_\star(N) O_\star(N).$$

Hence

$$(M, N) \mapsto \big[(Q_\star(M), K_\star(M), V_\star(N), O_\star(N))\big]$$

is a smooth local inverse of $\bar{\psi}_{\mathrm{att}}$. Therefore $\bar{\psi}_{\mathrm{att}}$ is a diffeomorphism. □

Remark I.7 (Consistency with recent transformer-symmetry theory). Proposition I.6 verifies the signature-complete quotient structure needed in the present paper for the standard multi-head attention block on a full-rank stratum. This is fully consistent with the recent generic-stratum symmetry characterization of Wang & Wang (2025), which identifies the same continuous and discrete gauge factors for standard multi-head attention.

I.3. Pairwise head collisions and the local branch corollary

Define the ordered multi-head signature map

$$\Psi_{\mathrm{att},h} : \Theta^\sharp_{\mathrm{att},h} \to M^h_{\mathrm{att}}, \qquad \Psi_{\mathrm{att},h}(\xi_1, \ldots, \xi_h) := \big(\psi_{\mathrm{att}}(\xi_1), \ldots, \psi_{\mathrm{att}}(\xi_h)\big). \tag{53}$$

Define the regular multi-head locus by

$$\Theta^{\mathrm{reg}}_{\mathrm{att},h} := \Big\{(\xi_1, \ldots, \xi_h) \in \Theta^\sharp_{\mathrm{att},h} : \psi_{\mathrm{att}}(\xi_i) \neq \psi_{\mathrm{att}}(\xi_j) \text{ for all } i \neq j\Big\}. \tag{54}$$

Let

$$B_{\mathrm{reg}_{\mathrm{att},h}} := \Theta^{\mathrm{reg}}_{\mathrm{att},h} / G_{\mathrm{att},h}.$$

Definition I.8 (Simple pairwise head collision)**.** A point

$$\theta^\star = (\xi_1^\star, \ldots, \xi_h^\star) \in \Theta^\sharp_{\mathrm{att},h}$$

is called a *simple pairwise head-collision point* if it is a simple pairwise collision point in the sense of Definition C.3 for the block-signature map ψ_{att}. Equivalently, after relabeling if necessary:

1. $$\psi_{\mathrm{att}}(\xi_1^\star) = \psi_{\mathrm{att}}(\xi_2^\star);$$
2. the remaining head signatures are pairwise distinct and distinct from the colliding signature;
3. the resulting collision is transverse in the signature coordinates on M_{att}.

Corollary I.9 (Local branch normal form for standard multi-head attention)**.** *Let $\theta^\star \in \Theta^\sharp_{\mathrm{att},h}$ be a simple pairwise head-collision point. Then all conclusions of Theorem C.8 and of Appendices D–E apply to the multi-head-attention quotient*

$$q_{\mathrm{att},h} : \Theta^{\mathrm{reg}}_{\mathrm{att},h} \to B_{\mathrm{reg}_{\mathrm{att},h}},$$

with block signature manifold M_{att} and relative-collision dimension

$$D_{\mathrm{att}} = d_k(2d_{\mathrm{model}} - d_k) + d_v(2d_{\mathrm{model}} - d_v).$$

More explicitly, there exist local coordinates near the collision point in which the quotient takes the form

$$(u,v) \longmapsto (u,[v]), \qquad u \in \mathbb{R}^{(h-1)D_{\mathrm{att}}}, \quad v \in \mathbb{R}^{D_{\mathrm{att}}}, \quad v \sim -v.$$

Since $D_{\mathrm{att}} \geq 2$, every transverse two-dimensional slice is topologically equivalent to the branched double cover

$$z \longmapsto z^2.$$

Consequently:

1. *any compact transverse annulus linking the head-collision stratum admits no continuous global head-ordering gauge;*
2. *the exact local atlas complexity on such an annulus is* 2*;*
3. *every canonicalization in branched invariant coordinates obeys the same $\epsilon^{-1/2}$ conditioning lower bound proved in Appendix E.*

Proof. By Proposition I.6, the single-head full-rank stratum $\Xi^{\mathrm{fr}}_{\mathrm{att}}$ together with the H_{att}-action and the signature map ψ_{att} satisfies the signature-complete hypothesis of Assumption 2.2. The h-head parameter space

$$\Theta^{\sharp}_{\mathrm{att},h} = \left(\Xi^{\mathrm{fr}}_{\mathrm{att}}\right)^h$$

is therefore an exchangeable-block family with continuous block gauge group H_{att} and discrete permutation group S_h, exactly as in Assumption 2.1.

A simple pairwise head-collision point is, by Definition I.8, precisely a simple pairwise collision point for this exchangeable-block family. Hence Theorem C.8 applies directly, with

$$M = M_{\mathrm{att}}, \qquad d = \dim M_{\mathrm{att}} = D_{\mathrm{att}}.$$

Since, after quotienting the continuous gauge, the reduced multi-head manifold is M^h_{att}, its dimension is hD_{att}, and the branch theorem gives the local quotient model

$$(u,v) \mapsto (u,[v]), \qquad u \in \mathbb{R}^{(h-1)D_{\mathrm{att}}}, \quad v \in \mathbb{R}^{D_{\mathrm{att}}}.$$

It remains only to note that

$$D_{\mathrm{att}} = d_k(2d_{\mathrm{model}} - d_k) + d_v(2d_{\mathrm{model}} - d_v) \geq 2,$$

because $1 \leq d_k, d_v \leq d_{\mathrm{model}}$ implies

$$d_k(2d_{\mathrm{model}} - d_k) \geq 1, \qquad d_v(2d_{\mathrm{model}} - d_v) \geq 1.$$

Therefore the two-dimensional slice corollary (Corollary C.9) applies, and the conclusions on monodromy, no global gauge, exact local atlas complexity, and branched-coordinate conditioning follow from Corollary D.6, Corollary D.7, and Corollary E.13. □

Remark I.10 (Interpretation). Corollary I.9 says that even in standard multi-head attention, head permutation is not merely a benign bookkeeping symmetry. Near a transverse pairwise head collision, the quotient develops the same branched local geometry as in the abstract theory, so any attempt to order heads continuously through a linked neighborhood must fail. Thus head canonicalization is necessarily chart-dependent in exactly the sense quantified by the earlier appendices.

Remark I.11 (RoPE and other position-encoded variants). For RoPE-based models, recent work shows that the query–key gauge group is reduced from $\mathrm{GL}(d_k)$ to the commutant of the rotary action, while the value–output $\mathrm{GL}(d_v)$ factor and head permutations remain (Wang & Wang, 2025). The same local program should therefore go through after replacing the $QK^\top$-sector by the appropriate commutant-invariant signature. We do not pursue that extension here because the present paper only needs the standard-MHA local corollary.

J. Experimental Details and Additional Figures

The experiments in this paper are theorem-validation studies rather than benchmark-style comparisons. They are designed to isolate the quotient-space phenomena proved in Sections 3 and 4: monodromy, branch-induced conditioning blow-up, exact chart thresholds for invariant learning, and the corresponding section-valued / optimizer consequences. This design philosophy is deliberately aligned with recent quotient-geometric and weight-space learning viewpoints (Simsek et al., 2021; Navon et al., 2023; Dong & Cheng, 2026; Dayan et al., 2026).

Across all experiments, we use deterministic double-precision numerical linear algebra. Wherever randomness enters, it is confined to target-family sampling from a fixed master seed and not to stochastic training dynamics. Experiments 1 and 2 are fully analytic and deterministic. Experiments 3 and 4 use fixed feature maps and deterministic ridge regression, so all variability reported in the tables comes from the target-family draws rather than from optimizer noise. Raw CSV diagnostics are produced by every script, and the compact main-text figures/tables are strict compressions of the more detailed appendix artifacts shown below.

J.1. Experiment 1: Exact Monodromy in the Homogeneous Two-Unit Model

This experiment validates the topological content of Theorems 3.2 and 3.3. We use the explicit homogeneous two-unit model from Section B with $d = 2$, fix $b_0 = 1$ and $\rho = 0.09$, and define the ordered lift

$$z(t) = \sqrt{\rho}\, e^{\pi i t}, \qquad w(t) = z(t)^2 = \rho e^{2\pi i t}, \qquad t \in [0, 1].$$

Thus $w(t)$ is a closed loop in quotient coordinates encircling the collision once, while $z(t)$ is its lifted path on the double cover. The experiment is sampled on a uniform grid of 1201 time points. For the homogeneous network realization, the local signatures are converted back to parameters using the canonical section

$$a_i = b_i, \qquad w_i = (\cos\phi_i, \sin\phi_i),$$

and the realized network is evaluated on a dense 121×121 probe grid in $[-1.5, 1.5]^2$. The one-chart canonicalizer is the lexicographic ordering of the local signature coordinates $(b - b_0, \phi)$, which is continuous away from the branch cut and is therefore forced to jump when the loop links the collision stratum.

The diagnostics recorded are:

1. the endpoint residual to the original ordering,
2. the endpoint residual to the swapped ordering,
3. the maximum realized-function discrepancy on the probe grid,
4. the maximum ordered-lift step norm,
5. the maximum canonicalizer step norm and the resulting jump amplification.

The theorem predicts that the lifted endpoint coincides with the swapped representative, that the function is unchanged, and that any one-chart canonicalizer must incur a single macroscopic jump.

J.2. Experiment 2: Conditioning Blow-Up and the $\epsilon^{-1/2}$ Law

This experiment validates Theorem 3.4 and the exact constants derived in Section E. We work on slit sectors

$$S^{\delta}_{\epsilon,R} = \left\{ re^{i\theta} : \epsilon \le r \le R,\ |\theta| \le \pi - \delta \right\}$$

with

$$R = 0.3, \qquad \delta = \frac{\pi}{6}, \qquad \epsilon \in \{10^{-1}, 5 \cdot 10^{-2}, 2 \cdot 10^{-2}, 10^{-2}, 5 \cdot 10^{-3}, 2 \cdot 10^{-3}, 10^{-3}\}.$$

For each ϵ, we discretize the sector using a 300×420 polar grid. We then evaluate two local gauges:

1. the exact square-root branch $z(w) = \sqrt{w}$,

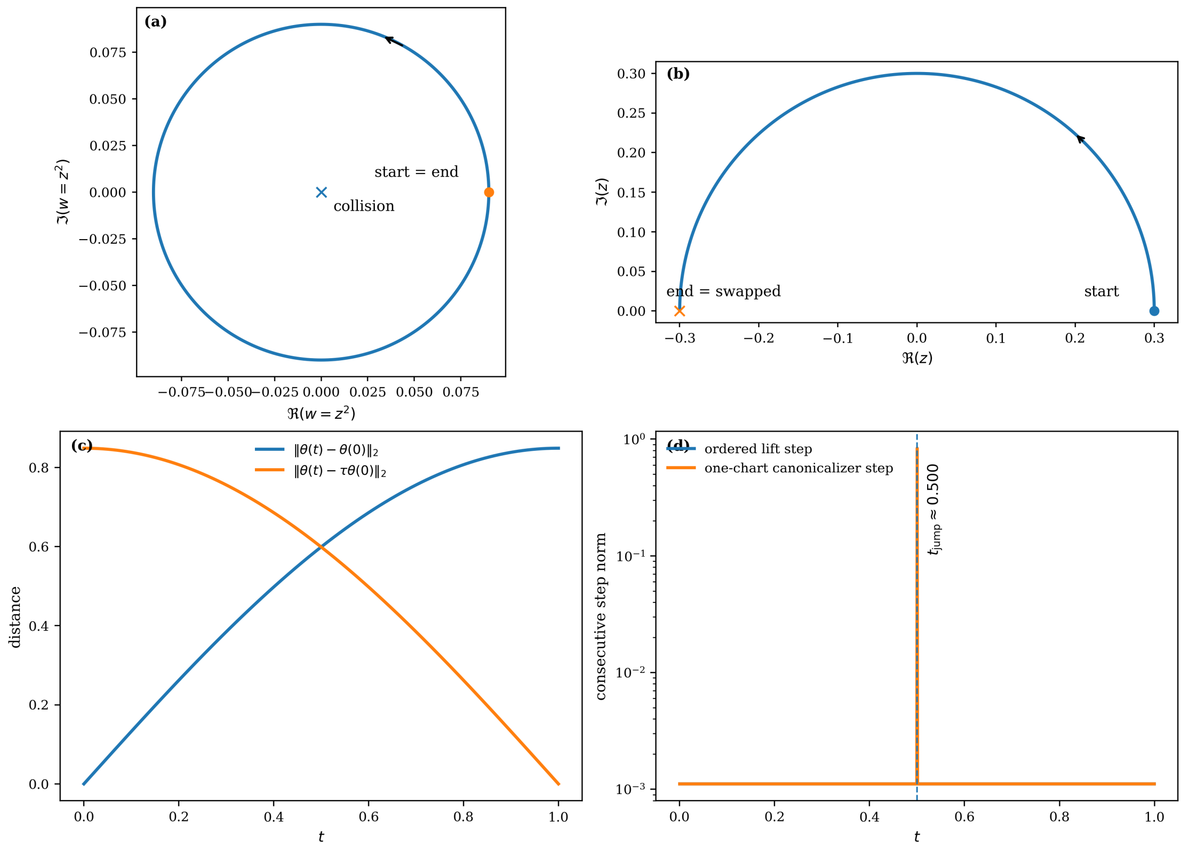

Figure 3. Expanded exact-monodromy diagnostics for the homogeneous two-unit model. **(a)** A closed loop in quotient coordinates $w = z^2$ winds once around the pairwise-collision stratum. **(b)** Its ordered lift on the double cover starts at $z(0) = +\sqrt{\rho}$ and ends at $z(1) = -\sqrt{\rho}$, i.e. at the swapped representative. **(c)** Along the loop, the ordered parameter path separates from the original ordering and converges to the swapped ordering. **(d)** The ordered lift remains smooth, whereas a one-chart canonicalizer incurs a single macroscopic jump.

2. the normalized homogeneous gauge induced by the same branch.

The branched-coordinate Lipschitz seminorm is estimated by a dense nearest-neighbor finite-difference scheme over radial, angular, and diagonal edges of the grid. The exact theory predicts

$$L_{\text{root}}(\epsilon) = \frac{1}{2\sqrt{\epsilon}}, \qquad L_{\text{hom}}(\epsilon) = \frac{1}{\sqrt{2\epsilon}}.$$

We fit a log-log power law

$$\log \widehat{L}(\epsilon) = \alpha + \beta \log \epsilon$$

and report both the fitted exponent $\hat{\beta}$ and the fitted leading constant $\hat{C}$. In addition, on the representative sector $\epsilon = 5 \times 10^{-3}$, we compute a dense Cartesian heatmap of the pointwise operator norm $\|Ds(w)\|_{\text{op}}$, which localizes the singularity near the branch point and the removed slit.

J.3. Experiment 3: Exact Chart Threshold for Invariant Learning

This experiment validates Theorem 4.1 on the linked annulus model from Section D. The quotient domain is the annulus

$$K_{\epsilon,R}, \qquad \epsilon = 10^{-2}, \qquad R = 9 \times 10^{-2},$$

represented on a dense evaluation grid with 84 radial levels and 240 angular levels. Training data consist of 3000 samples from the annulus, with deliberate oversampling near the principal branch cut to stress the one-chart learner. Targets are drawn from a smooth two-chart atlas class: for each target, local chart coordinates are transformed through degree-3 polynomial

Table 3. Expanded monodromy diagnostics for the homogeneous two-unit model.

Block	Metric	Value
Lift geometry	$\|\theta(1) - \theta(0)\|_2$	0.8485
Lift geometry	$\|\theta(1) - \tau\theta(0)\|_2$	5.2×10^{-17}
Lift geometry	$\max_x \lvert f_{\theta(1)}(x) - f_{\theta(0)}(x)\rvert$	1.11×10^{-16}
Lift geometry	$\max_x \lvert f_{\theta(1)}(x) - f_{\tau\theta(0)}(x)\rvert$	1.11×10^{-16}
Path smoothness	$\max_k \|\theta_{k+1} - \theta_k\|_2$	1.11×10^{-3}
Path smoothness	$\mathrm{med}_k \|\theta_{k+1} - \theta_k\|_2$	1.11×10^{-3}
Canonicalizer pathology	$\max_k \|\mathcal{C}(\theta_{k+1}) - \mathcal{C}(\theta_k)\|_2$	0.8359
Canonicalizer pathology	$\mathrm{med}_k \|\mathcal{C}(\theta_{k+1}) - \mathcal{C}(\theta_k)\|_2$	1.11×10^{-3}
Canonicalizer pathology	jump amplification	752.5
Canonicalizer pathology	detected jump time t_{jump}	0.5004

Table 4. Per-ϵ conditioning estimates and theory values for Experiment 2.

ϵ	$\widehat{L}_{\text{root}}$	$L^{\text{th}}_{\text{root}}$	root rel. err.	$\widehat{L}_{\text{hom}}$	$L^{\text{th}}_{\text{hom}}$	hom rel. err.
0.1	1.581	1.581	4.88×10^{-6}	2.236	2.236	4.88×10^{-6}
0.05	2.236	2.236	4.88×10^{-6}	3.162	3.162	4.88×10^{-6}
0.02	3.536	3.536	4.88×10^{-6}	5	5	4.88×10^{-6}
0.01	5	5	4.88×10^{-6}	7.071	7.071	4.88×10^{-6}
5×10^{-3}	7.071	7.071	4.88×10^{-6}	10	10	4.88×10^{-6}
2×10^{-3}	11.18	11.18	4.88×10^{-6}	15.81	15.81	4.88×10^{-6}
1×10^{-3}	15.81	15.81	4.88×10^{-6}	22.36	22.36	4.88×10^{-6}

coefficient maps and glued by the theorem-matched two-chart partition of unity. A fixed master seed 314159 is used to generate 40 target functions.

We compare three deterministic ridge-regression learners:

1. a one-chart learner using principal branch coordinates and degree-5 polynomial features (21 features total),
2. a theorem-matched two-chart learner with degree-3 local features ($10 + 10 = 20$ features),
3. a three-chart overcomplete control that contains the two-chart class exactly (30 features).

For each target and learner, we record:

1. test MSE on the full evaluation grid,
2. maximum absolute error,
3. cut-strip MSE on an angular strip of width 0.18 near the principal cut,
4. across-cut defect, measured by evaluating the learned function on paired points separated by a jump angle $\Delta\theta = 0.015$ across the cut.

The theorem predicts a sharp phase transition at chart count 2: one chart fails because of the branch cut, while two charts suffice and three charts do not materially improve over two.

J.4. Experiment 4: Bundle-Valued / Optimizer-Proposal Threshold and Local Attention Corollary

This experiment validates the section-valued universality theorem and the optimizer corollary, and then instantiates the same local obstruction in a modern attention-head slice.

Bundle-valued / optimizer-proposal task. The quotient domain is again the linked annulus with inner radius 10^{-2} and outer radius 9×10^{-2}. The target class is a smooth two-chart sign-bundle atlas class. In the reference chart, the target section has the form

$$c(z) = -(a_1 + a_3|z|^2 + a_5|z|^4)z,$$

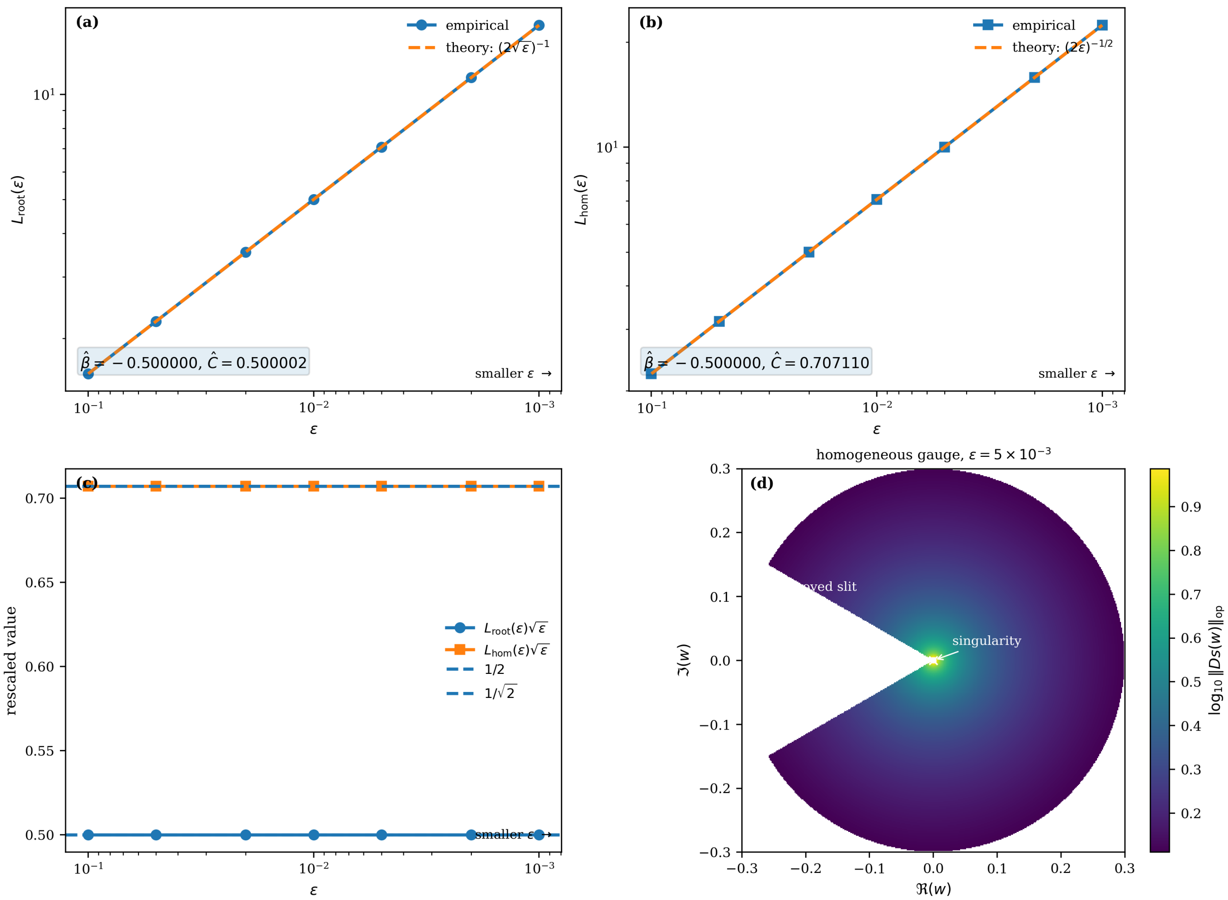

Figure 4. Expanded conditioning diagnostics. **(a)** Dense-grid estimates of $\widehat{L}(\epsilon)$ for the square-root branch and the normalized homogeneous gauge. **(b)** Rescaled curves $\widehat{L}(\epsilon)\sqrt{\epsilon}$ collapse onto the exact constants $1/2$ and $1/\sqrt{2}$. **(c)** Pointwise conditioning heatmap on a representative slit sector. **(d)** Per-ϵ theory-versus-estimate comparison on the same log-log scale.

Table 5. Expanded aggregate diagnostics for Experiment 3 over the sampled atlas-target family. Means and standard deviations are reported over target draws.

Model	Total features	Test MSE	Max abs. err.	Cut-strip MSE	Across-cut defect
1 chart	21	$6.54 \times 10^{-4} \pm 7.18 \times 10^{-4}$	0.1446 ± 0.1013	$1.02 \times 10^{-5} \pm 9.09 \times 10^{-6}$	$2.71 \times 10^{-3} \pm 6.48 \times 10^{-4}$
2 charts	20	$3.12 \times 10^{-20} \pm 1.5 \times 10^{-20}$	$9.1 \times 10^{-10} \pm 2.29 \times 10^{-10}$	$7.86 \times 10^{-21} \pm 4.52 \times 10^{-21}$	$2.68 \times 10^{-11} \pm 7.63 \times 10^{-12}$
3 charts	30	$1.99 \times 10^{-19} \pm 9.31 \times 10^{-20}$	$2.08 \times 10^{-9} \pm 5.28 \times 10^{-10}$	$4.77 \times 10^{-20} \pm 2.84 \times 10^{-20}$	$6.04 \times 10^{-11} \pm 1.97 \times 10^{-11}$

which is sign-covariant under $z \mapsto -z$, polynomial of total degree 5 in $(\Re z, \Im z)$, and points radially inward, so it is also a natural memoryless optimizer-proposal field. The coefficient family consists of the five deterministic triples

$$(1.00, 8, 0),\ (0.90, 6, 30),\ (1.10, 10, 20),\ (0.80, 4, 40),\ (1.20, 7, 10).$$

Training uses 7000 annulus samples with cut oversampling (seed 1); test evaluation uses 16000 annulus samples with the same sampling rule (seed 2). We compare one-chart, two-chart, and three-chart section learners with feature degrees and atlas structures matching the implementation in Sections G and H. The one-step optimizer metric uses the local potential

$$L(z) = \frac{1}{2}|z|^2$$

and step size $\eta = 0.25$. The recorded metrics are section MSE, maximum absolute section error, cut-strip MSE, across-cut defect, one-step gap, and cosine similarity.

Local attention-head corollary. For the attention slice, we use a synthetic two-head local model with

$$d_{\text{model}} = d_k = d_v = 2, \qquad h = 2, \qquad \rho = 0.05, \qquad \alpha = 0.4.$$

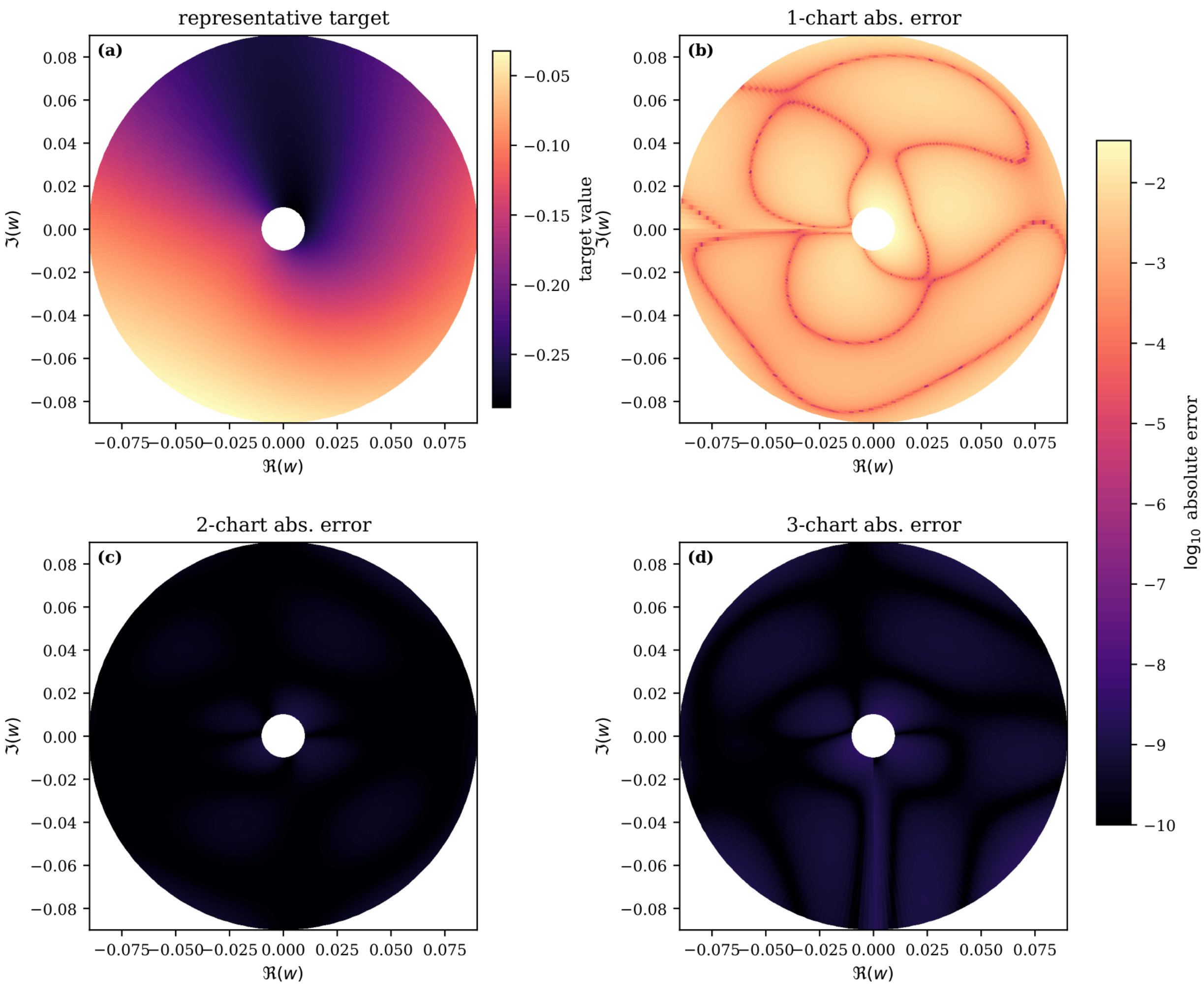

Figure 5. Expanded invariant-learning threshold diagnostics. **(a)** Representative target from the smooth two-chart atlas class. **(b)** Mean absolute error heatmap for the one-chart learner, showing a persistent ridge near the cut. **(c)** Mean absolute error heatmap for the two-chart learner, which collapses to numerical precision. **(d)** Aggregate error metrics versus chart count, showing the sharp threshold at 2.

The ordered lift is parameterized by

$$z(t) = \sqrt{\rho}\, e^{\pi i t}, \qquad w(t) = z(t)^2, \qquad t \in [0, 1],$$

and the two head signatures are

$$Q_1 = I + \Delta(z), \qquad Q_2 = I - \Delta(z), \qquad V_1 = V_2 = I,$$

with

$$\Delta(z) = \begin{pmatrix} \alpha\, \Re z & \alpha\, \Im z \\ 0 & 0 \end{pmatrix}.$$

This produces a full-rank local head-collision loop with the same monodromy as the abstract branch model. We record the endpoint residual to the original ordering, the endpoint residual to the swapped ordering, the maximum ordered head-step norm, the maximum canonicalizer step norm, the jump amplification, and the detected jump time.

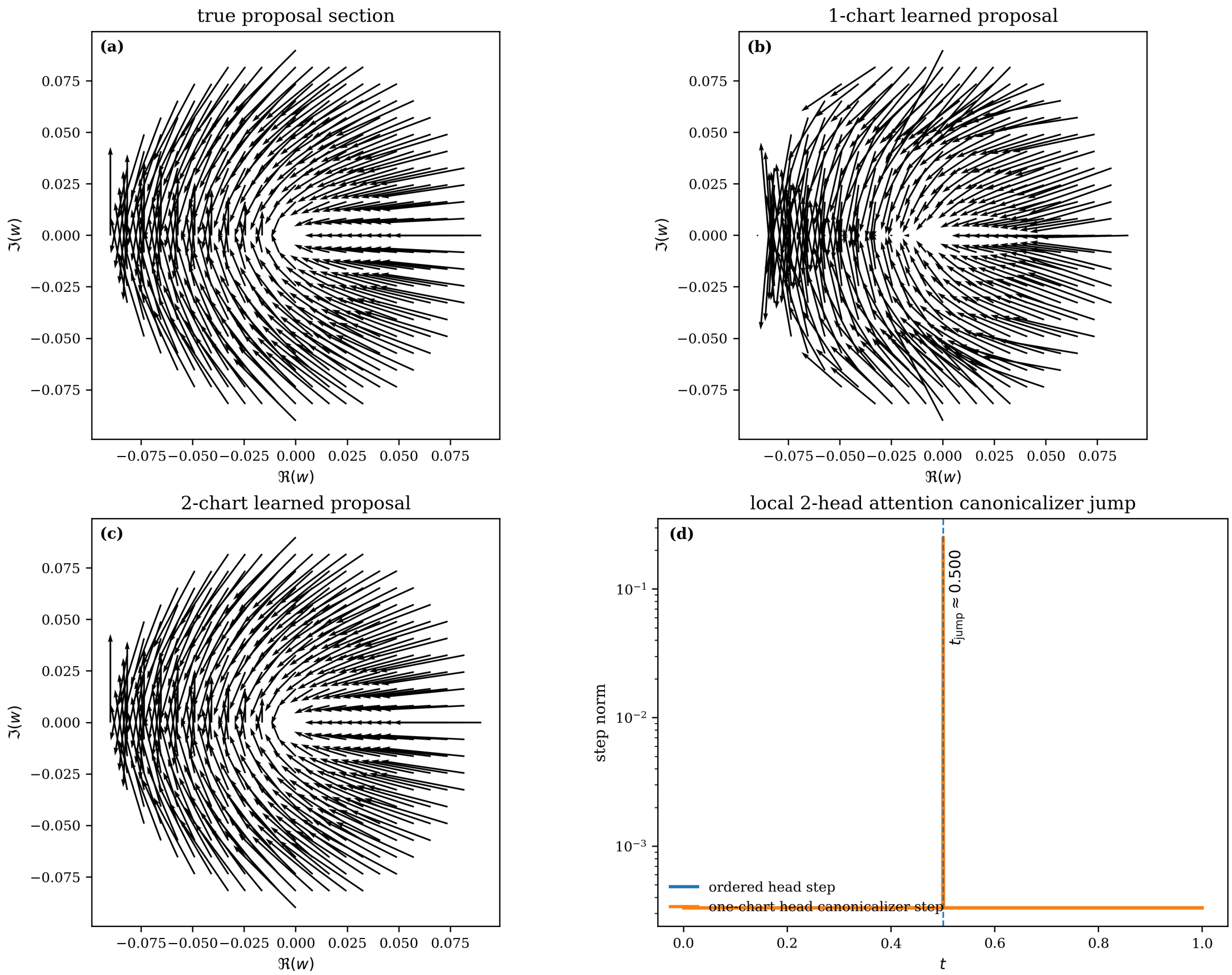

Figure 6. Bundle-valued / optimizer-proposal threshold and local attention-head corollary. **(a)** True proposal section on the linked annulus. **(b)** One-chart learned proposal, which exhibits cut-related distortion. **(c)** Two-chart learned proposal, which matches the target to numerical precision. **(d)** Local two-head attention canonicalizer jump induced by a linked head-collision loop.

J.5. Reproducibility Notes

All experimental assets are generated directly from the four scripts used in the paper:

```
experiment1_final_monodromy.py,
experiment2_final_conditioning.py,
experiment3_final_threshold_updated.py,
experiment4_final_bundle_attention_updated.py.
```

Each script writes both compact main-text artifacts and expanded appendix artifacts, together with raw CSV diagnostics and a plain-text summary. The experiments are deterministic conditional on the explicit seeds and target families stated above. Consequently, numerical reproduction should match the tables in the paper up to standard floating-point roundoff.

Table 6. Expanded bundle-valued / optimizer-proposal diagnostics on the linked annulus.

Model	Section MSE	Max abs. err.	Cut-strip MSE	Across-cut defect	One-step gap	Cosine sim.
1 chart	0.05435 ± 0.0161	0.5395 ± 0.07728	0.106 ± 0.03141	$0.02707 \pm 4.25 \times 10^{-3}$	$8.02 \times 10^{-3} \pm 7.67 \times 10^{-4}$	$0.9358 \pm 3 \times 10^{-4}$
2 charts	$5.34 \times 10^{-18} \pm 5.79 \times 10^{-18}$	$9.59 \times 10^{-9} \pm 7.14 \times 10^{-9}$	$2.6 \times 10^{-18} \pm 2.79 \times 10^{-18}$	$3.17 \times 10^{-7} \pm 4.85 \times 10^{-8}$	$4.82 \times 10^{-11} \pm 4.02 \times 10^{-11}$	$1 \pm 3.83 \times 10^{-12}$
3 charts	$2.6 \times 10^{-17} \pm 2.82 \times 10^{-17}$	$3.11 \times 10^{-8} \pm 2.39 \times 10^{-8}$	$6.49 \times 10^{-18} \pm 7.04 \times 10^{-18}$	$3.17 \times 10^{-7} \pm 4.85 \times 10^{-8}$	$1.05 \times 10^{-10} \pm 8.87 \times 10^{-11}$	$1 \pm 3.83 \times 10^{-12}$

Table 7. Local two-head attention loop diagnostics. The ordered lift ends at the swapped head ordering while the one-chart head-ordering canonicalizer incurs a macroscopic jump.

Metric	Value
$\|\Theta(1) - \Theta(0)\|_2$	0.253
$\|\Theta(1) - \tau\Theta(0)\|_2$	1.55×10^{-17}
$\max_k \|\Theta_{k+1} - \Theta_k\|_2$	3.31×10^{-4}
$\max_k \|\mathcal{C}(\Theta_{k+1}) - \mathcal{C}(\Theta_k)\|_2$	0.253
jump amplification	763.9
detected jump time t_{jump}	0.5004